

FORCING IN MODEL THEORY

A DISSERTATION SUBMITTED TO THE UNIVERSITY OF MANCHESTER
FOR THE DEGREE OF MASTER OF SCIENCE
IN THE FACULTY OF SCIENCE AND ENGINEERING

2025

Sophie Marshall
Department of Mathematics

Contents

Abstract	3
Declaration	4
Intellectual Property Statement	5
Acknowledgements	6
1 Introduction	7
1.1 Structure and content	11
1.2 Conventions and notation	13
1.3 Prerequisites and audience	15
1.4 Forcing literature	16
2 Existential Closure	18
2.1 Existentially closed structures	18
2.2 Existential closure and language	24
2.3 Inductive structures, classes and theories	28
2.4 Resultants	32
3 Term Algebras	40
3.1 The term algebra of a theory	40
3.2 The canonical model of a theory.	43
3.3 Hintikka theories	45
4 Forcing	54
4.1 Forcing structures	54
4.2 Enforceable properties	58
4.3 The compiled structure	72
5 The Forcing Relation	81
5.1 Strong forcing	82
5.2 Weak forcing	84
6 Finite Forcing	96
6.1 Finite forcing structures	96
6.2 Omitting universal types	101
6.3 The finite forcing companion	105

Abstract

This dissertation serves as an introduction to **forcing** in model theory, and surveys and compares the various approaches to model-theoretic forcing taken in the literature. Model-theoretic forcing is a technique for building models of a given first-order theory, while retaining a great deal of control over the finer details of the structure being built, called the **compiled structure**. The technique of forcing involves building ascending chains within a specified partially ordered set of ‘**conditions**’, which are partial pieces of information (usually sets of sentences) which we would like to be true of the compiled structure. Once an infinite chain of conditions has been built, we show how the **term algebra** construction can be used to obtain the compiled structure from the union of the chain of conditions. Then, we consider **enforceable properties**, which are properties that can be enforced of the compiled structure if we are only able to choose some of the conditions in the chain. We develop Wilfred Hodges’ **games** based approach to forcing, where players take turns to choose conditions in the chain, and a property is enforceable if one of the players has a strategic way to play which will ensure that the compiled structure has that property by the end of the chain. We then compare this semantic, games-based approach to the original, more syntactic approach to model-theoretic forcing taken by Abraham Robinson, where we syntactically define the **weak and strong forcing relations**. Finally, we look at the applications of forcing by focusing on a particular type of forcing called **finite forcing**, where conditions are finite sets of atomic and negated-atomic sentences. We use finite forcing to build **existentially closed models of inductive theories**, prove the **Omitting Universal Types** theorem, and construct **finite forcing companions**, which can serve as a well-behaved counterpart theory for theories which do not have a model companion.

Declaration

No portion of the work referred to in the dissertation has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

Intellectual Property Statement

- i. The author of this dissertation (including any appendices and/or schedules to this dissertation) owns certain copyright or related rights in it (the “Copyright”) and she has given The University of Manchester certain rights to use such Copyright, including for administrative purposes.
- ii. Copies of this dissertation, either in full or in extracts and whether in hard or electronic copy, may be made **only** in accordance with the Copyright, Designs and Patents Act 1988 (as amended) and regulations issued under it or, where appropriate, in accordance with licensing agreements which the University has entered into. This page must form part of any such copies made.
- iii. The ownership of certain Copyright, patents, designs, trade marks and other intellectual property (the “Intellectual Property”) and any reproductions of copyright works in the dissertation, for example graphs and tables (“Reproductions”), which may be described in this dissertation, may not be owned by the author and may be owned by third parties. Such Intellectual Property and Reproductions cannot and must not be made available for use without the prior written permission of the owner(s) of the relevant Intellectual Property and/or Reproductions.
- iv. Further information on the conditions under which disclosure, publication and commercialisation of this dissertation, the Copyright and any Intellectual Property and/or Reproductions described in it may take place is available in the University IP Policy (see <http://documents.manchester.ac.uk/DocuInfo.aspx?DocID=487>), in any relevant dissertation restriction declarations deposited in the University Library, The University Library’s regulations (see <http://www.manchester.ac.uk/library/ab-outus/regul-ations>) and in The University’s Guidance on Presentation of Dissertations.

Acknowledgements

I am extremely grateful for the guidance of my supervisor, Marcus Tressl, who has given invaluable feedback, advice, and support throughout the writing of both this dissertation and my main project, as well as during my last four years of undergraduate and postgraduate studies.

I am also very thankful for all the enthusiastic support and encouragement from my family, friends, and peers, as well as all of the academic staff in Manchester's mathematics and philosophy departments who have contributed to my learning and helped to develop my passion for pure mathematics and logic.

1 Introduction

The technique of forcing was pioneered by Paul Cohen in the 1960s in order to prove the independence of the Continuum Hypothesis (CH) from the axioms of Zermelo-Fraenkel set theory with the axiom of Choice (ZFC). It had already been shown in the 1930s by Kurt Gödel that CH was consistent with ZFC. Cohen set out to show that the negation of CH was also consistent with ZFC, and thus that CH was independent from ZFC. Cohen achieved this by using forcing to construct a model of set theory satisfying the negation of CH. Beyond the continuum hypothesis, set-theoretic forcing proved to be a powerful technique for proving the independence of various statements from the axioms of ZFC.¹

In the 1970s, model theorists such as Abraham Robinson set out to adapt the powerful technique of forcing to the more general setting of model theory. Rather than forcing with the specific axioms of set theory, model-theoretic forcing is a tool for constructing models of general first-order theories. Unlike the staple model-theoretic tools for producing models, such as the Löwenheim-Skolem theorems which allow one to produce models of different cardinalities but do not allow for much further control over the properties of the model obtained, forcing is a technique for producing models while retaining a great deal of control over the properties of each individual element, as well as the structure as a whole. An important focus in the study of forcing is finding out exactly what properties can be completely or partially controlled when building models with forcing.

The intuition behind the process of forcing is as follows. We always work with a countable first-order language $\mathcal{L}$, and often start with an $\mathcal{L}$ -theory T . We then extend $\mathcal{L}$ to a larger language $\mathcal{L}^+$ which contains a countably infinite number of new constants, called **witnesses**. We work within a partial order of **conditions**, each of which is a set of $\mathcal{L}^+$ -sentences which represents a fragment of information about finitely many of the elements in the structure we are building. By constructing an infinite ascending chain of these conditions and taking their union, we gradually build up a complete picture of a particular $\mathcal{L}^+$ -structure, which we call the **compiled structure**. The exact procedure for obtaining the compiled structure from the union of a chain of conditions is a technical task which we present in detail in sections 3 and 4. This is the process of building models with forcing.

The question becomes: if one is building a model with forcing and can choose

¹See [Koe23] for an overview of the Continuum Hypothesis and the contribution of Cohen's set-theoretic forcing to showing its independence from ZFC.

infinitely many steps in the chain, but there are also infinitely many steps which we cannot choose, what properties can we still ensure the compiled structure will have? We study this question using Wilfred Hodges' approach to forcing with **games**. We suppose that two players, $\exists$ (she) and $\forall$ (he), are taking turns to choose successive conditions in the chain. We are usually on the side of $\exists$. On their turn, each player can add some information about finitely many elements of the structure being built, subject to certain rules and limitations. The rules and limitations of a particular game are determined by the choice of **forcing structure**, which is a partially ordered set of conditions in which the players must build the chain. Some of these rules apply in every game / forcing structure. For example, a player cannot add any information which contradicts what has already been chosen.

Sometimes, we specify that a player must only choose information which is consistent with a particular theory. For example, if we are forcing with the theory of groups, we might want to make sure that the structure we are building is itself a group. In that case, the players cannot add any information to the chain which contradicts the group axioms. The uniqueness of inverses is an example of this: if the chain already contains the equation $gh_1 = e$, then adding the following would **not** be a legal move: $gh_2 = e, h_1 \neq h_2$ (here, e is the neutral element of the group and a constant in the smaller language and g, h_1, h_2 are witnesses from the extended language).²

Additionally, we sometimes place limitations on the quantity and complexity of the information which the players are allowed to add. These restrictions give rise to many different kinds of forcing. We will look at one example of this, called Robinson's **finite forcing**. In finite forcing, each player on their turn may only add finitely many atomic or negated-atomic sentences. In a language with no relation symbols, such as the language of groups, this means that players can only add a finite number of equations and inequations on their turn. Other kinds of forcing include Robinson's infinite forcing and Henkin forcing, though there are many more.³

Once the rules of the game have been established, we investigate the properties that are **enforceable** of the compiled structure. We consider a game where $\forall$ is allowed to go first, and then the players take turns to choose conditions from a particular forcing structure to form an ascending chain. Given a particular feature

²Our use of the example of forcing with groups where the players take turns to add a finite number of equations and inequations draws on the same example used in the introduction of [Hod85].

³See [Hod85, pp. 275–276] for a detailed list of different kinds of forcing in model theory.

or **property** of structures $\mathcal{P}$, if $\exists$ has a systematic strategy to ensure that, by the end of the game, the compiled structure has the property $\mathcal{P}$, then we say that $\mathcal{P}$ is **enforceable** (with respect to the rules of the game i.e. the choice of forcing structure).

We will see several properties which are always enforceable in every forcing structure. We write T^+ for the $\mathcal{L}^+$ -theory which is the union of the chain chosen over the course of the game. A property which is always enforceable in any forcing structure is that the compiled structure is a model of T^+ , ensuring that each piece of information added will indeed be true in compiled structure. Furthermore, it is always enforceable that every element of the compiled structure is named by infinitely many witnesses.

However, as a non-example of an enforceable property, we return to the scenario where we are forcing with groups, in a game where the players take turns to add a finite number of equations and inequations to the chain. It turns out that, while it is enforceable that the compiled structure is a group, the property of being **abelian** is not enforceable. In fact, being non-abelian is enforceable. Suppose $\exists$ is trying to ensure that the compiled structure is an abelian group, while $\forall$ opposes her. On her turn, $\exists$ may add at most finitely many equations of the form $gh = hg$, which can only ensure that at most finitely many pairs of elements commute. However, when it comes to $\forall$'s turn, only finitely many of the infinite number of witnesses will have been mentioned so far. He may therefore choose two witnesses a, b not yet mentioned in the game and add $ab \neq ba$ without contradicting either the group axioms or any of the previous moves. He thus succeeds in ensuring that, if the compiled structure is a group, then it will be a non-abelian group, as the witnesses a and b must name elements which do not commute. On the other hand, it is always enforceable that a group is non-trivial by adding the sentence $g \neq h$ for witnesses g, h which have not yet been mentioned.

Finite forcing is a particular kind of forcing corresponding to a particular choice of forcing structure for a given theory T . In fact, the forcing we have just described is an example of finite forcing with the theory of groups. When we look at finite forcing specifically, we will show that in finite forcing with certain kinds of theories T - those axiomatised by sentences of the form $\forall \bar{x} \exists \bar{y} \chi(\bar{x}, \bar{y})$ (with χ quantifier-free and $\bar{x}$ and $\bar{y}$ possibly empty tuples of variables) - it is enforceable that the compiled structure is an **existentially closed model** of T .

Existentially closed models of a theory T are a particularly useful and well-

behaved class of structures. For example, the existentially closed models of the theory of fields are exactly the algebraically closed fields (such as the algebraic numbers $\overline{\mathbb{Q}}$ and the complex numbers $\mathbb{C}$). Algebraically closed fields F have the property that if F is contained in a larger field K , every finite system of equations and inequations with parameters from F which has a solution in K already has a solution in F . We generalise this property from algebra to model theory, defining existential closure precisely in section 2. The fact that finite forcing allows us to produce existentially closed models of certain theories is one way in which forcing can be used in model theory to produce powerful results.

One of the most useful features of forcing, which we present in the [conjunction lemma](#), is that the conjunction of countably many enforceable properties is itself enforceable. It is often easier to handle each of the properties individually, so if we want to show that a structure has the property $\mathcal{P}$, it suffices to show that $\mathcal{P}$ is implied by a countable collection of properties which are all individually enforceable ([Hod85, p. vii]).

The aim of this dissertation is not just to give an introductory account of model-theoretic forcing, but also to survey the different approaches to model-theoretic forcing taken in the literature. For this reason, the focus of this dissertation will be on developing and comparing theories of forcing, with less of a focus on discussing examples or applications - though these will not be completely neglected. While we mainly discuss forcing within Hodges' approach with games, as described above, which is a more semantic approach and has the advantage of being fairly intuitive, we also consider the more syntactic approach taken by Robinson in his original adaptation of forcing from set theory to model theory. In Robinson's approach, forcing is based on two syntactically defined forcing relations (strong and weak) which hold between conditions in a forcing structure and sentences. For a condition P and a sentence σ , the strong forcing relation ' P strongly forces σ ' is defined by induction on the complexity of σ , and the weak forcing relation is then defined as follows: P weakly forces σ if and only if P strongly forces $\neg\neg\sigma$. A key result of this dissertation will be to compare and unite these two different perspectives and show that Hodges approach to forcing with enforceable properties and games is in fact equivalent to Robinson's weak forcing relation. Thus, the powerful intuition provided by understanding forcing through games is complemented by the formal rigour of the syntactic relation.

1.1 Structure and content

In sections 2 and 3, we set out some necessary prerequisite material required for our later discussion of forcing. Though some of this material does not find applications until much later in this dissertation, it has been removed to the beginning so that it does not interrupt the main mathematical focus of this dissertation.

In section 2, we define what it means for one structure to be **existentially closed** in another structure. If $\mathcal{M}$ is existentially closed in a superstructure $\mathcal{N}$ then any system of equations and inequations with parameters from $\mathcal{M}$ which has a solution in $\mathcal{N}$ must already have a solution in $\mathcal{M}$. The full definition is more general, extending to systems of atomic and negated atomic formulas. A model of an $\mathcal{L}$ -theory T is then an **existentially closed model** of T if it is existentially closed in all of its superstructures which are models of T .

Existential closure is a property which is closely related to a certain kind of theory: those which can be axiomatised entirely by sentences of the form $\forall \bar{x} \exists \bar{y} \chi(\bar{x}, \bar{y})$ where χ is quantifier-free and $\bar{x}$ and $\bar{y}$ are possibly empty tuples of variables, called **$\forall\exists$ -theories** or **inductive theories**. This material on existential closure and $\forall\exists$ -theories is developed for its application particularly in section 6, where we will use finite forcing to build existentially closed models of $\forall\exists$ -theories.

In section 3, we define the **term algebra** uniquely determined by a particular theory. Term algebras are structures in which every element is named by a constant (or iterated function of constants) in the language. They are vital to forcing as the structures which we build in forcing will be term algebras. However, the term algebra of a theory T is not guaranteed to be a model of T , so we introduce certain kinds of theories called **Hintikka theories**, and show that if T is a Hintikka theory then the term algebra of T will be a model of T .

We also discuss a similar, but subtly different construction to the term algebra: the **canonical model** of a theory. Since the canonical model is the preferred method of obtaining a structure uniquely determined by a theory in [Hod85], our main source, we compare the two construction methods. This allows us to use results in forcing originally stated using canonical models in [Hod85], but present them using term algebras instead.

In section 4, we begin to properly study forcing. We always work with a countable language $\mathcal{L}$, which we extend to $\mathcal{L}^+$ by the addition of a countably infinite number of new constants, called **witnesses**. We follow Hodges' approach in [Hod85], where

forcing is understood within the context of a game between two players. We define a **forcing structure**, a partially ordered set containing **conditions**, which are usually consistent sets of $\mathcal{L}^+$ -sentences in which at most finitely many witnesses are mentioned. In a forcing game, the players take turns choosing conditions to form an infinite ascending chain.

A property is then called **enforceable** if, no matter how one's opponent plays, one can play according to a systematic strategy which guarantees that the union of the chain - called the **compiled theory** - will have the desired property. We show that one such example of this is that it is enforceable in any forcing structure that the compiled theory is a Hintikka theory. This allows us to reliably produce a structure which is guaranteed to be a model of the compiled theory - its term algebra, which we call the **compiled structure**. We also show a central result in forcing: the [Conjunction Lemma](#), which states that if a countable collection of properties are all enforceable, then their conjunction is also enforceable.

In section [5](#), we study forcing using a completely different approach, following Robinson's original approach to model-theoretic forcing in [\[Rob71\]](#). Unlike the semantic, highly intuitive approach of forcing with games, in this section we axiomatise a forcing relation which holds between conditions in a forcing structure and sentences. We prove various properties of both the weak and strong forcing relations. This allows us to prove a key result of this dissertation: that Robinson's syntactic definition of weak forcing is equivalent to Hodges' more semantic, games based approach.

Finally, in section [6](#), we narrow our focus to a particular kind of forcing: **finite forcing**, also known as Robinson forcing. We define the finite forcing structure for a theory T , where conditions are finite sets of atomic and negated-atomic sentences consistent with T , and then investigate which properties are enforceable in finite forcing. We apply our earlier work on existential closure and $\forall\exists$ -theories from section [2](#) to prove one of the most useful results of finite forcing: that if T is an $\forall\exists$ -theory, it is enforceable that the compiled structure is an existentially closed model of T . We apply this result to prove the **Omitting Universal Types** theorem, a variant of the Classical Omitting Types theorem which was proved in [\[Mar25\]](#). We conclude this dissertation by defining the **finite forcing companion** of a theory, a well-behaved counterpart of a first-order theory T which can be used to study T .

1.2 Conventions and notation

We use conventional notation of model theory. We use $\mathcal{L}$ and $\mathcal{L}^+$ for languages. We use σ and τ for sentences. We will usually use φ and ψ for general formulas. We often use χ for quantifier-free formulas, and θ for basic formulas. We use x, y for logical variables, a, b for elements of structures, c, d for constant symbols of the language, and t, s for terms. We write, for example $\bar{x}, \bar{a}, \bar{c}, \bar{t}$ for tuples. When we write a formula as $\varphi(\bar{x})$, we mean that the variables which occur **freely** in φ are among the tuple $\bar{x}$ of variable symbols, and we assume that none of the $\bar{x}$ also occur bound in φ . If $\varphi(\bar{x})$ is a formula and $\bar{c}$ is a tuple of constant symbols (which we will often assume implicitly to be of matching length) then we write $\varphi(\bar{c})$ for the sentence resulting from replacing each free occurrence of each x_i from $\bar{x}$ in φ with the corresponding c_i from $\bar{c}$.

We use capital Greek letters to denote sets of formulas or sentences. Thus Φ and Ψ will usually denote sets of formulas, and we will often use Θ to denote a set of basic sentences. We write $\Phi(\bar{x})$ to indicate that every formula in Φ has free variables at most among the $\bar{x}$. If $\bar{c}$ is a tuple of constants, we write $\Phi(\bar{c})$ to denote the set of sentences $\varphi(\bar{c})$ for each $\varphi(\bar{x}) \in \Phi(\bar{x})$. We will use Σ for a set of sentences, which we do not necessarily assume to be consistent. We reserve the use of T for theories, which are assumed to be consistent, but not necessarily complete or deductively closed. We will write $\vec{T}$ for the deductive closure of T . We sometimes will use U to denote a subset of a theory T containing just atomic sentences.

Since we assume the completeness and soundness theorems of first order logic, and this is a dissertation focused on model theory, we will prefer to speak in semantic terms rather than syntactic, though the syntactic nature of forcing means that this dissertation will contain more syntactic content than the average work on model theory. Furthermore, ‘satisfiable’, ‘consistent’, and ‘has a model’ all mean the same thing. We write $\Sigma \models \sigma$ to mean that every model of Σ is a model of σ . We write $\Phi(\bar{x}) \models \varphi(\bar{x})$ to mean that in every model where Φ is satisfied at some assignment of variables, $\varphi(\bar{x})$ is also satisfied at exactly the same assignment of variables.

Structures are usually denoted with $\mathcal{M}, \mathcal{N}$, and sometimes $\mathcal{A}, \mathcal{B}$. The former usually stand for general structures, while the latter, $\mathcal{A}$ and $\mathcal{B}$, are usually reserved for a particular kinds of structures: canonical models and term algebras. We do not use separate notation for a structure and its domain (underlying set of elements) where it can be avoided, but where necessary for clarity, we may write $\mathcal{M}$ for a structure and $|\mathcal{M}|$ for its domain. If $\mathcal{M}$ and $\mathcal{N}$ are structures we write $\mathcal{M} \hookrightarrow \mathcal{N}$

for an embedding of structures, and we write $\mathcal{M} \leq \mathcal{N}$ if $\mathcal{M}$ is a substructure of $\mathcal{N}$ (meaning the identity map $\mathcal{M} \hookrightarrow \mathcal{N}$ is an embedding), and we write $\mathcal{M} \prec \mathcal{N}$ if $\mathcal{M}$ is an elementary substructure of $\mathcal{N}$. We write $\mathcal{M} \cong \mathcal{N}$ if the two structures are isomorphic as $\mathcal{L}$ -structures.

If $\varphi(\bar{x})$ is a formula and $\mathcal{M}$ is a structure, then we write $\bar{a} \in \mathcal{M}^{\bar{x}}$ to indicate that $\bar{a}$ is a tuple of elements from $\mathcal{M}$ of the same length as $\bar{x}$. We may then write $\mathcal{M} \models \varphi(\bar{a})$ if $\mathcal{M} \models \varphi[h]$ at some assignment of variables $h : \text{Vbl} \rightarrow \mathcal{M}$ which assigns each x_i in $\bar{x}$ to the corresponding a_i in $\bar{a}$. Similarly, if $\Phi(\bar{x})$ is a set of formulas and $\bar{a} \in \mathcal{M}^{\bar{x}}$ then we write $\mathcal{M} \models \Phi(\bar{a})$ if $\mathcal{M} \models \varphi(\bar{a})$ for every $\varphi \in \Phi$.

All languages are first order; we do not use any infinitary languages in this dissertation. The following symbols of first-order logic are taken as primitive: $=, \neg, \wedge, \vee$, and $\exists$ (and brackets). The remaining symbols $\rightarrow, \leftarrow, \leftrightarrow$, and $\forall$ are considered to be abbreviations for the relevant combinations of primitive symbols. For example, $\forall \bar{x} \varphi$ is considered an abbreviation of $\neg \exists \bar{x} \neg \varphi$. If Φ is a **finite** set of formulas then we may write $\bigwedge \Phi$ and $\bigvee \Phi$ for the formula obtained by taking the conjunction and disjunction respectively of all the formulas in Φ . We do not consider $\bigwedge$ or $\bigvee$ to be a formal part of our language; these are instead used as convenient abbreviations. Where possible, we avoid unnecessary technicalities and consider formulas and sentences to be ‘the same’ if they are logically equivalent. This will allow us to write formulas in prenex-normal form, and we will often put quantifier-free formulas in disjunctive or conjunctive normal form, whichever is more convenient.

We frequently work with languages extended by constants. We will write $\mathcal{L}^+ \supset \mathcal{L}$ for a language extension. We often write T for an $\mathcal{L}$ -theory and T^+ for an $\mathcal{L}^+$ -theory. If $\mathcal{M}$ is an $\mathcal{L}$ -structure, we will usually write $\mathcal{M}^+$ for its expansion to $\mathcal{L}^+$. We will write $\mathcal{M} = \mathcal{M}^+ \upharpoonright_{\mathcal{L}}$ for the reduct of $\mathcal{M}^+$ to $\mathcal{L}$.

However, in the specific context of forcing, where we construct both theories and structures, we are usually given an $\mathcal{L}$ -theory T and construct a theory in the extended language. We will call this constructed theory in $\mathcal{L}^+$ the compiled theory, and will use T^+ to denote not just any $\mathcal{L}^+$ -theory, but usually the specific compiled theory constructed during forcing. Similarly, the $\mathcal{L}^+$ -structure which we build during forcing will be called the compiled structure. Since the compiled structure will be defined as a particular term algebra, we will denote this with $\mathcal{A}^+$ and its reduct with $\mathcal{A}$.

All forcing notation is introduced in the text. However, to briefly summarise, we use N for a forcing structure and P, Q, R, S for the elements of the forcing structure

called N -conditions (we also sometimes use T in addition to P, Q, R, S when we run out of letters! It will always be clear in context when T refers specifically to a condition in a forcing structure rather than a first-order theory). We write $\overline{P}$ for a chain of conditions and will often write T^+ for $\bigcup \overline{P}$, the union of the chain. We write $\mathcal{P}$ for the properties of theories which we enforce. The symbol $\Vdash$ is used for the forcing relation. In section 5, we use $\Vdash^s$ and $\Vdash^w$ to distinguish between the two forms of the forcing relation, respectively strong and weak. However, after section 5, our focus is entirely on weak forcing, and we thus simplify our notation and simply use $\Vdash$ for the weak forcing relation in place of $\Vdash^w$.

In this dissertation, the natural numbers $\mathbb{N}$ start from 1.

1.3 Prerequisites and audience

The reader is assumed to be familiar with the basics of model theory, including first-order logic, languages, structures, morphisms, and Tarski's definition of truth in a structure. We assume knowledge of the soundness, completeness, and compactness theorems of first-order logic, as well as the Löwenheim-Skolem theorems. Introductions to model theory can be found in [Hod93] and [Tre10], which are two of our primary sources for model theory, in addition to [Mar02].

This dissertation continues the work of the main project [Mar25], in which we proved the Omitting Types theorem using techniques which resemble the forcing that will be explored in this dissertation. All results which are central to this text are restated where relevant, and indeed we give an account of several topics which appeared in the main project in this dissertation; in particular, we introduce both term algebras and Hintikka theories from scratch with significant improvements and corrections. Some material on language extensions and existentially closed structures is given a more brief treatment in this dissertation, particularly in areas which have already been explored in detail in [Mar25].

We will on several occasions refer to the theory of ordered sets, including partial, total, and well orders; this knowledge is generally assumed rather than introduced. We use dense linear orders without endpoints in examples, as well as groups, rings, and (algebraically closed) fields. However, knowledge of these objects is not necessary to understand the core material of the dissertation. Only minimal knowledge of cardinal arithmetic is required, as we will almost entirely be working with countable languages and structures. The relevant ordered set theory and cardinal arithmetic can be found in [JT24]. The relevant algebra can be found in [Fra14].

1.4 Forcing literature

Since this dissertation is intended to serve as a survey of the different approaches taken to forcing in the literature, we now give a brief summary of the literature which contributed to the content on forcing in this dissertation. We also mention a few texts which were not directly used but are relevant to the topic. The following description of the literature is likely not accessible to a reader who has not yet read the dissertation, but is intended as a useful overview of the literature which a reader may wish to refer back to, or as a resource to direct further exploration of the topic.

The primary source used in this dissertation is Hodges' *Building Models By Games* ([Hod85]). This text frames the construction methods of forcing within the context of games. In this dissertation, we make use of where the text has developed a general theory of forcing notions (called forcing *structures* in this dissertation) and enforceable properties, before looking specifically at finite forcing, especially the construction of existentially closed models of $\forall\exists$ -theories in finite forcing. The text continues to look at finite generic structures in finite forcing, before later exploring several other kinds of forcing. The text also contains a comprehensive list of the different types of forcing in model theory (such as Robinson finite forcing, Robinson infinite forcing, Henkin forcing, etc.) and their main features ([Hod85, pp. 275-276]).

A secondary source is Hodges' *Model Theory* ([Hod93]). There is some forcing in this text which we have used in one or two proofs, such as the proof of the Omitting Universal Types theorem in subsection 6.2. However, this text is mainly a wide and detailed survey of the subject of model theory as a whole, and many of the foundational model-theoretic results presented in this dissertation are taken directly from this text.

We also use Chang and Keisler's *Model Theory* ([CK73]). This, like [Hod93], is a comprehensive treatment of the subject of model theory as a whole. However, in its discussion of model completeness and model companions, both the finite forcing companion and infinite forcing companion of a theory are discussed, and the axioms for the weak and strong relations for both finite and infinite forcing are given as part of exercises ([CK73, pp. 209-210]).

We also refer to Robinson's original paper *Forcing in Model Theory* ([Rob71]), in which Robinson's model-theoretic finite forcing was originally presented. This is a very early example of forcing in model theory, and the subject was greatly developed between the publication of this paper and Hodges' book on the subject ([Hod85]).

Forcing also appears in briefly in Ziegler’s paper *Algebraisch Abgeschlossene Gruppen* (Algebraically Closed Groups) ([Zie80]). This paper, which applies forcing only within the narrow context of groups, is credited by Hodges as one of the earliest examples to identify that Robinson’s weak forcing relation could be studied with games ([Hod85, p. 34]).

In addition to the sources above, all of which were used directly in this dissertation, we mention two further sources which were not directly used, but may be of interest for further research. One is Bianconi’s *Lectures on Model-Theoretic Forcing* ([Bia18]), which gives a short introduction to the topic, covering forcing notions, a version of the weak and strong forcing relations, generic models, and Robinson infinite forcing.

The other is Saracino’s *Generic Structures and Model Companions* [Sar71], which studies Robinson’s infinite forcing, infinite generic structures, and the infinite forcing companion of a theory.

2 Existential Closure

Existential closure is a particularly desirable property of structures in model theory, and is a property which has been studied extensively. A structure may only ever be meaningfully said to be existentially closed *inside another structure*. If we have structures $\mathcal{M} \leq \mathcal{N}$ and $\mathcal{M}$ is existentially closed in $\mathcal{N}$, then these structures will have the property that given any finite system of equations and inequations with parameters from $\mathcal{M}$, if the system has a solution in $\mathcal{N}$ then the system must already have a solution in $\mathcal{M}$. This captures the idea of existential closure in a language with no relation symbols, but we will give a more general definition in terms of quantifier-free formulas.

We may still meaningfully describe a single structure $\mathcal{M}$ as an **existentially closed model** of a theory T . An existentially closed model of T is a structure which is existentially closed in every one of its superstructures which is also a model of T . The classic example of existentially closed models are the algebraically closed fields, which are the existentially closed models of the field axioms in the language of rings.

First, we classify certain types of formulas, sentences, and theories based on the presence of quantifiers. It turns out that theories which can be axiomatised entirely by sentences of the form $\forall \bar{x} \exists \bar{y} \chi(\bar{x}, \bar{y})$ where χ is quantifier-free, called $\forall\exists$ -theories are particularly relevant to existentially closed models. Returning to our example of algebraically closed fields, one of their most useful features is that any field can be embedded into an algebraically closed field (a proof of this can be found in [Mar24]). We will see that it is not a coincidence that the existentially closed fields have this property and that the theory of fields is an $\forall\exists$ -theory. See [Hod93, pp. 37-38] to see that the theory of fields is an $\forall\exists$ -theory.

The significance of existentially closed models in model theory is somewhat similar to the significance of algebraically closed fields in not just field theory, but ring theory and algebra in general. For this reason, one of the most useful features of **finite forcing**, which will be explored in detail in section 6, is that it allows us to construct existentially closed models of $\forall\exists$ -theories.

2.1 Existentially closed structures

First, we introduce a substantial amount of new terminology. This will allow us to more precisely differentiate between different types of formulas, sentences, and theories by the quantifiers which appear in them (when in prenex-normal form).

This material is adapted from [Hod93, pp. 47-49, 361-369] and [Tre10, pp. 58-60]. Some of the most basic, but fundamental content appeared in [Mar25] and has been reproduced without much modification.

2.1.1 Definition. Let $\mathcal{L}$ be a language.

- (i) An $\exists$ -formula is an $\mathcal{L}$ -formula which is logically equivalent to one of the form $\exists \bar{x} \chi(\bar{x})$, where χ is quantifier-free.
- (ii) An $\forall$ -formula is an $\mathcal{L}$ -formula which is logically equivalent to one of the form $\forall \bar{x} \chi(\bar{x})$, where χ is quantifier-free.
- (iii) An $\forall\exists$ -formula is an $\mathcal{L}$ -formula which is logically equivalent to one of the form $\forall \bar{x} \exists \bar{y} \chi(\bar{x}, \bar{y})$, where χ is quantifier-free.
- (iv) We denote subsets of $\text{Fml}(\mathcal{L})$ defined by these classifications with $\forall\text{-Fml}(\mathcal{L})$, $\exists\text{-Fml}(\mathcal{L})$, and $\forall\exists\text{-Fml}(\mathcal{L})$.
- (v) The similarly defined subsets of $\text{Sen}(\mathcal{L})$ are denoted with $\forall\text{-Sen}(\mathcal{L})$, $\exists\text{-Sen}(\mathcal{L})$, and $\forall\exists\text{-Sen}(\mathcal{L})$.

We also call $\forall$ -formulas **universal**, and $\exists$ -formulas **existential**.

2.1.2 Remark. Note that we allow the tuples within the range of quantifiers in (i)-(iii) above to be empty. Thus, quantifier-free formulas/sentences are contained in the sets of all higher tier formulas/sentences. As a consequence, quantifier-free, universal, and existential formulas are all $\forall\exists$ -formulas as well.

Note that the notation used by Hodges in [Hod85] and [Hod93] is different from ours. In these texts, $\forall_1$ is used instead of $\forall$ and $\exists_1$ is used instead of $\exists$. Furthermore, $\forall_2$ is used where we use $\forall\exists$. Therefore, these texts contain various results about ‘ $\forall_2$ -theories’, and these should be compared to our results about $\forall\exists$ -theories.

Hodges’ notation allows for a more expansive hierarchy, where for example, the set $\forall_{n+1}\text{-Fml}(\mathcal{L})$ is the smallest set containing all $\forall_n$ and $\exists_n$ formulas which is closed (up to logical equivalence) under conjunction, disjunction, and adding universal quantifiers to the front. See [Hod93, pp. 47-48] for further explanation of this hierarchy. However, in this dissertation, we will only need the classifications $\forall$, $\exists$, and $\forall\exists$.

2.1.3 Definition. Let $\mathcal{L}$ be a language.

- (i) Let $\mathcal{M} \leq \mathcal{N}$ be $\mathcal{L}$ -structures. We say that $\mathcal{M}$ is **existentially closed (e.c.)**

in $\mathcal{N}$ if for every quantifier free formula $\chi(\bar{x}, \bar{y})$ and every tuple of parameters $\bar{a} \in \mathcal{M}^{\bar{y}}$ we have:

$$\mathcal{N} \models \exists \bar{x} \chi(\bar{x}, \bar{a}) \implies \mathcal{M} \models \exists \bar{x} \chi(\bar{x}, \bar{a})$$

- (ii) Let K be a class of $\mathcal{L}$ -structures and let $\mathcal{M} \in K$. Then, $\mathcal{M}$ is said to be **existentially closed in K** if whenever $\mathcal{N} \in K$ with $\mathcal{M} \leq \mathcal{N}$, then $\mathcal{M}$ is existentially closed in $\mathcal{N}$.
- (iii) Let T be an $\mathcal{L}$ -theory and $\mathcal{M} \models T$ an $\mathcal{L}$ -structure. If K is the class of all models of T (in $\mathcal{L}$), and $\mathcal{M}$ is e.c. in K , then we say that $\mathcal{M}$ is an **existentially closed model of T** .

We present the following useful lemma which considers the behaviour of universal and existential formulas with respect to embeddings and superstructures. The proof is omitted, but can be found in [Mar25, pp. 43-44].

2.1.4 Lemma. *Let $\mathcal{L}$ be a language and let $\mathcal{M}$ be an $\mathcal{L}$ -structure. Then:*

- (i) *Existential formulas are preserved by embeddings. That is, if $f : \mathcal{M} \hookrightarrow \mathcal{N}$ then for all $\varphi(\bar{x}) \in \exists\text{-Fml}(\mathcal{L})$ and $\bar{a} \in \mathcal{M}^{\bar{x}}$ if $\mathcal{M} \models \varphi(\bar{a})$ then $\mathcal{N} \models \varphi(f(\bar{a}))$.*
- (ii) *Universal formulas are preserved in substructures. That is, if $\mathcal{N} \leq \mathcal{M}$, then for all $\varphi(\bar{x}) \in \forall\text{-Fml}(\mathcal{L})$ and $\bar{a} \in \mathcal{N}^{\bar{x}}$ if $\mathcal{M} \models \varphi(\bar{a})$ then $\mathcal{N} \models \varphi(\bar{a})$.*

An application of this lemma to the definition of e.c. structures is the following proposition:

2.1.5 Proposition. *Let $\mathcal{M} \leq \mathcal{N}$ be $\mathcal{L}$ -structures. Then, the following are equivalent:*

- (i) *$\mathcal{M}$ is existentially closed in $\mathcal{N}$*
- (ii) *The inclusion map $\mathcal{M} \hookrightarrow \mathcal{N}$ preserves all $\forall$ -formulas*

Proof. Recall that we defined $\forall$ -formulas and $\exists$ -formulas up to logical equivalence. Thus, if φ is an $\exists$ -formula, then $\neg\varphi$ is an $\forall$ -formula and vice versa.

By definition, $\mathcal{M}$ is e.c. in $\mathcal{N}$ iff for all $\varphi(\bar{x}) \in \exists\text{-Fml}(\mathcal{L})$ and $\bar{a} \in \mathcal{M}^{\bar{x}}$ we have:

$$\mathcal{N} \models \varphi(\bar{a}) \implies \mathcal{M} \models \varphi(\bar{a})$$

The equivalent, contrapositive statement to this is that for all $\varphi(\bar{x}) \in \exists\text{-Fml}(\mathcal{L})$ (equivalently, for all $\neg\varphi(\bar{x}) \in \forall\text{-Fml}(\mathcal{L})$) and all $\bar{a} \in \mathcal{M}^{\bar{x}}$, we have:

$$\mathcal{M} \models \neg\varphi(\bar{a}) \implies \mathcal{N} \models \neg\varphi(\bar{a})$$

This is precisely the statement that the inclusion map $\mathcal{M} \hookrightarrow \mathcal{N}$ preserves all $\forall$ -formulas. $\square$

Like how we defined the classifications of $\forall$ and $\exists$ -formulas, we now define the classifications of **primitive** and **positive primitive** formulas.

2.1.6 Definition. A formula $\varphi(\bar{x}) \in \text{Fml}(\mathcal{L})$ is said to be **primitive** if it is equivalent to one of the form:

$$\exists \bar{y} \bigwedge_{i=1}^n \theta_i(\bar{x}, \bar{y})$$

where each θ_i is a basic $\mathcal{L}$ -formula.⁴ Furthermore, we call φ a **positive primitive** (p.p.) formula if it is equivalent to one of the form above, except we impose the additional requirement that every θ_i is an atomic formula (that is, none of the θ_i are negated atomic).

2.1.7 Observation. Existential formulas and primitive formulas are closely related. Clearly, every primitive formula is an existential formula. However, we also note that every existential $\mathcal{L}$ -formula is of the following form, where we put the quantifier-free part in disjunctive normal form, and where each θ_{ij} is a basic $\mathcal{L}$ -formula:

$$\exists \bar{y} \bigvee_{j=1}^m \bigwedge_{i=1}^n \theta_{ij}(\bar{x}, \bar{y})$$

As existential quantifiers distribute over disjunctions, the existential formula above is logically equivalent to:

$$\bigvee_{j=1}^m \exists \bar{y} \bigwedge_{i=1}^n \theta_{ij}(\bar{x}, \bar{y})$$

It follows that every existential formula is logically equivalent to the disjunction of finitely many primitive formulas.

Applying this, we can characterise existential closure almost identically in terms of **primitive** formulas instead of existential ones.

2.1.8 Proposition. *The following are equivalent for $\mathcal{L}$ -structures $\mathcal{M}$ and $\mathcal{N}$:*

⁴Recall that a **basic formula**, is either an atomic formula or a negated atomic formula.

(i) $\mathcal{M}$ is e.c. in $\mathcal{N}$

(ii) For every existential formula $\varphi(\bar{x})$ and $\bar{a} \in \mathcal{M}^{\bar{x}}$ we have:

$$\mathcal{N} \models \varphi(\bar{a}) \implies \mathcal{M} \models \varphi(\bar{a})$$

(iii) For every primitive $\mathcal{L}$ -formula $\psi(\bar{x})$ and $\bar{a} \in \mathcal{M}^{\bar{x}}$, we have

$$\mathcal{N} \models \psi(\bar{a}) \implies \mathcal{M} \models \psi(\bar{a})$$

Proof. We already know that we have (i) $\Leftrightarrow$ (ii). We show (ii) $\Leftrightarrow$ (iii).

(ii) $\Rightarrow$ (iii): Suppose every $\exists$ -formula $\varphi(\bar{x})$ and $\bar{a} \in \mathcal{M}^{\bar{x}}$ satisfies that if $\mathcal{N} \models \varphi(\bar{a})$ then $\mathcal{M} \models \varphi(\bar{a})$.

Further suppose we have a primitive formula $\psi(\bar{x})$ and $\bar{a} \in \mathcal{M}^{\bar{x}}$ with $\mathcal{N} \models \psi(\bar{a})$. Then, as every primitive formula is an $\exists$ -formula, it follows that $\mathcal{M} \models \psi(\bar{a})$.

(iii) $\Rightarrow$ (ii): In the other direction, suppose every primitive formula $\varphi(\bar{x})$ and $\bar{a} \in \mathcal{M}^{\bar{x}}$ satisfies $\mathcal{N} \models \varphi(\bar{a}) \Rightarrow \mathcal{M} \models \varphi(\bar{a})$. Then, suppose we have the following existential formula, written as the disjunction of primitive $\mathcal{L}$ -formulas, where each θ_{ij} is a basic $\mathcal{L}$ -formula:

$$\bigvee_{j=1}^m \exists \bar{y} \bigwedge_{i=1}^n \theta_{ij}(\bar{x}, \bar{y})$$

For $\bar{a} \in \mathcal{N}^{\bar{x}}$ we have:

$$\begin{aligned} \mathcal{N} &\models \bigvee_{j=1}^m \exists \bar{y} \bigwedge_{i=1}^n \theta_{ij}(\bar{a}, \bar{y}) \\ \iff \mathcal{N} &\models \exists \bar{y} \bigwedge_{i=1}^n \theta_{ij}(\bar{a}, \bar{y}) && \text{for some } 1 \leq j \leq m \\ \implies \mathcal{M} &\models \exists \bar{y} \bigwedge_{i=1}^n \theta_{ij}(\bar{a}, \bar{y}) && \text{for some } 1 \leq j \leq m \\ \iff \mathcal{M} &\models \bigvee_{j=1}^m \exists \bar{y} \bigwedge_{i=1}^n \theta_{ij}(\bar{a}, \bar{y}) && \text{as required} \end{aligned}$$

□

In the proof of the following theorem, we first need to recall a fact of first order logic.

2.1.9. Lemma on constants ([Hod93, Lemma 2.3.2]). Let T be an $\mathcal{L}$ -theory and $\varphi(\bar{x})$ an $\mathcal{L}$ -formula. Let $\bar{c}$ be a tuple of constants, none of which appear in T . Then:

$$T \models \varphi(\bar{c}) \implies T \models \forall \bar{x} \varphi(\bar{x})$$

2.1.10 Theorem ([Tre10, Proposition 3.6.3, pp. 58-59]). Let $\mathcal{M} \leq \mathcal{N}$ be $\mathcal{L}$ -structures. Then, the following are equivalent:

- (i) $\mathcal{M}$ is e.c. in $\mathcal{N}$
- (ii) There is an $\mathcal{L}$ -structure $\mathcal{A}$ such that $\mathcal{M} \prec \mathcal{A}$ and $\mathcal{N} \leq \mathcal{A}$. In other words, $\mathcal{M}$ is a substructure of an elementary extension of $\mathcal{N}$.

Proof.

(ii) $\implies$ (i) We show that the map $\mathcal{M} \hookrightarrow \mathcal{N}$ preserves all $\forall$ -formulas, which by 2.1.5 suffices to show that $\mathcal{M}$ is e.c. in $\mathcal{N}$. Let $\varphi(\bar{x}) \in \forall\text{-Fml}(\mathcal{L})$ and let $\bar{a} \in \mathcal{M}^{\bar{x}}$, and suppose $\mathcal{M} \models \varphi(\bar{a})$. Then, as the elementary embedding $\mathcal{M} \hookrightarrow \mathcal{A}$ preserves all formulas by definition, we have $\mathcal{A} \models \varphi(\bar{a})$. Finally, since $\forall$ -formulas are preserved in substructures (2.1.4) and $\mathcal{N} \leq \mathcal{A}$, from both $\bar{a} \in \mathcal{N}^{\bar{x}}$ (as $\mathcal{M} \leq \mathcal{N}$) and $\mathcal{A} \models \varphi(\bar{a})$ it follows that also $\mathcal{N} \models \varphi(\bar{a})$. Thus $\mathcal{M}$ is e.c. in $\mathcal{N}$.

(i) $\implies$ (ii) For any structure $\mathcal{M}$, we write $\text{Diag}(\mathcal{M})$ for the **quantifier-free** diagram of $\mathcal{M}$ and $\text{Diag}_{\infty}(\mathcal{M})$ for the **complete** diagram of $\mathcal{M}$. These are sets of $\mathcal{L}(\mathcal{M})$ -sentences⁵ defined as follows:

$$\begin{aligned} \text{Diag}(\mathcal{M}) &= \{\varphi(\bar{a}) \mid \mathcal{M} \models \varphi(\bar{a}), \bar{a} \in \mathcal{M}^{\bar{x}}, \varphi(\bar{x}) \in \text{QF-Fml}(\mathcal{L})\} \\ \text{Diag}_{\infty}(\mathcal{M}) &= \{\varphi(\bar{a}) \mid \mathcal{M} \models \varphi(\bar{a}), \bar{a} \in \mathcal{M}^{\bar{x}}, \varphi(\bar{x}) \in \text{Fml}(\mathcal{L})\} \end{aligned}$$

If we have an $\mathcal{L}$ -structure $\mathcal{M}$ and its quantifier-free diagram $\text{Diag}(\mathcal{M})$, then for any $\mathcal{L}(\mathcal{M})$ -structure with $\mathcal{N} \models \text{Diag}(\mathcal{M})$ there is a map $\mathcal{M} \hookrightarrow \mathcal{N}$ defined by $a \mapsto c_a^{\mathcal{N}}$ for each $a \in \mathcal{M}$ which, by the definition of the quantifier-free diagram of $\mathcal{M}$, must preserve every quantifier-free $\mathcal{L}$ -formula, and is thus an embedding.

Furthermore, if we have an $\mathcal{L}$ -structure $\mathcal{M}$ and its complete diagram $\text{Diag}_{\infty}(\mathcal{M})$ then any $\mathcal{L}(\mathcal{M})$ -structure with $\mathcal{N} \models \text{Diag}_{\infty}(\mathcal{M})$ is an elementary extension of $\mathcal{M}$ - the same map $a \mapsto c_a^{\mathcal{N}}$ is an elementary embedding as by the definition of the complete diagram of $\mathcal{M}$, it must preserve all $\mathcal{L}$ -formulas.

⁵The language $\mathcal{L}(\mathcal{M})$ is the extension by constants of $\mathcal{L}$, containing for each $a \in \mathcal{M}$ a new constant symbol c_a , as was covered in [Mar25].

We let $\Sigma = \text{Diag}_\infty(\mathcal{M}) \cup \text{Diag}(\mathcal{N}) \subseteq \text{Sen}(\mathcal{L}(\mathcal{N}))$ (notice that $\text{Sen}(\mathcal{L}(\mathcal{M})) \subseteq \text{Sen}(\mathcal{L}(\mathcal{N}))$ as the domains have $\mathcal{M} \subseteq \mathcal{N}$) and we show that Σ is consistent. From this, we will obtain a model $\mathcal{A} \models \text{Diag}_\infty(\mathcal{M}) \cup \text{Diag}(\mathcal{N})$, which we have shown will satisfy $\mathcal{M} \prec \mathcal{A}$ and $\mathcal{N} \leq \mathcal{A}$.

If Σ is not consistent then there are $\mathcal{L}$ -formulas $\varphi(\bar{x})$ and $\chi(\bar{x}, \bar{y})$ with $\bar{x}, \bar{y}$ disjoint and χ quantifier-free, and elements $\bar{a} \in \mathcal{M}^{\bar{x}}$ and $\bar{b} \in \mathcal{N}^{\bar{y}}$ such that $\mathcal{M} \models \varphi(\bar{a})$ and $\mathcal{N} \models \chi(\bar{a}, \bar{b})$ but $\{\varphi(\bar{a}), \chi(\bar{a}, \bar{b})\}$ is inconsistent.

Then we have $\models \varphi(\bar{a}) \rightarrow \neg\chi(\bar{a}, \bar{b})$ in $\mathcal{L}(\mathcal{N})$. By 2.1.9, we get $\models \forall \bar{x}, \bar{y} (\varphi(\bar{x}) \rightarrow \neg\chi(\bar{x}, \bar{y}))$. Since the tuples are disjoint, we get $\models \forall \bar{x} (\varphi(\bar{x}) \rightarrow \forall \bar{y} \neg\chi(\bar{x}, \bar{y}))$, and in particular $\mathcal{M} \models \forall \bar{x} (\varphi(\bar{x}) \rightarrow \forall \bar{y} \neg\chi(\bar{x}, \bar{y}))$. But then for $\bar{a} \in \mathcal{M}^{\bar{x}}$ it follows that $\mathcal{M} \models \varphi(\bar{a}) \rightarrow \forall \bar{y} \neg\chi(\bar{a}, \bar{y})$, and as $\mathcal{M} \models \varphi(\bar{a})$ we have $\mathcal{M} \models \forall \bar{y} \neg\chi(\bar{a}, \bar{y})$.

But $\forall \bar{y} \neg\chi(\bar{x}, \bar{y})$ is an $\forall$ -formula as $\neg\chi$ is quantifier-free. Therefore, as $\mathcal{M}$ is e.c. in $\mathcal{N}$, we get $\mathcal{N} \models \forall \bar{y} \neg\chi(\bar{a}, \bar{y})$, a contradiction to our assumption that there is $\bar{b} \in \mathcal{N}$ with $\mathcal{N} \models \chi(\bar{a}, \bar{b})$.

□

2.2 Existential closure and language

Existential closure is an example of a **language dependent** property. Let $\mathcal{L} \subset \mathcal{L}^+$ be a language extension⁶ and let T be an $\mathcal{L}$ -theory, consisting entirely of sentences in the smaller language. Suppose $\mathcal{M}^+$ is an $\mathcal{L}^+$ -structure which is an e.c. model of T . That is, $\mathcal{M}^+$ is existentially closed in the class of all $\mathcal{L}^+$ -structures which are a model of T . Then, we will show that while it is the case that the reduct $\mathcal{M}$ of $\mathcal{M}^+$ to $\mathcal{L}$ will also be an e.c. model of T , the reverse does not necessarily hold. If we take an e.c. model of T , and consider some arbitrary expansion $\mathcal{M}^+$ of $\mathcal{M}$ to $\mathcal{L}^+$, then it is *not* the case that we will always get $\mathcal{M}^+$ an e.c. model of T in $\mathcal{L}^+$.

The material in this short subsection is not included explicitly in any of the main source texts. Instead, it is usually implicitly assumed to be true. The following is not particularly deep or difficult to observe given the basics of language extensions and existential closure. We have included an explicit explanation as we will frequently be looking at existential closure alongside expansions and reducts of structures, and the detail of existential closure's language dependence can be easily overlooked.

First, we will need the following lemma.

⁶See [Mar25] for more details on language extensions.

2.2.1 Lemma. *Let $\mathcal{L}^+ \supset \mathcal{L}$ be a language extension. Let $\mathcal{M}^+$ be an $\mathcal{L}^+$ -structure and let $\mathcal{M}$ be its reduct to $\mathcal{L}$. If $\mathcal{N}$ is an $\mathcal{L}$ -structure with $\mathcal{M} \leq \mathcal{N}$ then there is an expansion $\mathcal{N}^+$ of $\mathcal{N}$ to $\mathcal{L}^+$ such that $\mathcal{M}^+ \leq \mathcal{N}^+$.*

Proof. Suppose we have $\mathcal{M} = \mathcal{M}^+ \upharpoonright_{\mathcal{L}}$ and $\mathcal{N} \geq \mathcal{M}$. To expand $\mathcal{N}$ to $\mathcal{N}^+$ we need to give interpretations for all the constant, function, and relation symbols in $\mathcal{L}^+$ which are not in $\mathcal{L}$.

Since we already know that the identity map $\mathcal{M} \hookrightarrow \mathcal{N}$ is an embedding, we have $\mathcal{M}^+ \subseteq \mathcal{N}^+$ and that the identity map $\mathcal{M}^+ \rightarrow \mathcal{N}^+$ is an injection. We need to choose how the symbols in $\mathcal{L}^+ \setminus \mathcal{L}$ are interpreted in $\mathcal{N}^+$ to ensure that this map is also an embedding of $\mathcal{L}^+$ -structures.

For constant symbols, we need to ensure that every constant symbol c in $\mathcal{L}^+ \setminus \mathcal{L}$ satisfies $c^{\mathcal{M}^+} = c^{\mathcal{N}^+}$. Therefore, for such a c there is $a = c^{\mathcal{M}^+} \in \mathcal{M}^+ \subseteq \mathcal{N}^+$ and we interpret $c^{\mathcal{N}^+} = a \in \mathcal{N}^+$.

For relation symbols, we need to ensure that for every n -ary relation symbol R in $\mathcal{L}^+ \setminus \mathcal{L}$ and every $\bar{a} \in \mathcal{M}^+$ we have $\bar{a} \in R^{\mathcal{M}^+}$ iff $\bar{a} \in R^{\mathcal{N}^+}$. Therefore, for such an R we can simply choose to set $R^{\mathcal{N}^+} = R^{\mathcal{M}^+}$. The fact that this will result in $\mathcal{N}^+ \models \neg R(\bar{a})$ for every $\bar{a} \in (\mathcal{N}^+)^{\bar{x}}$ not from $\mathcal{M}^+$ is not a problem, as this will not affect whether $\mathcal{M}^+ \leq \mathcal{N}^+$.

For function symbols, we need to ensure that for every n -ary function symbol F in $\mathcal{L}^+ \setminus \mathcal{L}$ and every $\bar{a} \in \mathcal{M}^+$ we have $F^{\mathcal{M}^+}(\bar{a}) = F^{\mathcal{N}^+}(\bar{a})$. Therefore, for such an F we can choose its interpretation $F^{\mathcal{N}^+}$ to be a function $F^{\mathcal{N}^+} : (\mathcal{N}^+)^n \rightarrow \mathcal{N}^+$ such that its restriction to $\mathcal{M}^+$ is equal to $F^{\mathcal{M}^+}$, that is $F^{\mathcal{N}^+}|_{\mathcal{M}^+} = F^{\mathcal{M}^+}$.

To summarise, the interpretation of $\mathcal{L}^+$ -constant symbols in $\mathcal{N}^+$ is completely determined by their interpretation in $\mathcal{M}^+$. Then, for function and relation symbols, we ensure that their interpretations in $\mathcal{N}^+$ agree with their interpretation in $\mathcal{M}^+$ specifically with respect to the elements of $\mathcal{M}^+$ - but their interpretation with respect to elements of $\mathcal{N}^+$ which are not in $\mathcal{M}^+$ can essentially be chosen arbitrarily.

An embedding is, by definition, an injective map satisfying exactly the properties described above for all constant, relation, and function symbols in $\mathcal{L}^+$. We already know this is satisfied for all the symbols in $\mathcal{L}$ as $\mathcal{M} \leq \mathcal{N}$, and we have shown that it can be satisfied for those in $\mathcal{L}^+$ but not in $\mathcal{L}$. It follows that there is an expansion $\mathcal{N}^+$ such that $\mathcal{M}^+ \hookrightarrow \mathcal{N}^+$. $\square$

2.2.2 Proposition. *Let $\mathcal{L}^+ \supset \mathcal{L}$ be a language extension. Let T be an $\mathcal{L}$ -theory, and let $\mathcal{M}^+$ be an $\mathcal{L}^+$ -structure. Then, if $\mathcal{M}^+$ is an e.c. model of T in $\mathcal{L}^+$ then its reduct $\mathcal{M}$ to $\mathcal{L}$ will also be an e.c. model of T in $\mathcal{L}$.*

Proof. Suppose $\mathcal{M}^+$ is an e.c. model of T . To show $\mathcal{M}$ is an e.c. model of T , we take some $\exists \bar{y} \chi(\bar{x}, \bar{y}) \in \exists\text{-Fml}(\mathcal{L})$ where χ is quantifier-free and let $\bar{a} \in \mathcal{M}^{\bar{x}}$. Then, let $\mathcal{N}$ be an $\mathcal{L}$ -structure with $\mathcal{N} \models T$ and $\mathcal{N} \geq \mathcal{M}$. Suppose $\mathcal{N} \models \exists \bar{y} \chi(\bar{a}, \bar{y})$. Then, by 2.2.1, there is an expansion $\mathcal{N}^+$ of $\mathcal{N}$ to $\mathcal{L}^+$ with $\mathcal{N}^+ \geq \mathcal{M}^+$.

We use Theorem 2.1.7 from [Mar25][pp. 7-8] which we restate: suppose we have an $\mathcal{L}^+$ -structure $\mathcal{N}^+$ and its reduct $\mathcal{N}$ to $\mathcal{L}$. If T is an $\mathcal{L}$ -theory and $\varphi(\bar{x})$ is an $\mathcal{L}$ -formula, then for all $\bar{a} \in (\mathcal{N}^+)^{\bar{x}}$ we have:

$$\mathcal{N} \models T \iff \mathcal{N}^+ \models T \quad \text{and} \quad \mathcal{N} \models \varphi(\bar{a}) \iff \mathcal{N}^+ \models \varphi(\bar{a})$$

Applying this, we have both $\mathcal{N}^+ \models T$ and $\mathcal{N}^+ \models \exists \bar{y} \chi(\bar{a}, \bar{y})$.

Then as $\mathcal{M}^+$ is an e.c. model of T and $\mathcal{M}^+ \leq \mathcal{N}^+$ with $\mathcal{N}^+ \models T$, it follows that $\mathcal{M}^+$ is e.c. in $\mathcal{N}^+$. Thus as $\mathcal{N}^+ \models \exists \bar{y} \chi(\bar{a}, \bar{y})$ we get also $\mathcal{M}^+ \models \exists \bar{y} \chi(\bar{a}, \bar{y})$. Finally, as $\exists \bar{y} \chi(\bar{x}, \bar{y})$ is an $\mathcal{L}$ -sentence, we obtain $\mathcal{M} \models \exists \bar{y} \chi(\bar{a}, \bar{y})$ as required. Then as $\mathcal{N}$ was an arbitrary model of T containing $\mathcal{M}$, we get that $\mathcal{M}$ is an e.c. model of T . $\square$

2.2.3 Example. In this example, we show that if we start from an e.c. model of T in a smaller language that its expansion to a larger language is not guaranteed to be an e.c. model of T in the extended language. The following statement is **false**:

- Let T be an $\mathcal{L}$ -theory and let $\mathcal{M}^+$ be an $\mathcal{L}^+$ -structure with the reduct $\mathcal{M}$ to $\mathcal{L}$. If $\mathcal{M}$ is an e.c. model of T then $\mathcal{M}^+$ is an e.c. model of T

We demonstrate this with the following counterexample, which was created with the assistance of Generative AI (Google Gemini).

Take the theory DLO, which axiomatises the class of dense linear orders without endpoints in the language $\mathcal{L} = \{<\}$ (see [Hod93, p. 39] for the axioms of DLO). Then $\mathbb{Q}$ is an e.c. model of DLO. This holds because DLO has a property called **model-completeness**, which states that any embedding between any two models must be elementary. It turns out that a theory T is model-complete if and only if every model of T is an e.c. model of T . For more on DLO, model-completeness, and existential closure, see [Hod93, pp. 374-376].

Now, consider the extended language $\mathcal{L}^+ = \mathcal{L} \cup \{Z\}$, containing an additional unary

relation symbol. We interpret Z in $\mathbb{Q}$ to be $Z^{\mathbb{Q}} = \mathbb{Z} \subset \mathbb{Q}$. That is, the symbol Z picks out the integers from within the rational numbers.

Then consider the following $\mathcal{L}^+$ -formula $\varphi(x, 0, 1)$ with parameters 0 and 1 from $\mathbb{Q}$:

$$0 < x \wedge x < 1 \wedge Z(x)$$

As $\mathbb{Q}$ contains no integer strictly between 0 and 1, there is no $a \in \mathbb{Q}$ with $\mathbb{Q} \models \varphi(a, 0, 1)$. However, we will show that there is an $\mathcal{L}^+$ -superstructure of $\mathbb{Q}$ in which φ has a solution.

Let $\mathbb{Q} \oplus \mathbb{Q}\varepsilon$ denote the set of formal sums $a + b\varepsilon$ for $a, b \in \mathbb{Q}$.⁷ We make $\mathbb{Q} \oplus \mathbb{Q}\varepsilon$ into an $\mathcal{L}^+$ -structure by providing interpretations for Z and $<$ in $\mathbb{Q} \oplus \mathbb{Q}\varepsilon$.

We interpret $Z^{\mathbb{Q} \oplus \mathbb{Q}\varepsilon} = \mathbb{Z} \cup \{\varepsilon\}$.⁸ We interpret $<$ in $\mathbb{Q} \oplus \mathbb{Q}\varepsilon$ lexicographically, using the usual order on $\mathbb{Q}$. That is:

$$a_1 + b_1\varepsilon < a_2 + b_2\varepsilon \iff a_1 < a_2 \text{ or } (a_1 = a_2 \text{ and } b_1 < b_2)$$

Then, there is a copy of $\mathbb{Q}$ inside $\mathbb{Q} \oplus \mathbb{Q}\varepsilon$, which is the set $\mathbb{Q} \oplus 0\varepsilon = \{a + 0\varepsilon \mid a \in \mathbb{Q}\}$, and the obvious map is an embedding of $\mathcal{L}^+$ -structures. This follows from observing that:

$$<^{\mathbb{Q} \oplus \mathbb{Q}\varepsilon} \cap \mathbb{Q}^2 = <^{\mathbb{Q}} \text{ and } Z^{\mathbb{Q} \oplus \mathbb{Q}\varepsilon} \cap \mathbb{Q} = Z^{\mathbb{Q}}$$

By identifying $\mathbb{Q}$ with this copy $\mathbb{Q} \oplus 0\varepsilon$, we can view $\mathbb{Q}$ as a substructure of $\mathbb{Q} \oplus \mathbb{Q}\varepsilon$. Furthermore, it is clear that $\mathbb{Q} \oplus \mathbb{Q}\varepsilon$ with a lexicographic order is a model of DLO. However, it is clear that we also have:

$$\mathbb{Q} \oplus \mathbb{Q}\varepsilon \models 0 < \varepsilon \wedge \varepsilon < 1 \wedge Z(\varepsilon)$$

Thus the quantifier-free formula $\varphi(x, 0, 1)$ has a solution in $\mathbb{Q} \oplus \mathbb{Q}\varepsilon$ but not in the substructure $\mathbb{Q}$, and thus $\mathbb{Q}$ is not e.c. in $\mathbb{Q} \oplus \mathbb{Q}\varepsilon$ as $\mathcal{L}^+$ -structures. Thus, despite the fact that $\mathbb{Q}$ is an e.c. model of DLO when considered as an $\mathcal{L}$ -structure, we have found a language extension of $\mathcal{L}$ and expansion of $\mathbb{Q}$ to $\mathcal{L}^+$ which is not an e.c. model of DLO.

⁷The phrase ‘formal sums’ is used to indicate that the addition sign in each element $a + b\varepsilon$ has no inherent meaning and should not be thought of as an algebraic operation. The elements of $\mathbb{Q} \oplus \mathbb{Q}\varepsilon$ could easily be thought of as ordered pairs (a, b) in $\mathbb{Q}^2$ instead, and $\mathbb{Q}$ identified with the subset of $\mathbb{Q}^2$ consisting of elements of the form $(a, 0)$ for $a \in \mathbb{Q}$.

⁸More formally, we should write $0 + 1\varepsilon$ where we have used the more readable ε , and similarly we should write the subset $\mathbb{Z} \cup \{\varepsilon\}$ as $(\mathbb{Z} \oplus 0\varepsilon) \cup \{0 + 1\varepsilon\} \subset \mathbb{Q} \oplus \mathbb{Q}\varepsilon$.

Thus, with this counterexample, we have shown that being an e.c. model of a theory is language dependent, and not always preserved by expanding the structure to an extended language.

2.3 Inductive structures, classes and theories

We now show an important connection between existential closure and $\forall\exists$ -theories. We prove the **elementary chain lemma** (2.3.4), which we will refer back to when we look at model companions in section 6. This content can be found in [Tre10, pp. 35, 59–60].

2.3.1 Definition. Let T an $\mathcal{L}$ -theory.

- (i) We write $T_{\forall\exists}$ for the set of all $\forall\exists$ -consequences (in $\mathcal{L}$) of T . That is:

$$T_{\forall\exists} = \{\sigma \in \forall\exists\text{-Sen}(\mathcal{L}) \mid T \models \sigma\}$$

- (ii) We write $T_{\forall}$ and $T_{\exists}$ respectively for the sets of all $\forall$ and $\exists$ -consequences of T .
- (iii) We say that T is an $\forall\exists$ -**theory** if it can be axiomatised by a set of $\forall\exists$ -sentences.

$\forall\exists$ -theories are closely related to the study of existentially closed models. The next proposition is a first step towards understanding this relationship.

2.3.2 Proposition ([Tre10, Proposition 3.6.5, p. 59]). *Let T be an $\mathcal{L}$ -theory and $\mathcal{M}$ an $\mathcal{L}$ -structure. Then $\mathcal{M} \models T_{\forall\exists}$ iff $\mathcal{M}$ is existentially closed in a model of T .*

Proof. ($\Leftarrow$) This direction is more simple. Suppose $\mathcal{M}$ is e.c. in some $\mathcal{N} \models T$. Then certainly $\mathcal{N} \models T_{\forall\exists}$. Then, let $\forall\bar{x}\exists\bar{y} \chi(\bar{x}, \bar{y}) \in T_{\forall\exists}$ with χ quantifier-free, so we have $\mathcal{N} \models \forall\bar{x}\exists\bar{y} \chi(\bar{x}, \bar{y}) \in T_{\forall\exists}$.

Now, let $\bar{a} \in \mathcal{M}^{\bar{x}}$, so we also have $\bar{a} \in \mathcal{N}^{\bar{x}}$. Then $\mathcal{N} \models \exists\bar{y} \chi(\bar{a}, \bar{y})$. By the definition of $\mathcal{M}$ being existentially closed in $\mathcal{N}$, we obtain that $\mathcal{M} \models \exists\bar{y} \chi(\bar{a}, \bar{y})$. Since $\bar{a}$ was an arbitrary tuple in $\mathcal{M}$, it follows that $\mathcal{M} \models \forall\bar{x}\exists\bar{y} \chi(\bar{x}, \bar{y}) \in T_{\forall\exists}$. Thus $\mathcal{M} \models T_{\forall\exists}$.

($\Rightarrow$) Conversely, suppose $\mathcal{M} \models T_{\forall\exists}$. Then, we define the **universal diagram** of $\mathcal{M}$ to be:

$$\text{Diag}_{\forall}(\mathcal{M}) = \{\varphi(\bar{a}) \mid \mathcal{M} \models \varphi(\bar{a}), \bar{a} \in \mathcal{M}^{\bar{x}}, \varphi(\bar{x}) \in \forall\text{-Fml}(\mathcal{L})\}$$

For the remainder of this proof, we will write $\mathcal{M}^+$ for the expansion of $\mathcal{M}$ to $\mathcal{L}(\mathcal{M})$,

with each new constant symbol of $\mathcal{L}(\mathcal{M})$ interpreted as itself. This structure is sometimes written $(\mathcal{M}, M)$.

We claim that $T \cup \text{Diag}_\forall(\mathcal{M})$ is a consistent $\mathcal{L}(\mathcal{M})$ -theory. Otherwise, there must be some $\varphi(\bar{x}) \in \forall\text{-Fml}(\mathcal{L})$ and $\bar{a} \in \mathcal{M}^{\bar{x}}$ with $\mathcal{M} \models \varphi(\bar{a})$ such that $T \cup \{\varphi(\bar{a})\}$ is inconsistent, i.e. $T \models \neg\varphi(\bar{a})$ (in $\mathcal{L}(\mathcal{M})$). By the lemma on constants, it follows that $T \models \forall\bar{x} \neg\varphi(\bar{x})$. Then, from $\varphi(\bar{x}) \in \forall\text{-Fml}(\mathcal{L})$ we get $\neg\varphi(\bar{x}) \in \exists\text{-Fml}(\mathcal{L})$. Thus $\forall\bar{x} \neg\varphi(\bar{x}) \in \forall\exists\text{-Sen}(\mathcal{L})$. Therefore, $\forall\bar{x} \neg\varphi(\bar{x}) \in T_{\forall\exists}$ and it follows that $\mathcal{M} \models \forall\bar{x} \neg\varphi(\bar{x})$, a contradiction. Thus, $T \cup \text{Diag}_\forall(\mathcal{M})$ is a consistent $\mathcal{L}(\mathcal{M})$ -theory.

From this claim, it follows that there exists an $\mathcal{L}(\mathcal{M})$ -structure $\mathcal{N}^+ \models T \cup \text{Diag}_\forall(\mathcal{M})$. Since the quantifier-free diagram of $\mathcal{M}$ has $\text{Diag}(\mathcal{M}) \subseteq \text{Diag}_\forall(\mathcal{M})$, then by the same reasoning as in 2.1.10, there is an embedding $f : \mathcal{M}^+ \hookrightarrow \mathcal{N}^+$. Therefore, taking $\mathcal{N} = \mathcal{N}^+ \upharpoonright_{\mathcal{L}}$ we have the embedding $f : \mathcal{M} \hookrightarrow \mathcal{N}$.

By identifying each $a \in \mathcal{M}$ with its isomorphic image $f(a) \in f(\mathcal{M})$ inside $\mathcal{N}$, we can consider $\mathcal{M}$ to be a substructure of $\mathcal{N}$. We now show that $\mathcal{M}$ is e.c. in $\mathcal{N}$. This follows from Proposition 2.1.5, as the fact that $\mathcal{N}^+ \models \text{Diag}_\forall(\mathcal{M})$ ensures that the embedding $f : \mathcal{M} \hookrightarrow \mathcal{N}$ preserves all $\forall$ -formulas in $\mathcal{L}$. $\square$

Given a chain of structures, we now show how we can form a new structure with a domain equal to the union of the chain. The material on chains can be found in [Tre10, p. 35].

2.3.3 Definition (The union of a chain of structures). Let I be a totally ordered index set. Let $(\mathcal{M}_i)_{i \in I}$ be a sequence of $\mathcal{L}$ -structures. We call $(\mathcal{M}_i)_{i \in I}$ a **chain** of $\mathcal{L}$ -structures if for all $i \leq j$ we have $\mathcal{M}_i \leq \mathcal{M}_j$. Similarly, $(\mathcal{M}_i)_{i \in I}$ is an **elementary chain** if for all $i \leq j < \gamma$ we have $\mathcal{M}_i \prec \mathcal{M}_j$.

Given a chain $(\mathcal{M}_i)_{i \in I}$ of $\mathcal{L}$ -structures, we define a new $\mathcal{L}$ -structure $\mathcal{M}$, the **union of the chain** denoted $\bigcup \mathcal{M}_i$ as follows:

- The domain of $\mathcal{M}$ is $\bigcup \mathcal{M}_i$.
- For each relation symbol R we define:

$$R^{\mathcal{M}} = \bigcup R^{\mathcal{M}_i}$$

Equivalently, for each n -ary relation symbol R and $\bar{a} \in \mathcal{M}^n$:

$$\mathcal{M} \models R(\bar{a}) \iff \text{there is } i \in I \text{ and } \bar{a} \in \mathcal{M}_i \text{ such that } \mathcal{M}_i \models R(\bar{a})$$

- For each n -ary function symbol F and $\bar{a} \in \mathcal{M}^n$, we define:

$$F^{\mathcal{M}}(\bar{a}) = F^{\mathcal{M}_i}(\bar{a}) \text{ for all } i \in I \text{ such that } \bar{a} \in \mathcal{M}_i^n$$

This is well defined as whenever $i < j$ and $\bar{a} \in \mathcal{M}_i^n$ we have $\mathcal{M}_i \leq \mathcal{M}_j$ and therefore $F^{\mathcal{M}_i}(\bar{a}) = F^{\mathcal{M}_j}(\bar{a})$.

- For c a constant symbol, we define $c^{\mathcal{M}} = c_i^{\mathcal{M}}$ for all $i \in I$.

This is well defined as whenever $i < j$ we have $\mathcal{M}_i \leq \mathcal{M}_j$, and thus $c^{\mathcal{M}_i} = c^{\mathcal{M}_j}$.

In fact, any structure in the chain will already contain the interpretation of every $\mathcal{L}$ -symbol. There will always be an $\mathcal{L}$ -structure which is a lower bound for the chain and is generated by the $\mathcal{L}$ -constant symbols.

2.3.4. Elementary Chain Lemma ([Tre10, Lemma 2.5.1, p. 35]). *Let $(\mathcal{M}_i)_{i \in I}$ be a chain of $\mathcal{L}$ -structures and let $\mathcal{M} = \bigcup \mathcal{M}_i$ be the union of the chain. Then, for all $i \in I$ we have:*

$$(i) \quad \mathcal{M}_i \leq \mathcal{M}$$

$$(ii) \quad \text{If, in addition, } \mathcal{M}_j \prec \mathcal{M}_k \text{ for all } j \leq k, \text{ then } \mathcal{M}_i \prec \mathcal{M}.$$

Proof. (i) To show $\mathcal{M}_i \leq \mathcal{M}$, we just have to show that the inclusion map $\mathcal{M}_i \hookrightarrow \mathcal{M}$ is an embedding. This follows from how we defined the interpretations of constant, function, and relation symbols in 2.3.3.

In particular, the inclusion map is not just a homomorphism but an embedding as, by the latter way of defining relation symbols in $\mathcal{M}$, for each n -ary relation symbol R and $\bar{a} \in \mathcal{M}_i^n$ we have:

$$\mathcal{M} \models R(\bar{a}) \iff \mathcal{M}_i \models R(\bar{a})$$

- (ii) Now, suppose that for each $\mathcal{M}_j \leq \mathcal{M}_k$ in the chain that the embedding is elementary, so $\mathcal{M}_j \prec \mathcal{M}_k$. We already know that $\mathcal{M}_i \leq \mathcal{M}$, so we just need to show the embedding $\mathcal{M}_i \leq \mathcal{M}$ is elementary - that is, that it preserves all $\mathcal{L}$ -formulas.

We know this holds for quantifier-free formulas. Therefore, by prenex-normal form, it suffices to show this by induction on quantifiers.

Suppose $\varphi(\bar{y})$ is of the form $\exists x \psi(x, \bar{y})$, where $\psi(x, \bar{y})$ is a formula preserved

in the embedding $\mathcal{M}_i \hookrightarrow \mathcal{M}$. Then suppose there is some $\bar{b} \in \mathcal{M}_i^{\bar{y}}$ such that $\mathcal{M}_i \models \exists x \psi(x, \bar{b})$. Then there exists $a \in \mathcal{M}_i$ with $\mathcal{M}_i \models \psi(a, \bar{b})$, and thus also $\mathcal{M} \models \psi(a, \bar{b})$. It follows that $\mathcal{M} \models \exists x \psi(x, \bar{a})$, and the formula is preserved.

For the more complex case, suppose $\varphi(\bar{y})$ is of the form $\forall x \psi(x, \bar{y})$, where $\psi(x, \bar{y})$ is a formula preserved in the embedding $\mathcal{M}_i \hookrightarrow \mathcal{M}$. Then suppose we have $\bar{b} \in \mathcal{M}_i^{\bar{y}}$ such that $\mathcal{M}_i \models \forall x \psi(x, \bar{b})$. Let $a \in \mathcal{M}$. We need to show $\mathcal{M} \models \psi(a, \bar{b})$.

Since $a \in \mathcal{M}$, there is some $j \in I$ with $a \in \mathcal{M}_j$. If $j \leq i$ then also $a \in \mathcal{M}_i$ and we may take $j = i$. Thus we may assume without loss of generality that $i \leq j$ and that $\mathcal{M}_i \leq \mathcal{M}_j$.

Thus, by assumption $\mathcal{M}_i \prec \mathcal{M}_j$, and since $\mathcal{M}_i \models \forall x \psi(x, \bar{b})$ it follows that $\mathcal{M}_j \models \forall x \psi(x, \bar{b})$. In particular, we have that $\mathcal{M}_j \models \psi(a, \bar{b})$, and therefore also $\mathcal{M} \models \psi(a, \bar{b})$. Since a was arbitrary, it follows that $\mathcal{M} \models \forall x \psi(x, \bar{b})$, as required.

□

2.3.5 Definition. Let T be an $\mathcal{L}$ -theory and let K be a class of $\mathcal{L}$ -structures.

- (i) A class of structures K is said to be **inductive** if it is closed under taking unions of chains of substructures.
- (ii) We call T an **inductive theory** if the class of all models of T is inductive.

It is common in the literature for $\forall\exists$ -theories to be called **inductive** theories. Sometimes, an inductive theory is, by definition, a theory axiomatised by $\forall\exists$ -sentences. However, we define an inductive theory differently, as a theory where the union of any chain of its models is again a model. In the theorem below, we show that these two descriptions are equivalent: the inductive theories are exactly the $\forall\exists$ -theories.

2.3.6 Theorem. *Let T be an $\mathcal{L}$ -theory. Then T is an inductive theory iff it is an $\forall\exists$ -theory, i.e. can be axiomatised by a set of $\forall\exists$ -sentences.*

Proof. Suppose T is inductive. To show it is $\forall\exists$, it suffices to show that $T_{\forall\exists} \models T$. Therefore, assume $\mathcal{M} \models T_{\forall\exists}$.

By 2.3.2, $\mathcal{M}$ is e.c. in some $\mathcal{N}_0 \models T$. Then, by 2.1.10 there is an elementary extension $\mathcal{M}_1 \succ \mathcal{M}$ such that $\mathcal{M}_1 \geq \mathcal{N}_0$. Thus $\mathcal{M}_1 \models T_{\forall\exists}$.

We repeat this process and generate a chain of $\mathcal{L}$ -structures:

$$\mathcal{M} = \mathcal{M}_0 \leq \mathcal{N}_0 \leq \mathcal{M}_1 \leq \mathcal{N}_1 \leq \mathcal{M}_2 \leq \mathcal{N}_2 \leq \dots$$

The chain has the property that for each $i \geq 0$ we have $\mathcal{M}_i \prec \mathcal{M}_{i+1}$ as well as $\mathcal{N}_i \models T$. It also contains the two subchains: $(\mathcal{M}_i)_i$ and $(\mathcal{N}_i)_i$.

We take $\mathcal{N}$ be the union of this chain. By assumption applied to the chain $(\mathcal{N}_i)_{i < \omega}$ we have that $\mathcal{N} \models T$. By the **elementary chain lemma** applied to the chain $(\mathcal{M}_i)_{i < \omega}$ we have that $\mathcal{M} \prec \mathcal{N}$. Thus $\mathcal{M} \equiv \mathcal{N}$, and so $\mathcal{M} \models T$ as required.

Conversely, suppose T is $\forall\exists$, and let $(\mathcal{M}_i)_{i \in I}$ be a chain of $\mathcal{L}$ -structures that model T . We let $\mathcal{M} = \bigcup \mathcal{M}_i$ and show $\mathcal{M} \models T_{\forall\exists}$, and thus $\mathcal{M} \models T$. Let $\varphi \in T_{\forall\exists}$ be the formula $\forall \bar{x} \exists \bar{y} \psi(\bar{x}, \bar{y})$ with ψ quantifier-free. Then $\mathcal{M}_i \models \varphi$ for all $i \in I$.

We let $\bar{a} \in \mathcal{M}^{\bar{x}}$, and aim to show $\mathcal{M} \models \exists \bar{y} \psi(\bar{a}, \bar{y})$. Then there is some $i \in I$ with $\bar{a} \in \mathcal{M}_i^{\bar{x}}$. Then $\mathcal{M}_i \models \exists \bar{y} \psi(\bar{a}, \bar{y})$, and so there is $\bar{b} \in \mathcal{M}_i^{\bar{y}}$ such that $\mathcal{M}_i \models \psi(\bar{a}, \bar{b})$.

Since $\mathcal{M}_i \leq \mathcal{M}$, we obtain that $\mathcal{M} \models \psi(\bar{a}, \bar{b})$. Thus, $\mathcal{M} \models \exists \bar{y} \psi(\bar{a}, \bar{y})$. Since $\bar{a}$ was arbitrary, we have that $\mathcal{M} \models \forall \bar{x} \exists \bar{y} \psi(\bar{x}, \bar{y})$ as required. $\square$

2.4 Resultants

Later in this dissertation, we use forcing to construct existentially closed models of $\forall\exists$ -theories. A useful tool in these constructions will be particular sets of universal formulas, called the **resultant** of an existential formula with respect to a theory T . Our goal is to use resultants to characterise the existentially closed models of an $\forall\exists$ -theory.

This material on resultants can be found in [Tre10, p. 62], [Hod85, pp. 36-37, 49-51], and [Hod93, pp. 368, 391-392].

2.4.1. Notation. Given a set of $\mathcal{L}$ -formulas $\Phi(\bar{x})$, an $\mathcal{L}$ -structure $\mathcal{M}$ and $\bar{a} \in \mathcal{M}^{\bar{x}}$, we write $\mathcal{M} \models \Phi(\bar{a})$ if for every $\varphi \in \Phi$ we have $\mathcal{M} \models \varphi(\bar{a})$. In this case we say that $\bar{a}$ **realises** Φ in $\mathcal{M}$, or that Φ is **realised** in $\mathcal{M}$ by $\bar{a}$. If for all $\bar{a} \in \mathcal{M}^{\bar{x}}$ we have $\mathcal{M} \not\models \Phi(\bar{a})$ (i.e. $\mathcal{M} \models \neg\varphi(\bar{a})$ for some $\varphi \in \Phi$) then we say that $\mathcal{M}$ **omits** Φ .

2.4.2 Definition. Let T be an $\mathcal{L}$ -theory and let $\varphi(\bar{x})$ be an existential $\mathcal{L}$ -formula. We define the **resultant** of φ for T to be the following set of universal formulas:

$$\text{Res}_{\varphi}^T(\bar{x}) = \{\psi(\bar{x}) \in \forall\text{-Fml}(\mathcal{L}) \mid T \models \forall \bar{x} (\varphi \rightarrow \psi)\}$$

2.4.3 Observation. Notice that the sentence $\forall \bar{x} (\varphi \rightarrow \psi)$ is logically equivalent to $\forall \bar{x} (\neg \varphi \vee \psi)$. Already ψ is universal by definition, and since φ is existential, also $\neg \varphi$ is universal. Therefore, as $\forall\text{-Fml}(\mathcal{L})$ is closed under conjunction and adding universal quantifiers, we get that $\forall \bar{x} (\varphi \rightarrow \psi)$ is itself a universal formula.

It follows that $T \models \forall \bar{x} (\varphi \rightarrow \psi)$ iff $T_{\forall} \models \forall \bar{x} (\varphi \rightarrow \psi)$, and therefore:

$$\text{Res}_{\varphi}^T(\bar{x}) = \text{Res}_{\varphi}^{T_{\forall}}(\bar{x})$$

This means that the resultant of a formula for T only depends on $T_{\forall}$, the universal consequences of T .

2.4.4 Lemma ([Tre10][Theorem 3.6.16] and [Hod85][Theorem 3.1.1]). *Let T be an $\mathcal{L}$ -theory and let $\mathcal{M}$ be an $\mathcal{L}$ -structure. Then for any $\varphi(\bar{x}) \in \exists\text{-Fml}(\mathcal{L})$ and $\bar{a} \in \mathcal{M}^{\bar{x}}$, the following are equivalent:*

- (i) *There is $\mathcal{N} \geq \mathcal{M}$ such that $\mathcal{N} \models T$ and $\mathcal{N} \models \varphi(\bar{a})$*
- (ii) $\mathcal{M} \models \text{Res}_{\varphi}^T(\bar{a})$

Proof. We prove that the negations of (i) and (ii) are equivalent:

- (i)* For all $\mathcal{N} \geq \mathcal{M}$ with $\mathcal{N} \models T$, we have $\mathcal{N} \models \neg \varphi(\bar{a})$.
 - (ii)* $\mathcal{M} \not\models \text{Res}_{\varphi}^T(\bar{a})$, i.e. there is $\psi(\bar{x}) \in \text{Res}_{\varphi}^T(\bar{x})$ such that $\mathcal{M} \models \neg \psi(\bar{a})$.
- (i)* $\Rightarrow$ (ii)*: Let $\varphi(\bar{x}) \in \exists\text{-Fml}(\mathcal{L})$ and $\bar{a} \in \mathcal{M}^{\bar{x}}$ and assume every $\mathcal{N} \geq \mathcal{M}$ with $\mathcal{N} \models T$ has $\mathcal{N} \models \neg \varphi(\bar{a})$.

We take the quantifier-free diagram $\text{Diag}(\mathcal{M})$ and note that the superstructures of $\mathcal{M}$ are exactly the models of $\text{Diag}(\mathcal{M})$ which contain $\mathcal{M}$. Therefore, by our assumption, it follows that every model $\mathcal{N} \models \text{Diag}(\mathcal{M})$ also satisfying $\mathcal{N} \models T$ has $\mathcal{N} \models \neg \varphi(\bar{a})$. In short, $T \cup \text{Diag}(\mathcal{M}) \models \neg \varphi(\bar{a})$.

By the compactness theorem, there is some finite subset $\Sigma \subseteq \text{Diag}(\mathcal{M})$ with $T \cup \Sigma \models \neg \varphi(\bar{a})$. From Σ , we get a quantifier-free $\mathcal{L}$ -formula $\chi(\bar{x}, \bar{y})$ and tuple of elements $\bar{b} \in \mathcal{M}^{\bar{y}}$ disjoint from $\bar{a}$, such that $\chi(\bar{a}, \bar{b})$ is logically equivalent to $\bigwedge \Sigma$. Thus we

have $T \cup \{\chi(\bar{a}, \bar{b})\} \models \neg\varphi(\bar{a})$. Then:

$$\begin{aligned}
T \cup \{\chi(\bar{a}, \bar{b})\} \models \neg\varphi(\bar{a}) &\iff T \models \chi(\bar{a}, \bar{b}) \rightarrow \neg\varphi(\bar{a}) \\
&\iff T \models \varphi(\bar{a}) \rightarrow \neg\chi(\bar{a}, \bar{b}) \\
&\implies T \models \varphi(\bar{a}) \rightarrow \neg\exists\bar{y} \chi(\bar{a}, \bar{y}) \\
&\iff T \models \varphi(\bar{a}) \rightarrow \forall\bar{y} \neg\chi(\bar{a}, \bar{y})
\end{aligned}$$

Then, it cannot be the case that $\mathcal{M} \models \forall\bar{y} \neg\chi(\bar{a}, \bar{y})$, as this would entail $\mathcal{M} \models \neg\chi(\bar{a}, \bar{b})$, contradicting our choice of χ and $\bar{b}$.

Therefore, writing $\psi(\bar{x})$ for $\forall\bar{y} \neg\chi(\bar{x}, \bar{y})$ we have found $\psi(\bar{x}) \in \text{Res}_\varphi^T(\bar{x})$ with $\mathcal{M} \models \neg\psi(\bar{a})$.

(ii)* $\Rightarrow$ (i)*: Assume $\mathcal{M} \not\models \text{Res}_\varphi^T(\bar{a})$ and suppose we have $\mathcal{N} \geq \mathcal{M}$ with $\mathcal{N} \models T$.

Then there is $\psi(\bar{x}) \in \text{Res}_\varphi^T(\bar{x})$ with $\mathcal{M} \models \neg\psi(\bar{a})$. Since $\mathcal{M} \models \forall\bar{x}(\varphi \rightarrow \psi)$ also $\mathcal{M} \models \forall\bar{x}(\neg\psi \rightarrow \neg\varphi)$. Thus $\mathcal{M} \models \neg\varphi(\bar{a})$.

Then as $\neg\varphi(\bar{x})$ is a universal formula, it is preserved by embeddings by 2.1.4. Therefore $\mathcal{N} \models \neg\varphi(\bar{a})$. $\square$

2.4.5 Corollary ([Hod85][Theorem 3.1.2]). *Let $\mathcal{M}$ be an $\mathcal{L}$ -structure. Then $\mathcal{M} \models T_\forall$ iff there is $\mathcal{N} \models T$ with $\mathcal{M} \leq \mathcal{N}$.*

In short, the models of $T_\forall$ are exactly the substructures of the models of T .

Proof. Take φ to be a trivially true $\exists$ -**sentence** (e.g. $\exists x (x = x)$), so its resultant will consist entirely of sentences with no free variables. Then:

$$\text{Res}_\varphi^T = \{\sigma \in \forall\text{-Sen}(\mathcal{L}) \mid T \models (\varphi \rightarrow \sigma)\} = \{\sigma \in \forall\text{-Sen}(\mathcal{L}) \mid T \models \sigma\} = T_\forall$$

Our claim then follows from 2.4.4, since $\exists x (x = x)$ is true in any structure. $\square$

2.4.6 Theorem ([Hod85, Theorem 3.2.3] and [Hod93, Theorem 8.2.4]). *Let T be an $\forall\exists$ -theory in $\mathcal{L}$ and $\mathcal{M}$ an $\mathcal{L}$ -structure. Then TFAE:*

- (i) $\mathcal{M}$ is an e.c. model of T
- (ii) $\mathcal{M} \models T_\forall$ and for every $\varphi(\bar{x}) \in \exists\text{-Fml}(\mathcal{L})$ and tuple $\bar{a} \in \mathcal{M}^{\bar{x}}$, if $\mathcal{M} \models \text{Res}_\varphi^T(\bar{a})$ then $\mathcal{M} \models \varphi(\bar{a})$
- (iii) $\mathcal{M}$ is an e.c. model of $T_\forall$

Proof.

(i) $\Rightarrow$ (ii): Assume $\mathcal{M}$ is an e.c. model of T . Let $\varphi(\bar{x}) \in \exists\text{-Fml}(\mathcal{L})$ and $\bar{a} \in \mathcal{M}^{\bar{x}}$ and suppose $\mathcal{M} \models \text{Res}_{\varphi}^T(\bar{a})$.

By 2.4.4, there is $\mathcal{N} \geq \mathcal{M}$ with $\mathcal{N} \models T$ satisfying $\mathcal{N} \models \varphi(\bar{a})$. As $\mathcal{M}$ is an e.c. model of T , it is e.c. in $\mathcal{N}$. Then we have:

$$\mathcal{N} \models \varphi(\bar{a}) \implies \mathcal{M} \models \varphi(\bar{a})$$

(ii) $\Rightarrow$ (iii): We assume (ii). Immediately we have $\mathcal{M} \models T_{\forall}$.

To show $\mathcal{M}$ is an e.c. model of $T_{\forall}$, let $\mathcal{N} \models T_{\forall}$ with $\mathcal{N} \geq \mathcal{M}$. Let $\varphi(\bar{x}) \in \exists\text{-Fml}(\mathcal{L})$ and $\bar{a} \in \mathcal{M}^{\bar{x}}$, and suppose $\mathcal{N} \models \varphi(\bar{a})$.

By 2.4.4, this is equivalent to $\mathcal{M} \models \text{Res}_{\varphi}^{T_{\forall}}(\bar{a})$. By 2.4.3, also $\mathcal{M} \models \text{Res}_{\varphi}^T(\bar{a})$. Then by our assumption of (ii), we obtain $\mathcal{M} \models \varphi(\bar{a})$.

(iii) $\Rightarrow$ (i): Assume that $\mathcal{M}$ is an e.c. model of $T_{\forall}$. First, we show $\mathcal{M} \models T$. Since T is an $\forall\exists$ -theory, it suffices to show $\mathcal{M} \models T_{\forall\exists}$.

Let $\forall\bar{x} \exists\bar{y} \chi(\bar{x}, \bar{y}) \in T_{\forall\exists}$ where χ is quantifier-free. We take an arbitrary $\bar{a} \in \mathcal{M}^{\bar{x}}$ and show that $\mathcal{M} \models \exists\bar{y} \chi(\bar{a}, \bar{y})$.

Then, by 2.4.5, as $\mathcal{M} \models T_{\forall}$ there is $\mathcal{N} \models T$ with $\mathcal{M} \leq \mathcal{N}$. Then, we have $\mathcal{N} \models \forall\bar{x} \exists\bar{y} \chi(\bar{x}, \bar{y})$, and we get $\mathcal{N} \models \exists\bar{y} \chi(\bar{a}, \bar{y})$. Since $\exists\bar{y} \chi(\bar{x}, \bar{y}) \in \exists\text{-Fml}(\mathcal{L})$ and $\mathcal{M}$ is e.c. in $\mathcal{N}$, we obtain $\mathcal{M} \models \exists\bar{y} \chi(\bar{a}, \bar{y})$.

Thus $\mathcal{M} \models T$. Since $\mathcal{M}$ is e.c. in every model of $T_{\forall}$, this includes all the models of T , and thus $\mathcal{M}$ is an e.c. model of T . $\square$

We will need the following fact, set as an exercise in [Hod93, Exercise 3, p. 372]. This exercise allows us to state 2.4.6 (ii) in a few equivalent ways.

2.4.7. Exercise. *Let T be an $\forall\exists$ -theory in $\mathcal{L}$ and $\mathcal{M} \models T_{\forall}$. Then TFAE:*

- (i) *For every primitive $\mathcal{L}$ -formula $\varphi(\bar{x})$ and $\bar{a} \in \mathcal{M}^{\bar{x}}$, if $\mathcal{M} \models \neg\varphi(\bar{a})$ then there is a primitive $\mathcal{L}$ -formula $\psi(\bar{x})$ such that $\mathcal{M} \models \psi(\bar{a})$ and $T \models \forall\bar{x} (\psi \rightarrow \neg\varphi)$.*
- (ii) *For every $\exists$ -formula $\varphi(\bar{x})$ of $\mathcal{L}$ and $\bar{a} \in \mathcal{M}^{\bar{x}}$, if $\mathcal{M} \models \neg\varphi(\bar{a})$ then there is an $\exists$ -formula $\psi(\bar{x})$ of $\mathcal{L}$ such that $\mathcal{M} \models \psi(\bar{a})$ and $T \models \forall\bar{x} (\psi \rightarrow \neg\varphi)$.*
- (iii) *For every $\exists$ -formula $\varphi(\bar{x})$ of $\mathcal{L}$ and $\bar{a} \in \mathcal{M}^{\bar{x}}$, if $\mathcal{M} \models \neg\varphi(\bar{a})$ then $\mathcal{M} \not\models \text{Res}_{\varphi}^T(\bar{a})$*

Proof. Throughout this proof, we will use the fact every primitive formula is an $\exists$ -formula and, given an existential formula $\varphi(\bar{x})$, by disjunctive normal form, we may assume it to be of the form:

$$\exists \bar{y} \bigvee_{j=1}^m \bigwedge_{i=1}^n \chi_{ij}(\bar{x}, \bar{y}) \quad \text{which is equivalent to} \quad \bigvee_{j=1}^m \exists \bar{y} \bigwedge_{i=1}^n \chi_{ij}(\bar{x}, \bar{y})$$

where the χ_{ij} are basic formulas. It follows that every $\exists$ -formula can be written as the disjunction of primitive formulas.

(i) $\Rightarrow$ (ii): Let $\varphi(\bar{x}) \in \exists\text{-Fml}(\mathcal{L})$ and $\bar{a} \in \mathcal{M}^{\bar{x}}$ and suppose $\mathcal{M} \models \neg\varphi(\bar{a})$. Then, we may assume that φ is equivalent to the disjunction of the primitive formulas $\varphi_1, \dots, \varphi_m$. Then we get:

$$\begin{aligned} \mathcal{M} \models \neg\varphi(\bar{a}) &\iff \mathcal{M} \models \neg \bigvee_{j=1}^m \varphi_j(\bar{a}) \\ &\iff \mathcal{M} \models \bigwedge_{j=1}^m \neg\varphi_j(\bar{a}) \\ &\iff \mathcal{M} \models \neg\varphi_j(\bar{a}) \quad \text{for all } 1 \leq j \leq m \end{aligned}$$

Then by our assumption of (i), for each of the primitive formulas $\varphi_j(\bar{x})$ for $1 \leq j \leq m$ we get a primitive formula $\psi_j(\bar{x})$ satisfying $\mathcal{M} \models \psi_j(\bar{a})$ and $T \models \forall \bar{x} (\psi_j(\bar{x}) \rightarrow \neg\varphi(\bar{x}))$. Then, for each $1 \leq j \leq m$, the primitive formula ψ_j is of the form:

$$\exists \bar{y}_j \bigwedge_{i=1}^n \chi_{ij}(\bar{x}, \bar{y}_j)$$

where the $\bar{y}_1, \dots, \bar{y}_m$ are pairwise disjoint tuples of distinct variables.⁹ We will write $\bar{y}$ for $(\bar{y}_1 \bar{y}_2 \dots \bar{y}_m)$, the tuple of variables resulting from the concetenation of the $\bar{y}_j$.

We claim that the conjunction of the ψ_j is the required $\exists$ -formula to prove (ii). The

⁹Introducing distinct variables allows us to bring the existential quantifier to the front of a conjunction. To see why, note that the following is true for any φ and ψ :

$$\models (\exists \bar{x} \varphi(\bar{x}) \wedge \exists \bar{y} \psi(\bar{y})) \leftrightarrow (\exists \bar{x} \bar{y} (\varphi(\bar{x}) \wedge \psi(\bar{y})))$$

However, the following may fail for some choice of φ and ψ :

$$\models (\exists \bar{x} \varphi(\bar{x}) \wedge \exists \bar{x} \psi(\bar{x})) \leftrightarrow (\exists \bar{x} (\varphi(\bar{x}) \wedge \psi(\bar{x})))$$

following are all logically equivalent:

$$\begin{aligned}
& \bigwedge_{j=1}^m \psi_j(\bar{x}) \\
& \bigwedge_{j=1}^m \exists \bar{y}_j \bigwedge_{i=1}^n \chi_{ij}(\bar{x}, \bar{y}_j) \\
& \exists \bar{y} \bigwedge_{j=1}^m \bigwedge_{i=1}^n \chi_{ij}(\bar{x}, \bar{y}_j) \\
& \exists \bar{y} \bigwedge_{\substack{1 \leq i \leq n \\ 1 \leq j \leq m}} \chi_{ij}(\bar{x}, \bar{y}_j)
\end{aligned}$$

Thus, the conjunction of the ψ_j is not only an $\exists$ -formula, but a primitive formula as well.¹⁰ Finally, we have:

$$\mathcal{M} \models \psi_j(\bar{a}) \text{ for all } 1 \leq j \leq m \iff \mathcal{M} \models \bigwedge_{j=1}^m \psi_j(\bar{a})$$

And:

$$\begin{aligned}
& T \models \forall \bar{x} (\psi_j \rightarrow \neg \varphi) && \text{for all } 1 \leq j \leq m \\
& \iff T \models \bigwedge_{j=1}^m \forall \bar{x} (\psi_j \rightarrow \neg \varphi) \\
& \iff T \models \forall \bar{x} \bigwedge_{j=1}^m (\psi_j \rightarrow \neg \varphi) \\
& \iff T \models \forall \bar{x} \left(\bigwedge_{j=1}^m \psi_j \rightarrow \neg \varphi \right) && \text{by distributivity laws for } \vee \text{ and } \wedge
\end{aligned}$$

Thus $\bigwedge_{j=1}^m \psi_j \in \exists\text{-Fml}(\mathcal{L})$ is the required formula satisfying:

$$\mathcal{M} \models \bigwedge_{j=1}^m \psi_j(\bar{a}) \text{ and } T \models \forall \bar{x} \left(\bigwedge_{j=1}^m \psi_j \rightarrow \neg \varphi \right)$$

(ii) $\Rightarrow$ (i): Conversely, let $\varphi(\bar{x})$ be a primitive formula and $\bar{a} \in \mathcal{M}^{\bar{x}}$ and suppose $\mathcal{M} \models \neg \varphi(\bar{a})$.

Then, as $\varphi(\bar{x})$ is also an $\exists$ -formula, by our hypothesis, there is an $\exists$ -formula $\psi(\bar{x})$

¹⁰It is a fact that $\exists\text{-Fml}(\mathcal{L})$ is closed under conjunction, so we could have used this to bypass the more explicit demonstration above.

satisfying $\mathcal{M} \models \psi(\bar{a})$ and $T \models \forall \bar{x} (\psi(\bar{x}) \rightarrow \neg \varphi(\bar{x}))$.

Write ψ as the disjunction of primitive $\psi_1, \dots, \psi_m$ as follows:

$$\bigvee_{j=1}^m \psi_j(\bar{x})$$

Then, since we know that $\mathcal{M} \models \psi(\bar{a})$, it follows that there is some $1 \leq j \leq m$ with $\mathcal{M} \models \psi_j(\bar{a})$. Let ψ' be such a ψ_j , so $\mathcal{M} \models \psi'(\bar{a})$.

We have:

$$\models \forall \bar{x} (\psi' \rightarrow \psi) \quad \text{and} \quad T \models \forall \bar{x} (\psi \rightarrow \neg \varphi) \quad \text{therefore} \quad T \models \forall \bar{x} (\psi' \rightarrow \neg \varphi)$$

Thus, ψ' is the required primitive formula satisfying:

$$\mathcal{M} \models \psi'(\bar{a}) \quad \text{and} \quad T \models \forall \bar{x} (\psi' \rightarrow \neg \varphi)$$

(ii) $\Leftrightarrow$ (iii): Both of these statements are conditionals with the same antecedent. We show that their consequents are equivalent.

First observe that for any $\psi(\bar{x}) \in \forall\text{-Fml}(\mathcal{L})$ we have $\neg\psi(\bar{x}) \in \exists\text{-Fml}(\mathcal{L})$ and:

$$\psi(\bar{x}) \in \text{Res}_\varphi^T(\bar{x}) \iff T \models \forall \bar{x} (\varphi \rightarrow \psi) \iff T \models \forall \bar{x} (\neg\psi \rightarrow \neg\varphi)$$

Then for any $\varphi(\bar{x}) \in \exists\text{-Fml}(\mathcal{L})$ and $\bar{a} \in \mathcal{M}^{\bar{x}}$, the following are equivalent:

$$\begin{aligned} & \mathcal{M} \not\models \text{Res}_\varphi^T(\bar{a}) \\ \iff & \text{There is } \psi(\bar{x}) \in \text{Res}_\varphi^T(\bar{x}) \text{ with } \mathcal{M} \models \neg\psi(\bar{a}) \\ \iff & \text{There is } \psi(\bar{x}) \in \forall\text{-Fml}(\mathcal{L}) \text{ with } \mathcal{M} \models \neg\psi(\bar{a}) \text{ and } T \models \forall \bar{x} (\varphi \rightarrow \psi) \\ \iff & \text{There is } \psi'(\bar{x}) \in \exists\text{-Fml}(\mathcal{L}) \text{ with } \mathcal{M} \models \psi'(\bar{a}) \text{ and } T \models \forall \bar{x} (\psi' \rightarrow \neg\varphi) \end{aligned}$$

□

2.4.8 Corollary. *Let T be an $\forall\exists$ -theory and let $\mathcal{M}$ be an $\mathcal{L}$ -structure. Then the following are equivalent:*

- (i) $\mathcal{M}$ is an e.c. model of T
- (ii) $\mathcal{M} \models T_\forall$ and for every primitive $\mathcal{L}$ -formula $\varphi(\bar{x})$ and $\bar{a} \in \mathcal{M}^{\bar{x}}$, if $\mathcal{M} \models \neg\varphi(\bar{a})$

then there is a primitive $\mathcal{L}$ -formula $\psi(\bar{x})$ such that $\mathcal{M} \models \psi(\bar{a})$ and $T \models \forall \bar{x}(\psi \rightarrow \neg\varphi)$

Proof. This follows from applying 2.4.7 to the equivalence of (i) and (ii) in 2.4.6. $\square$

3 Term Algebras

The **term algebra** of an $\mathcal{L}$ -theory is a particular $\mathcal{L}$ -structure which is completely determined by a first-order theory T and in which every element is named by a closed term. In this section, we will define the term algebra of a theory and discuss some of its particularly useful general properties. Furthermore, we will see that the term algebra of a theory T is not necessarily a model of T , but that there are certain conditions which can be imposed on a theory T to ensure it is modelled by its term algebra.

It is often useful to have a procedure to produce a uniquely determined structure from a theory in which every element is named by a constant. Different authors have different approaches to this, resulting in two slightly different constructions. The term algebra is used in [JT24, pp.98-104], while in [Hod85, p. 18], the **canonical model** is used instead (see also [Hod93, pp. 44-45] for the canonical model).

The goal of this section is not only to develop the theory of term algebras, which are constructions that will be integral to the whole of the rest of the dissertation, but also to compare their construction to that of canonical models. This will allow us to state Hodges' forcing results, which use canonical models, in terms of term algebras instead.

3.1 The term algebra of a theory

3.1.1 Definition. Recall that an $\mathcal{L}$ -term is either a variable x , a constant c , or a function of terms $F(t_1, \dots, t_n)$.

A **closed $\mathcal{L}$ -term** is a term in which no variable symbols appear. That is, t is a closed $\mathcal{L}$ -term iff t is the symbol c for some constant c , or is $F(t_1, \dots, t_n)$ for F a function symbol and $t_1, \dots, t_n$ closed terms of lower complexity.

3.1.2 Definition. Let $\mathcal{L}$ be a language with at least one constant symbol and let T be an $\mathcal{L}$ -theory. The **term algebra of T** denoted $\text{TmAlg}(T)$, is defined as follows:

- (i) The domain of $\text{TmAlg}(T)$ consists of equivalence classes of closed $\mathcal{L}$ -terms, with respect to an equivalence relation of equality modulo T . For closed $\mathcal{L}$ -terms t_1, t_2 we define:

$$t_1 \sim_T t_2 \iff T \models t_1 = t_2$$

The equivalence class of a closed $\mathcal{L}$ -term t under $\sim_T$ is denoted $[t]$ (or $[t]_{\sim_T}$).

(ii) For a constant symbol c (which is a closed term), we define $c^{\text{TmAlg}(T)} = [c]$.

(iii) For an n -ary function symbol F and closed terms $t_1, \dots, t_n$, we define:

$$F^{\text{TmAlg}(T)}([t_1], \dots, [t_n]) = [F(t_1, \dots, t_n)]$$

(iv) For an n -ary relation symbol R and closed terms $t_1, \dots, t_n$, we define:

$$\text{TmAlg}(T) \models R([t_1], \dots, [t_n]) \text{ iff } T \models R(t_1, \dots, t_n)$$

For a proof that $\sim_T$ is an equivalence relation and that each of these definitions is well-defined, see [JT24, Lemma 2.8.2, pp. 98-100].

The term algebra has the following properties:

3.1.3 Proposition. *Let T be an $\mathcal{L}$ -theory in a language with at least one constant symbol.*

(i) *Every $a \in \text{TmAlg}(T)$ is of the form $t^{\text{TmAlg}(T)}$ for some closed $\mathcal{L}$ -term t*

(ii) *Every atomic $\mathcal{L}$ -sentence σ satisfies:*

$$\text{TmAlg}(T) \models \sigma \iff T \models \sigma$$

(iii) *If $\mathcal{B}$ is another $\mathcal{L}$ -structure satisfying (i) and (ii) then $\text{TmAlg}(T) \cong \mathcal{B}$*

(iv) *If additionally for every closed $\mathcal{L}$ -term t there is a constant symbol c such that $T \models t = c$, then every element $a \in \text{TmAlg}(T)$ is of the form $c^{\text{TmAlg}(T)}$ for some constant symbol c*

Proof. (i), (ii) and (iv) are immediate from the definition of the term algebra.

For (iii), let $\mathcal{B}$ be such a structure. For each $a \in \text{TmAlg}(T)$, there is a closed $\mathcal{L}$ -term t_a such that $t_a^{\text{TmAlg}(T)} = a$. We define a map $f : \text{TmAlg}(T) \rightarrow \mathcal{B}$ by $a \mapsto t_a^{\mathcal{B}}$ for each

$a \in \text{TmAlg}(T)$. This is well defined, as well as injective, since:

$$\begin{aligned}
& a_1 = a_2 \in \text{TmAlg}(T) \\
\iff & t_{a_1}^{\text{TmAlg}(T)} = t_{a_2}^{\text{TmAlg}(T)} \\
\iff & \text{TmAlg}(T) \models t_{a_1} = t_{a_2} \\
\iff & T \models t_{a_1} = t_{a_2} && \text{by (ii)} \\
\iff & \mathcal{B} \models t_{a_1} = t_{a_2} && \text{since } \mathcal{B} \text{ satisfies (ii)} \\
\iff & t_{a_1}^{\mathcal{B}} = t_{a_2}^{\mathcal{B}}
\end{aligned}$$

Furthermore, this map is surjective as if $b \in \mathcal{B}$ then by assumption, $b = t^{\mathcal{B}}$ for some closed $\mathcal{L}$ -term t . Then $t^{\text{TmAlg}(T)} \in \text{TmAlg}(T)$. Therefore, $f(t^{\text{TmAlg}(T)}) = t_{t^{\text{TmAlg}(T)}}^{\mathcal{B}} = t^{\mathcal{B}} = b$.

Finally, we show that f is an embedding of $\mathcal{L}$ -structures. For this, let $\varphi(x_1, \dots, x_n)$ be an atomic formula and let $a_1, \dots, a_n \in \text{TmAlg}(T)$ and the corresponding tuple of closed $\mathcal{L}$ -terms $t_{a_1}, \dots, t_{a_n}$. Since we know that:

$$\text{TmAlg}(T) \models \varphi(t_{a_1}, \dots, t_{a_n}) \iff T \models \varphi(t_{a_1}, \dots, t_{a_n}) \iff \mathcal{B} \models \varphi(t_{a_1}, \dots, t_{a_n})$$

Using the fact that $f(t_{a_i}^{\text{TmAlg}(T)}) = t_{a_i}^{\mathcal{B}}$ for each a_i , we have:

$$\text{TmAlg}(T) \models \varphi(a_1, \dots, a_n) \iff \mathcal{B} \models \varphi(f(a_1), \dots, f(a_n))$$

Thus, f is an isomorphism and $\mathcal{A} \cong \mathcal{B}$. □

3.1.4 Remark. From 3.1.3 (ii), it follows that if T is an $\mathcal{L}$ -theory containing only atomic sentences, then we will have $\text{TmAlg}(T) \models T$. However, in general, the term algebra of an $\mathcal{L}$ -theory T may not be a model of T . For example, T may contain a sentence which implies that every model of T has at least two elements, for example the sentence $\exists xy (x \neq y)$. However, if $\mathcal{L}$ contains only a single constant symbol c and no function symbols, then the only closed $\mathcal{L}$ -term will be c itself, and the term algebra of T will contain only one element, and therefore will certainly not be a model of T .

3.2 The canonical model of a theory.

In this dissertation, we will always use term algebras as our preferred way to choose a structure determined by a theory. However, since in [Hod85] the **canonical model** is used where we would like to use the term algebra, we will now compare the two constructions.

3.2.1 Definition. Let T be an $\mathcal{L}$ -theory. Then T is called **=-closed** if it satisfies the following:

- (i) For every closed $\mathcal{L}$ -term t , we have $t = t$ in T
- (ii) For every atomic $\mathcal{L}$ -formula $\varphi(x)$ and all closed $\mathcal{L}$ -terms s, t if $\varphi(s) \in T$ and $s = t \in T$ or $t = s \in T$ then $\varphi(t) \in T$

3.2.2 Definition. Let $\mathcal{L}$ be a language with at least one constant symbol and let T be an =-closed $\mathcal{L}$ -theory. The **canonical model of T** , which in this section we will generally denote with $\mathcal{A}$ (though we will later use $\mathcal{A}$ for term algebras), is defined as follows:

- (i) The domain of $\mathcal{A}$ consists of equivalence classes of closed $\mathcal{L}$ -terms, with respect to an equivalence relation determined by the equations in T . For closed $\mathcal{L}$ -terms t_1, t_2 we define:

$$t_1 \equiv_T t_2 \iff t_1 = t_2 \in T$$

The equivalence class of a closed $\mathcal{L}$ -term t under $\equiv_T$ is denoted $[t]$ (or $[t]_{\equiv_T}$).

- (ii) For a constant symbol c (which is a closed term), we define $c^{\mathcal{A}} = [c]$.
- (iii) For an n -ary function symbol F and closed terms $t_1, \dots, t_n$, we define:

$$F^{\mathcal{A}}([t_1], \dots, [t_n]) = [F(t_1, \dots, t_n)]$$

- (iv) For an n -ary relation symbol R and closed terms $t_1, \dots, t_n$, we define:

$$\mathcal{A} \models R([t_1], \dots, [t_n]) \text{ iff } R(t_1, \dots, t_n) \in T$$

3.2.3 Remark. Note that, in [Hod85], canonical models are defined only in terms of =-closed sets of **atomic** sentences. Since the definition of =-closure only concerns the atomic sentences of a theory T , if $U \subseteq T$ is the set of all atomic sentences in T ,

the set U will be $=$ -closed if and only if T is $=$ -closed.

Then, the canonical model it is defined strictly as the canonical model of $U \subseteq T$. However, since the definition of the canonical model of T depends only on the atomic sentences in T , these two constructions produce exactly the same structure. The difficulty is that, while the canonical model of a set of atomic sentences U is guaranteed to be a model of U , the canonical model of a general $=$ -closed theory T may not be a model of T .

3.2.4 Proposition. *The canonical model of an $=$ -closed $\mathcal{L}$ -theory T is a well-defined structure.*

Proof. Note that $\equiv_T$ is an equivalence relation on closed terms exactly because T is $=$ -closed. Reflexivity follows from 3.2.1 (i). Symmetry and transitivity follow from (ii) for atomic formulas of the form $x = t$. Constant symbols are closed terms, so $\equiv_T$ is an equivalence relation on constant symbols as well.

Then the interpretations of function and relations symbols are well-defined as well.

We show that if $[t_1] = [s_1], \dots, [t_n] = [s_n]$ then $F^{\mathcal{A}}([t_1], \dots, [t_n]) = F^{\mathcal{A}}([s_1], \dots, [s_n])$.

First, we claim that:

$$\text{if } (t_1 = s_1), \dots, (t_n = s_n) \in T \text{ then } F(t_1, \dots, t_n) = F(s_1, \dots, s_n) \in T$$

We have $F(s_1, \dots, s_n) = F(s_1, \dots, s_n) \in T$ by 3.2.1 (i). Then for $(t_1 = s_1) \in T$ and the atomic formula $F(x, s_2, \dots, s_n) = F(s_1, \dots, s_n)$ we have $F(t_1, s_2, \dots, s_n) \in T$ by 3.2.1 (ii). We repeat for $F(t_1, x, s_3, \dots, s_n) = F(s_1, \dots, s_n), \dots, F(t_1, \dots, t_{n-1}, x) = F(s_1, \dots, s_n)$ and obtain that $F(t_1, \dots, t_n) = F(s_1, \dots, s_n) \in T$.

Thus we have:

$$\begin{aligned} [t_1] = [s_1], \dots, [t_n] = [s_n] &\implies (t_1 = s_1), \dots, (t_n = s_n) \in T \\ &\implies F(t_1, \dots, t_n) = F(s_1, \dots, s_n) \in T \\ &\implies [F(t_1, \dots, t_n)] = [F(s_1, \dots, s_n)] \\ &\implies F^{\mathcal{A}}([t_1], \dots, [t_n]) = F^{\mathcal{A}}([s_1], \dots, [s_n]) \end{aligned}$$

Finally, we show:

$$\text{if } [t_1] = [s_1], \dots, [t_n] = [s_n] \text{ then } \mathcal{A} \models R([t_1], \dots, [t_n]) \iff \mathcal{A} \models R([s_1], \dots, [s_n])$$

For this, it suffices to show that:

$$\text{if } (t_1 = s_1), \dots, (t_n = s_n) \in T \text{ then } R(t_1, \dots, t_n) \in T \iff R(s_1, \dots, s_n) \in T$$

The claim is symmetrical between the t_i and s_i so we suppose $(t_1 = s_1), \dots, (t_n = s_n) \in T$ and assume $R(s_1, \dots, s_n) \in T$. Then by 3.2.1 (ii) for $R(x, s_2, \dots, s_n)$ and $t_1 = s_1$ we have $R(t_1, s_2, \dots, s_n) \in T$. We repeat this for the atomic formulas $R(t_1, x, s_3, \dots, s_n), \dots, R(t_1, \dots, t_{n-1}, x)$ and get $R(t_1, \dots, t_n) \in T$ as well.

□

3.2.5 Remark. If T is not $=$ -closed, the relation $\equiv_T$ is unlikely to be well-defined. However, given any $\mathcal{L}$ -theory T , we can still obtain a canonical model by taking T' to be the smallest $=$ -closed $\mathcal{L}$ -theory containing T , and taking $\mathcal{A}$ to be the canonical model of T' .

Before we say more and begin to compare the canonical model with the term algebra, we state the analogues of 3.1.3 (i)-(iii) for canonical models.

3.2.6 Proposition. *Let T be an $=$ -closed $\mathcal{L}$ -theory in a language with at least one constant symbol. Write $\mathcal{A}$ for the canonical model of T .*

(i) *Every $a \in \mathcal{A}$ is of the form $t^{\mathcal{A}}$ for some closed $\mathcal{L}$ -term t*

(ii) *Every atomic $\mathcal{L}$ -sentence σ satisfies:*

$$\mathcal{A} \models \sigma \iff \sigma \in T$$

(iii) *If $\mathcal{B}$ is another $\mathcal{L}$ -structure satisfying (i) and (ii) then $\mathcal{A} \cong \mathcal{B}$*

Proof. (i) and (ii) are immediate from the definition. The proof of (iii) follows exactly the same argument as 3.1.3 (iii). □

3.3 Hintikka theories

Despite being defined very similarly, the term algebra and canonical model of a theory T can be very different. We will now work towards a particular condition on T which is sufficient to eliminate this difference: if T is a **Hintikka Theory** then the canonical model and term algebra are identical. We will conclude this section by showing that under these conditions, not only are the structures identical, but also a model of T .

In fact, a theory T does not need to be a Hintikka theory to ensure that the term algebra and canonical model of T coincide. We can impose a weaker condition on just the atomic sentences of T to ensure this. However, the crucial utility of Hintikka theories is that if T is a Hintikka theory, the term algebra and canonical model will not just coincide but also be models of T . The second useful property of Hintikka theories is that any consistent set of sentences T can be methodically extended to a Hintikka theory T' containing T - and the definition of a Hintikka theory gives a very clear picture of exactly how to perform this algorithmically.

3.3.1 Observation. Suppose T is an $\mathcal{L}$ -theory which is not $=$ -closed and consider the procedure needed to produce T' , the smallest $=$ -closed $\mathcal{L}$ -theory containing T . We show that $T \models T'$, with an argument that will closely resemble the forcing constructions later in this dissertation. First, take $T_1 = T \cup \{t = t \mid t \text{ is a closed } \mathcal{L}\text{-term}\}$. Then certainly $T \models T_1$.

Next, enumerate the triples $((\varphi, t, s)_i \mid i \in \mathbb{N})$ for each atomic $\mathcal{L}$ -formula $\varphi(x)$ and closed terms t, s . Given the i 'th triple $(\varphi, t, s)_i$, consider whether T_i contains $\varphi(t)$ and one of $t = s$ or $s = t$. If so, put $T_{i+1} = T_i \cup \{\varphi(s)\}$, and certainly $T_i \models T_{i+1}$ as $\varphi(t) \wedge (t = s \vee s = t) \models \varphi(s)$. Then T' will be the union of the chain of T_i . Since $T \models T_1$ and $T_i \models T_{i+1}$ for all $i \in \mathbb{N}$, we get $T \models T_i$ for all $i \in \mathbb{N}$, and obtain $T \models T'$.

Thus, since the term algebra of T depends on the deductive closure of T , it follows that if T' is the smallest $=$ -closed set containing T then we will have $\text{TmAlg}(T) = \text{TmAlg}(T')$. For this reason, in the following results, we can assume T is $=$ -closed to compare the term algebra and canonical model of T without any loss of generality.

3.3.2 Remark. By comparing 3.1.3 and 3.2.6 we can see the key difference between the two constructions. Suppose T is an $=$ -closed $\mathcal{L}$ -theory. Take the term algebra $\text{TmAlg}(T)$ of T and the canonical model $\mathcal{A}$ of T . Let σ be an atomic $\mathcal{L}$ -sentences. Then we have:

$$\text{TmAlg}(T) \models \sigma \iff T \models \sigma \quad (3.1.3)$$

$$\mathcal{A} \models \sigma \iff \sigma \in T \quad (3.2.6)$$

Thus if $\sigma \in \overrightarrow{T}$ (the deductive closure of T) but $\sigma \notin T$, then we will have $\text{TmAlg}(T) \models \sigma$ but $\mathcal{A} \not\models \sigma$.

This motivates our next proposition.

3.3.3 Proposition. *Let T be an $=$ -closed $\mathcal{L}$ -theory. Let $\text{TmAlg}(T)$ be the term*

algebra of T and let $\mathcal{A}$ be the canonical model of T . Then, writing $\vec{T}$ for the deductive closure of T and $\text{At-Sen}(\mathcal{L})$ for the set of all atomic $\mathcal{L}$ -sentences, we have:

$$\text{TmAlg}(T) = \mathcal{A}^{11} \quad \text{iff} \quad \vec{T} \cap \text{At-Sen}(\mathcal{L}) \subseteq T$$

Proof. Take an $=$ -closed $\mathcal{L}$ -theory T and let $\text{TmAlg}(T)$ and $\mathcal{A}$ be as above. Then suppose $\mathcal{A} = \text{TmAlg}(T)$. Then for all atomic $\sigma \in \text{At-Sen}(\mathcal{L})$:

$$\begin{aligned} \sigma &\in \vec{T} \\ \iff T &\models \sigma && \text{by Proposition 3.1.3} \\ \iff \text{TmAlg}(T) &\models \sigma && \text{by Definition 3.1.2} \\ \iff \mathcal{A} &\models \sigma && \text{since } \mathcal{A} = \text{TmAlg}(T) \\ \iff \sigma &\in T && \text{by Proposition 3.2.6} \end{aligned}$$

Therefore, $\vec{T} \cap \text{At-Sen}(\mathcal{L}) \subseteq T$ - every atomic consequence of T is contained in T .

Conversely, if $\vec{T} \cap \text{At-Sen}(\mathcal{L}) \subseteq T$ then for all $\sigma \in \text{At-Sen}(\mathcal{L})$ we have:

$$T \models \sigma \iff \sigma \in T$$

It follows that the equivalence relations $\sim_T$ and $\equiv_T$ are identical as for any closed terms t_1, t_2 we have:

$$t_1 \equiv_T t_2 \iff t_1 = t_2 \in T \iff T \models t_1 = t_2 \iff t_1 \sim_T t_2$$

Thus $\sim_T$ and $\equiv_T$ determine the exact same equivalence classes of closed $\mathcal{L}$ -terms, and $\text{TmAlg}(T)$ and $\mathcal{A}$ have exactly the same domain.

Since these equivalence classes also determine the interpretations of constant and function symbols, for all constants c we have:

$$c^{\text{TmAlg}(T)} = [c]_{\sim_T} = [c]_{\equiv_T} = c^{\mathcal{A}}$$

¹¹Here, we are claiming *equality* between structures, not just isomorphism. An $\mathcal{L}$ -structure is completely determined by its domain and the interpretations of $\mathcal{L}$ -symbols in the structure. Therefore, structures which share these features are identical. Equivalently, equality between structures holds iff the identity map between the structures is an isomorphism.

And for all n -ary functions F and all closed terms $t_1, \dots, t_n$:

$$\begin{aligned} F^{\text{TmAlg}(T)}([t_1]_{\sim_T}, \dots, [t_n]_{\sim_T}) &= [F(t_1, \dots, t_n)]_{\sim_T} \\ &= [F(t_1, \dots, t_n)]_{\equiv_T} \\ &= F^{\mathcal{A}}([t_1]_{\equiv_T}, \dots, [t_n]_{\equiv_T}) \end{aligned}$$

Finally, for any n -ary relation R and closed terms $t_1, \dots, t_n$, note that $R(t_1, \dots, t_n)$ is atomic. Therefore:

$$\begin{aligned} \text{TmAlg}(T) \models R([t_1]_{\sim_T}, \dots, [t_n]_{\sim_T}) &\iff T \models R(t_1, \dots, t_n) \\ &\iff R(t_1, \dots, t_n) \in T \\ &\iff \mathcal{A} \models R([t_1]_{\equiv_T}, \dots, [t_n]_{\equiv_T}) \end{aligned}$$

It follows that the structures are identical. Note that we could also have applied either of the propositions 3.1.3 and 3.2.6 to show that $\text{TmAlg}(T) \cong \mathcal{A}$ and observed that, since the domains are equal, this isomorphism is the identity map. $\square$

3.3.4 Corollary. *Let T be an $=$ -closed $\mathcal{L}$ -theory such that $\vec{T} \cap \text{At-Sen}(\mathcal{L}) \subseteq T$, so every atomic consequence of T is already contained in T . Then the term algebra of T is identical to the canonical model of T .*

3.3.5 Example. The term algebra and the canonical model have almost identical definitions, but despite their apparent similarity, their differences can result in two completely different structures.

For example, consider the language $\mathcal{L} = \{c, d\}$, consisting of just two constant symbols and no function or relation symbols. Then let:

$$T = \{c = c, d = d, \forall xy (x = y)\}$$

This theory satisfies (H2) and (H3) and is therefore $=$ -closed. Then note that $T \models c = d$, but $c = d \notin T$.

Now, $c \sim_T d$ and therefore c and d will name the same element in the term algebra of T . Therefore the domain of the term algebra of T will be:

$$|\text{TmAlg}(T)| = \{[c]_{\sim_T}\} = \{[d]_{\sim_T}\} \quad \text{with } c^{\text{TmAlg}(T)} = d^{\text{TmAlg}(T)}$$

However, since $c = d \notin T$ it is not the case that $c \equiv_T d$, so c and d will name distinct elements in the canonical model $\mathcal{A}$ of the atomic sentences of T . Therefore the domain of the canonical model $\mathcal{A}$ will be:

$$|\mathcal{A}| = \{[c]_{\equiv_T}, [d]_{\equiv_T}\} \quad \text{with } c^{\mathcal{A}} \neq d^{\mathcal{A}}$$

From this it follows that $\mathcal{A} \neq \text{TmAlg}(T)$. Furthermore, it is clear that $\text{TmAlg} \models T$, but $\mathcal{A} \not\models T$ - the canonical model is only guaranteed to be a model of the atomic sentences in T , while the term algebra will also be a model of the atomic consequences of T .

3.3.6 Remark. We have shown how to ensure that the canonical model and term algebra of a theory T coincide. However, we have not yet explained how we can ensure that they are both a model of T .

We now define Hintikka theories, which will be very important for the rest of this dissertation. Their original definition comes from [Hod93, p. 44] (though they are called Hintikka *sets*). Interestingly, they are never explicitly defined in [Hod85], despite still being an important part of the machinery of the text.

This can be seen in the definition of a consistency structure in [Hod85, pp. 19-20] (which we will define in 4.1.1 as a partially ordered set containing certain sets of sentences), which has an obvious resemblance to the following definition of a Hintikka set - consistency structures are defined to allow one to build Hintikka theories out of ascending chains of the elements (conditions) in a consistency structure. Furthermore, in [Hod85, Theorem 2.1.2], Hodges proves that every element of a consistency structure is a consistent set of sentences. If one inspects the proof of this theorem closely, Hodges proves that a certain set of sentences is consistent just by showing that it is contained in a Hintikka theory, and that this Hintikka theory has a model (the canonical model and, it turns out, also the term algebra).

3.3.7 Definition. Let $\mathcal{L}$ be a language. A **Hintikka theory** in $\mathcal{L}$ is a set T of $\mathcal{L}$ -sentences satisfying all of the following properties:

- (H1) For every atomic $\mathcal{L}$ -sentence σ , if $\sigma \in T$ then $\neg\sigma \notin T$
- (H2) For every closed $\mathcal{L}$ -term t , we have $t = t$ in T
- (H3) For every atomic $\mathcal{L}$ -formula $\varphi(x)$ and all closed $\mathcal{L}$ -terms s, t if $\varphi(s) \in T$ and $s = t \in T$ or $t = s \in T$ then $\varphi(t) \in T$

For all $\mathcal{L}$ -sentences σ, τ :

(H4) If $\neg\neg\sigma \in T$ then $\sigma \in T$

(H5) If $\sigma \wedge \tau \in T$ then both $\sigma \in T$ and $\tau \in T$

If $\neg(\sigma \wedge \tau) \in T$ then at least one of $\neg\sigma \in T$ or $\neg\tau \in T$

(H6) If $\sigma \vee \tau \in T$ then at least one of $\sigma \in T$ or $\tau \in T$

If $\neg(\sigma \vee \tau) \in T$ then both $\neg\sigma \in T$ and $\neg\tau \in T$.

And for all $\mathcal{L}$ -formulas $\varphi(x)$:

(H7) If $\forall x \varphi \in T$ then for every closed $\mathcal{L}$ -term t , we have $\varphi(t) \in T$

If $\neg\forall x \varphi \in T$ then for some closed $\mathcal{L}$ -term t , we have $\neg\varphi(t) \in T$

(H8) If $\exists x \varphi \in T$ then for some closed $\mathcal{L}$ -term t , we have $\varphi(t) \in T$

If $\neg\exists x \varphi \in T$ then for every closed $\mathcal{L}$ -term t , we have $\neg\varphi(t) \in T$

3.3.8 Remark. It is important to note a theory T can only be described as a Hintikka theory with respect to a particular language. An important example of this is where T is a Hintikka theory in $\mathcal{L}$ and $\mathcal{L}^+ \supsetneq \mathcal{L}$ is an extension by constants; then T will not be a Hintikka theory in $\mathcal{L}^+$. To see this note that if $c \in \mathcal{L}^+ \setminus \mathcal{L}$ is a new constant symbol then $c = c \notin T$ contradicts (H2). However, if there is no ambiguity about the language in use, we say that an $\mathcal{L}$ -theory T a Hintikka theory to mean that T is a Hintikka theory in $\mathcal{L}$. However, it is otherwise important to be specific about the language.

3.3.9 Theorem ([Hod93, Theorem 2.3.3, pp. 44-45]). *Let T be a set of $\mathcal{L}$ -sentences. If T is a Hintikka theory then the canonical model of the atomic sentences in T is a model of T . Consequently, T is consistent.*¹²

Proof. Suppose T is a Hintikka theory. Then by (H2) and (H3), T is $=$ -closed. We now show that the canonical model $\mathcal{A}$ of T is a model of T . For this, we show that for every $\mathcal{L}$ -sentence σ we have:

$$(\dagger) \quad \sigma \in T \Rightarrow \mathcal{A} \models \sigma \quad \text{and} \quad \neg\sigma \in T \Rightarrow \mathcal{A} \models \neg\sigma$$

¹²The convention throughout this dissertation is to use ‘theory’ for a consistent set of $\mathcal{L}$ -sentences. By this theorem, the terminology for Hintikka ‘theories’ fits with this convention. Importantly, the definition of a Hintikka theory applies to a set of $\mathcal{L}$ -sentences without assuming consistency, so there is no circularity.

We prove this by induction on the complexity of σ . First, suppose σ is atomic. If $\sigma \in T$ then by 3.2.6 (ii) we have $\mathcal{A} \models \sigma$. Then, if $\neg\sigma \in T$ then $\sigma \notin T$ by (H1). Again by 3.2.6 (ii), we have $\mathcal{A} \not\models \sigma$, and thus $\mathcal{A} \models \neg\sigma$.

Suppose σ is of the form $\neg\tau$, for some sentence τ satisfying $(\dagger)$. Then if $\neg\tau \in T$, it follows that $\mathcal{A} \models \neg\tau$. If $\neg\neg\tau \in T$, then by (H4) we have $\tau \in T$, and thus $\mathcal{A} \models \tau$.

Now, suppose σ is of the form $\tau_1 \wedge \tau_2$, for sentences τ_1, τ_2 satisfying $(\dagger)$. If $\tau_1 \wedge \tau_2 \in T$, then by (H5), we have $\tau_1, \tau_2 \in T$ and therefore $\mathcal{A} \models \tau_1$ and $\mathcal{A} \models \tau_2$. Thus $\mathcal{A} \models \tau_1 \wedge \tau_2$. Similarly, by (H5), if $\neg(\tau_1 \wedge \tau_2) \in T$ then $\mathcal{A} \models \neg(\tau_1 \wedge \tau_2)$.

The disjunctive case, where σ is of the form $\tau_1 \vee \tau_2$ follows the same argument as the conjunctive case, instead using (H6).

For the final two cases, suppose $\varphi(x)$ is an $\mathcal{L}$ -formula such that for any closed $\mathcal{L}$ -term, the sentence $\varphi(t)$ satisfies $(\dagger)$.

Now, suppose σ is of the form $\forall x \varphi$. Then, if $\forall x \varphi \in T$ then by (H7) for all closed $\mathcal{L}$ -terms t we have $\varphi(t) \in T$. Thus, for all closed $\mathcal{L}$ -terms t we have $\mathcal{A} \models \varphi(t)$. Let $a \in A$ be an arbitrary element. By our observations above, $a = t^A$ for some closed term t and therefore $\mathcal{A} \models \varphi(a)$. Since a was arbitrary, it follows that $\mathcal{A} \models \forall x \varphi$. Similarly by (H7) if $\neg\forall x \varphi \in T$, then $\neg\varphi(t) \in T$ for some closed $\mathcal{L}$ -term t . Thus $\mathcal{A} \models \neg\varphi(t) \in T$, and we have $\mathcal{A} \models \neg\forall x \varphi$.

Finally, the existential case follows the same argument as the universal case, instead using (H8).

Having proved the claim, we conclude that $\mathcal{A} \models T$ and thus that the canonical model of the atomic sentences of a Hintikka theory is a model of that theory. Thus, every Hintikka theory is consistent. $\square$

3.3.10 Corollary. *Let T be a Hintikka theory in $\mathcal{L}$. Then:*

- (i) *If σ is an atomic $\mathcal{L}$ -sentence, then we have $\sigma \in T$ if and only if $T \models \sigma$*
- (ii) *Therefore $\vec{T} \cap \text{At-Sen}(\mathcal{L}) \subseteq T$. Consequently, the canonical model $\mathcal{A}$ of the atomic sentences of a Hintikka theory T is the same structure as $\text{TmAlg}(T)$.*

Proof.

- (i) Clearly $\sigma \in T$ entails $T \models \sigma$. So, for the converse, suppose $T \models \sigma$, and for contradiction suppose that $\sigma \notin T$. Consider the set $\Sigma = T \cup \{\neg\sigma\}$. We claim

Σ is a Hintikka theory. We know T satisfies (H1)-(H8). We just have to check that the addition of $\neg\sigma$ in Σ does not change this.

We know Σ satisfies (H1) as we have $\sigma \notin \Sigma$. For the remaining conditions, since T satisfies (H2)-(H8) and since $\neg\sigma$ is only a negated atomic sentence (and thus not of the relevant form to affect whether Σ satisfies the remaining conditions), it follows that Σ satisfies (H2)-(H8) as well.

Therefore, Σ is a Hintikka theory. However, as our assumption that $T \models \sigma$ implies that $\Sigma = T \cup \{\neg\sigma\}$ is inconsistent, this is a contradiction to 3.3.9.

- (ii) This follows from part (i), as we now know that for a Hintikka theory T , we have $\vec{T} \in \text{At-Sen}(\mathcal{L}) \subseteq T$. Thus by 3.3.4, it follows that $\text{TmAlg}(T) = \mathcal{A}$.

□

3.3.11 Corollary. *Let T be an $\mathcal{L}$ -theory. If T is a Hintikka theory in $\mathcal{L}$ then $\text{TmAlg}(T) \models T$.*

Proof. Let T be a Hintikka theory. By 3.3.10, the term algebra of T is a model of T iff the canonical model of the atomic sentences of T is a model of T . Then, the corollary follows from 3.3.9. □

3.3.12 Remark. In forcing, we will work within a partially ordered set of sets of sentences (‘conditions’) which are ordered by inclusion. Within these partially ordered sets, we will construct theories by building chains of conditions and taking their union. We will soon see that when building chains by forcing, it is always possible to ensure that the union of the chain is a Hintikka theory.

Since forcing is intended as a method to build structures, not just theories, we want the unions of these chains to determine a particular structure. In [Hod85], the structure determined by a chain of conditions determines is the canonical model. However, for the purposes of forcing in this dissertation, the structure determined by a chain of conditions will be a term algebra.

The work of this section allows us to translate Hodges’ results about canonical models constructed with forcing into results about term algebras built by forcing - since when forcing, we will always be able to ensure that the union of a chain of conditions is a Hintikka theory, the canonical model in Hodges’ forcing is exactly the same structure as the term algebra in the forcing in this dissertation.

These ideas are vaguely worded here. However, the next section will give explicit definitions for what it means to build a chain or structure 'by forcing' and exactly how we can 'ensure the union satisfies a certain property'.

4 Forcing

In this section, we will introduce the technique of **forcing**, which is a method of constructing models of first-order theories with certain properties. A key feature of forcing is the expansion of the working language by an infinite set of new constant symbols, called witnesses. Within a specified partially ordered set, we build a chain of sets of sentences called conditions, using the witnesses to control the properties of individual elements of the resulting model. However, at each step of the chain, only finitely many witnesses will have been mentioned. Roughly we build the chain by adding small, incomplete pieces of information about the structure we are building. After infinitely many steps, we will have constructed a complete picture of the structure.

In finite forcing, which will be looked at in section 6, each condition in the chain must contain at most finitely many basic sentences, and can therefore dictate the properties of only finitely many elements of the final model. However, the union of the chain may be infinite and contain occurrences of infinitely many witnesses. This allows us to construct infinite structures where the properties of each element have been tightly controlled.

Another form of forcing is Robinson infinite forcing, though we will not look at this in this dissertation. In infinite forcing, the conditions in the chain may be infinite objects, though each must still mention at most finitely many witnesses. The conditions in Robinson infinite forcing are sometimes thought of not as sets of sentences, but as finitely generated structures. These form a chain of structures, and the final structure produced is the union of this chain. We have defined the union of a chain in 2.3.3 and shown that if T is a $\forall\exists$ -theory then the union of a chain of models of T will again be a model of T . This is the approach taken in [Sar71].

4.1 Forcing structures

When forcing, we always work in a countable language $\mathcal{L}^+$ extended from a base language $\mathcal{L}$ by the addition of new constants, called witnesses, from a countably infinite set C . Within this extended language, we work within a partially ordered set of consistent sets of $\mathcal{L}^+$ -sentences, called N -conditions, and which form a partial order under inclusion. We call this set a **forcing structure**, provided it satisfies several properties essential for forcing.

Note that we use the terminology ‘forcing structure’ and ‘consistency structure’. In [Hod85], these objects are respectively called a ‘notion of forcing’ and a ‘notion of consistency’.

The goal will be to construct ascending chains within this forcing structure and then take their union. In this way, we can construct theories with certain helpful properties. The first of these helpful properties will be that the union of the chain is a Hintikka theory. However, as we continue, we will go on to explore what additional properties we can be sure to impose on the chain.

Throughout this section, all languages are countable. Even where we take languages extended by constants, there will be at most countably many constants, and these languages extensions will be countable as well. This is crucial, as many results (including the Omitting Types Theorem presented in [Mar25]) fail when we drop the assumption of countability.

This material on consistency structures and forcing structures can be found in [Hod85, pp. 19-22].

4.1.1 Definition. Let $\mathcal{L}$ be a first-order countable language. A **consistency structure in $\mathcal{L}$** is a set N of sets of $\mathcal{L}$ -sentences, known as N -conditions (or just conditions), such that the following are satisfied for every condition $P \in N$:

- (C1) For every atomic $\mathcal{L}$ -sentence σ , if $\sigma \in P$ then $\neg\sigma \notin P$
- (C2) For every closed $\mathcal{L}$ -term t , we have $P \cup \{t = t\} \in N$
- (C3) For every atomic $\mathcal{L}$ -formula $\varphi(x)$ and all closed $\mathcal{L}$ -terms s, t , if $\varphi(s) \in P$ and $s = t \in P$ or $t = s \in P$ then $P \cup \{\varphi(t)\} \in N$

And for all $\mathcal{L}$ -sentences σ, τ :

- (C4) If $\neg\neg\sigma \in P$ then $P \cup \{\sigma\} \in N$
- (C5) If $\sigma \wedge \tau \in P$ then $P \cup \{\sigma, \tau\} \in N$
 If $\neg(\sigma \wedge \tau) \in P$ then $P \cup \{\neg\sigma\} \in N$ or $P \cup \{\neg\tau\} \in N$
- (C6) If $\sigma \vee \tau \in P$ then $P \cup \{\sigma\} \in N$ or $P \cup \{\tau\} \in N$
 If $\neg(\sigma \vee \tau) \in P$ then $P \cup \{\neg\sigma, \neg\tau\} \in N$

And for all $\mathcal{L}$ -formulas $\varphi(x)$:

- (C7) If $\forall x \varphi \in P$ then for every closed $\mathcal{L}$ -term t , we have $P \cup \{\varphi(t)\} \in N$

If $\neg\forall x \varphi \in P$ then for some closed $\mathcal{L}$ -term t , we have $P \cup \{\neg\varphi(t)\} \in N$ ¹³

(C8) If $\exists x \varphi \in P$ then for some closed $\mathcal{L}$ -term t , we have $P \cup \{\varphi(t)\} \in N$

If $\neg\exists x \varphi \in P$ then for every closed $\mathcal{L}$ -term t , we have $P \cup \{\neg\varphi(t)\} \in N$

The following theorem validates the terminology ‘consistency structure’. The proof of this theorem is not difficult, but follows from an almost identical argument that we will use to prove the stronger result (4.3.2) in the next section.

4.1.2 Theorem. *Let N be a consistency structure in $\mathcal{L}$ and let $P \in N$. Then P is consistent.*

Proof. This is a consequence of 4.3.2, which shows that every $P \in N$ is contained in a Hintikka theory, which we know to be consistent as it is satisfied in its own term algebra. $\square$

4.1.3 Remark. Note the similarity in structure between the definition of a consistency structure and 3.3.7 where we defined Hintikka theories. From their similarity, we observe that if T is a Hintikka theory in $\mathcal{L}$ then all of the following are consistency structures in $\mathcal{L}$:

- $\{T\}$
- The set of all subsets of T
- The set of all finite subsets of T

Furthermore, we will see in the proof of 4.3.2 that the definition of a Hintikka theory gives us an obvious method for adding sentences to a theory in order to make it into a Hintikka theory. For example, based on (H5) in the definition of a Hintikka theory, we should take every sentence $\sigma \wedge \tau$ in T and then add σ, τ to T . The definition of a consistency structure N ensures that this algorithm can be carried out by building a chain in N - for example, (C5) ensures that there is always a condition available to ensure that (H5) holds.

¹³Since in this dissertation we take $\exists$ as primitive and $\forall x \varphi$ to be an abbreviation for $\neg\exists x \neg\varphi$, technically (C7) is unnecessary in the presence of (C8). We also take $\rightarrow, \leftarrow$ and $\leftrightarrow$ to be abbreviations for the equivalent combinations of $\vee, \wedge$, and $\neg$, and indeed statements for these connectives analogous to (C5) and (C6) can be derived from those already listed - (e.g. if $\sigma \rightarrow \tau \in P$ then $P \cup \{\neg\sigma\} \in N$ or $P \cup \{\tau\} \in N$). Though redundant, it will be useful to have a separate statement for $\forall$ to refer back to. On the other hand, additional statements for $\rightarrow, \leftarrow$ and $\leftrightarrow$ will not be any more useful than those we already have.

The following definition is not present in the literature, but is instead an *ad hoc* definition to simplify terminology for the remainder of this dissertation.

4.1.4 Definition. Let $\mathcal{L}$ be a language. If $\mathcal{L}$ is countable and contains an infinite subset $C \subseteq \mathcal{L}$ of distinguished constant symbols, called witnesses, then $\mathcal{L}$ is called a **language with witnesses**.

Now, to get a forcing structure, we impose just two more requirements on a consistency structure.

4.1.5 Definition. Let $\mathcal{L}$ be a language with witnesses C . A **forcing structure N in $\mathcal{L}$** is a consistency structure in $\mathcal{L}$ which satisfies the following additional properties for every condition $P \in N$:

- (F1) For every closed $\mathcal{L}$ -term t , if $c \in C$ is a witness which appears nowhere in P or t , then $P \cup \{t = c\} \in N$
- (F2) At most finitely many witnesses occur in P

4.1.6 Examples. Let $\mathcal{L}$ be a language with witnesses C . The following are some examples of forcing structures.

- (i) $N = \{P \subseteq \text{Sen}(\mathcal{L}) \mid P \text{ is consistent and contains finitely many witnesses}\}$
Claim: This forcing structure N is the largest forcing structure in $\mathcal{L}$, in that it contains all other forcing structures in $\mathcal{L}$ as subsets.

Proof. If this were not the case, and N' were another forcing structure in $\mathcal{L}$ with $N' \not\subseteq N$ then there must be $P' \in N'$ with $P' \notin N$. However, for this to be the case, P' would have to be either inconsistent or contain infinitely many witnesses, which cannot be the case without N' failing to be a forcing structure. Therefore no such N' exists. $\square$

- (ii) $N \cap \{P \subseteq \text{Sen}(\mathcal{L}) \mid P \text{ is a set of atomic sentences}\}$
- (iii) $N \cap \{P \subseteq \text{Sen}(\mathcal{L}) \mid P \text{ is a set of basic sentences}\}$
- (iv) $N \cap \{P \subseteq \text{Sen}(\mathcal{L}) \mid P \text{ is a set of quantifier-free sentences}\}$
- (v) $N \cap \{P \subseteq \text{Sen}(\mathcal{L}) \mid P \subseteq Q\text{-Sen}(\mathcal{L})\}$ where Q stands for $\forall$, $\exists$, or $\forall\exists$

It is routine to verify that (ii)-(iv) satisfy (C1)-(C8) and (F1)-(F2).

4.1.7 Examples. Let $\mathcal{L}$ be a language with witnesses C . Let N be a forcing structure. Then the following are also forcing structures:

(i) The set of all finite N -conditions

$$\{P \in N \mid P \text{ is finite}\}$$

(ii) The set of all subsets of N -conditions

$$\{P \subseteq Q \mid Q \in N\}$$

(iii) The set of all N -conditions containing a particular $Q \in N$. This is

$$\{P \in N \mid Q \subseteq P\}$$

4.1.8 Remark. Let $\mathcal{L}$ and $\mathcal{L}^+ = \mathcal{L}(C)$ be as above. Let T be a Hintikka theory in $\mathcal{L}^+$. Then by (H2), it follows that for every witness $c \in C$ we have $c = c \in T$. Then, due to the fact that T contains infinitely many witnesses, it follows that no Hintikka theory T in a language with infinitely many witnesses can ever be a forcing condition.

In the content that follows, we will construct chains of conditions within forcing structures. Part of the goal of each construction will be to ensure that the union of the chain of conditions is a Hintikka theory, despite the fact that no forcing structure (which, by definition, is only defined in a language with infinitely many witnesses) can actually contain a Hintikka theory.

The idea is that at each step of the chain, only finitely many witnesses have been mentioned. However, by iterating this process and constructing an infinite chain, the union will be a theory in which all of the witnesses have been mentioned.

We will go on to use these theories, constructed by taking the union of a chain of conditions, to construct structures where every element is named by a witness. At each stage of the construction of the chain, we will have only influenced the properties of finitely many elements of this structure. However, after iterating the construction, we will obtain an infinite structure where the properties of each individual element will have been influenced at some finite step(s) in the construction.

4.2 Enforceable properties

The following content can be found in [Hod85, pp. 23-25].

4.2.1 Definition. Let N be a forcing structure in a countable language $\mathcal{L}$ containing an infinite set of witnesses C . An **N -construction sequence** is a sequence $(P_i)_{i \in \mathbb{N}}$ of N -conditions which forms an ascending chain. We will write $\overline{P}$ as an abbreviation for the sequence $(P_i)_{i \in \mathbb{N}}$.

The union $\bigcup_{i \in \mathbb{N}} P_i$ of a chain of N -conditions is called the **union of the $\overline{P}$** or the **compiled theory**, which we will also abbreviate with $\bigcup \overline{P}$. We will also often use T^+ to denote the compiled theory.

4.2.2 (Forcing with games). Given a forcing structure N and a property $\mathcal{P}$ which could be true or false of a set of sentences (e.g. non-empty, consistent, deductively closed, complete). We can now define what it means for $\mathcal{P}$ to be enforceable with respect to the forcing structure N .

Hodges' approach in [Hod85] is to frame enforceable properties within a game, which is conceived to take place between two players $\exists$ (she) and $\forall$ (he). In this game, the players take turns to choose the next condition in an N -construction sequence.

Within the games framework, a property $\mathcal{P}$ is **enforceable** if, in a game where we allow $\forall$ to make the first choice, there is a way for $\exists$ to play the game which, no matter how $\forall$ plays for the rest of the game, will reliably ensure that $\mathcal{P}$ is true of the **compiled theory**. In this case, $\exists$ is said to have a **winning strategy**, where we consider $\exists$ to have won the game if $\mathcal{P}$ is true of the compiled theory at the end of the game.

The following definition is a formal presentation of this idea.

4.2.3 Definition. Let N be a forcing structure in $\mathcal{L}$, and let $\mathcal{P}$ be a property which an $\mathcal{L}$ -theory could have or fail to have.

- The property $\mathcal{P}$ is said to be **enforceable in N** (or N -enforceable) if, given any infinite subset $X \subseteq \mathbb{N}$, there exists a family of functions $(f_i : N \rightarrow N \mid i \in X)$ satisfying:

(S1) $P \subseteq f_i(P)$ for all $i \in X$ and all $P \in N$

(S2) If $\overline{P}$ is an N -construction sequence satisfying $P_{i+1} = f_i(P_i)$ for all $i \in X$, then $\bigcup \overline{P}$ has the property $\mathcal{P}$.

- If $(f_i : N \rightarrow N \mid i \in X)$ satisfies (S1), then the family $(f_i : N \rightarrow N \mid i \in X)$ as a whole is called a **strategy in X** . Furthermore, each of the functions are called **strategy functions** and the particular function f_i for $i \in X$ is called

the i 'th strategy function.

- If a strategy $(f_i : N \rightarrow N \mid i \in X)$ satisfies (S2) in addition to (S1), then it is called a **$\mathcal{P}$ -winning strategy for X** .

4.2.4 Remark. There are some technicalities in this definition which we discuss. Let $\mathcal{L}$ be a language with witnesses and fix a forcing structure N in $\mathcal{L}$.

- (i) First, to understand the idea behind a strategy $(f_i : N \rightarrow N \mid i \in X)$, we suppose that the player $\exists$ is given an infinite subset $X \subseteq \mathbb{N}$ and she chooses the component P_{i+1} of the N -construction sequence whenever $i \in X$ (and similarly, the player $\forall$ chooses P_1 as well as P_{i+1} whenever $i \in \mathbb{N} \setminus X$).

Then, $(f_i : N \rightarrow N \mid i \in X)$ acts as a guide for how $\exists$ should choose each P_{i+1} - given P_i on the i 'th turn of the game, she should use the i 'th strategy function to choose P_{i+1} to be $f_i(P_i)$. Condition (S1) that $P \subseteq f_i(P)$ ensures that her move is 'legal' i.e. that her choice always continues the construction of an ascending chain.

Then, condition (S2) says that if $\bar{P}$ represents a game where $\exists$ has played according to the strategy $(f_i : N \rightarrow N \mid i \in X)$, then the compiled theory $\bigcup \bar{P}$ will satisfy $\mathcal{P}$. For this reason, if we consider $\exists$ to have 'won' whenever the compiled theory has $\mathcal{P}$, we call a strategy which satisfies (S2) a winning strategy.

- (ii) In this dissertation, the natural numbers $\mathbb{N}$ are the set $\{1, 2, 3, \dots\}$ and do not include 0. Therefore, this definition implicitly assumes that $\exists$ is never able to choose P_1 . In order to have a winning strategy, it must be winning regardless of how the first move of the game is played. This can cause some confusion. Suppose $O \subset \mathbb{N}$ is the set of all odd numbers. If $(f_i : N \rightarrow N \mid i \in O)$ is a strategy, then if $\exists$ plays according to this strategy then she is choosing P_{i+1} for all $i \in O$ - that is, she is choosing the **even** components of the chain, based on what she is given as the **odd** components.

In 4.2.17, we will discuss properties which are enforceable in games where $\exists$ is allowed to make the first move.

- (iii) Note that according to 4.2.3 and the explanation just given, a property $\mathcal{P}$ is enforceable if the player $\exists$ has a $\mathcal{P}$ -winning strategy for any infinite $X \subseteq \mathbb{N}$, in a game where the P_{i+1} is chosen by $\exists$ if $i \in X$, and chosen by $\forall$ if $i \in \mathbb{N} \setminus X$. However, in our earlier description of the game between $\exists$ and $\forall$ in 4.2.2, we

described a property $\mathcal{P}$ as being enforceable if $\exists$ has a winning strategy in a game where the players *take turns* (and $\forall$ goes first) - that is, in the special case where X is the set of odd numbers.

In fact, these are equivalent. This follows as a consequence of the following theorem, which says that if a player has a winning strategy in just one infinite and coinfinite set $X \subset \mathbb{N}$, then that player has a winning strategy in all other infinite subsets of $\mathbb{N}$.

The proof that follows will prove this formally. However, this follows intuitively by the following reasoning, an argument which can be found in [Hod85, p. 28]. Supposing both players choose infinitely many components of the chain, each player will always have at most finitely many consecutive choices. If $\forall$ has the choice of $P_i, P_{i+1}, \dots, P_{i+k}$ then the outcome of the game will be no different if he chooses $P_i, \dots, P_{i+k-1}$ to all be equal to his final choice of P_{i+k} . In particular since if $\exists$ plays according to a strategy, she will make her choice of P_{i+k+1} based only on $\forall$'s final choice of P_{i+k} . The same logic works vice versa. The assumption that **both** players have infinite choices is necessary to ensure that no player is allowed to make infinitely many consecutive choices, and therefore that every sequence of consecutive choices has a 'final' component.

(iv) Finally, the reader should notice that we have defined a winning strategy in a particular way - as a family of functions which is indexed by a set X (satisfying certain conditions to be winning). Let us call this a **standard** winning strategy. There are two other ways we could have defined a winning strategy:

- Let N^* denote the set of all finite sequences $(P_1, \dots, P_n)$ of elements of N such that $P_i \subseteq P_{i+1}$ for all $1 \leq i < n$. A **strong** winning strategy for some property $\mathcal{P}$ is a function $F : N^* \rightarrow N$ such that whenever $\bar{P}$ is a chain satisfying $P_{i+1} = F(P_1, \dots, P_i)$ for all $i \in \mathbb{N}$, the union of the chain $\bigcup \bar{P}$ has the property $\mathcal{P}$. In this case, the strategy takes into account the entire history of the game played so far.
- A **weak** winning strategy is a single function $f : N \rightarrow N$ (satisfying certain conditions to be winning, which should be clear at this point), which is used repeatedly. This strategy does not take into account either the game's history, or how many times the strategy has already been applied - but can take into account the entire contents of the chain so

far.

According to [Zie80, p. 545], the notions of standard, strong, and weak strategies are all equivalent - that given a winning strategy of any kind, a winning strategy of any other kind can be obtained.

Before we move on to proving some more technical facts about winning strategies, we state a fact about enforceable properties which, though fairly obvious, is important enough to be stated.

4.2.5 Proposition. *Let N be a forcing structure in $\mathcal{L}$ and let $\mathcal{P}$ be a property of theories which is N -enforceable. Then there exists an $\mathcal{L}$ -theory T^+ which has the property $\mathcal{P}$.*

Proof. The property $\mathcal{P}$ is N -enforceable, so there is a winning strategy for any infinite subset of $\mathbb{N}$. Take the subset to be $\mathbb{N}$ itself, giving us the $\mathcal{P}$ -winning strategy $(f_i : N \rightarrow N \mid i \in \mathbb{N})$. Then take any $P \in N$ to be P_1 , and we have a chain:

$$P_1 \subseteq f_1(P_1) = P_2 \subseteq f_2(P_2) = P_3 \subseteq \dots$$

This chain is clearly an N -construction sequence satisfying the requirements for its union, the $\mathcal{L}$ -theory $T^+ := \bigcup \overline{P}$, to have the property $\mathcal{P}$. $\square$

4.2.6 Theorem. *Let $\mathcal{L}$ be a language with witnesses and let N be a forcing structure in $\mathcal{L}$. Let $\mathcal{P}$ be a property of sets of $\mathcal{L}$ -sentences. Let $O \subseteq \mathbb{N}$ be the set of odd natural numbers. Let $X \subset \mathbb{N}$ be an infinite and coinfinite subset. Then:*

There is a $\mathcal{P}$ -winning strategy for $O \iff$ There is a $\mathcal{P}$ -winning strategy for X

Proof. We will match up maximal sets of consecutive elements of X with odd numbers in O and construct strategy functions accordingly.

Let $X_1, X_2, X_3, \dots \subseteq X$ be non-empty, disjoint subsets of X , each consisting of a collection of consecutive natural numbers, and such that if $x \in X_{k_1}$ and $y \in X_{k_2}$ and $k_1 \neq k_2$ then x and y are not consecutive. Furthermore, reordering if necessary, we can suppose that if $x \in X_{k_1}$ and $y \in X_{k_2}$ and $k_1 < k_2$ then $x < y$. Then the X_i form a partition of X into maximally consecutive subsets.

We claim that, because X is coinfinite, each X_k must be finite. Suppose otherwise that there is some infinite X_k . By the well-ordering principle, $X_k \subseteq \mathbb{N}$ has a smallest element $n \in \mathbb{N}$ and can be written $\{n, n+1, \dots\}$. Then every $m \geq n$ must satisfy

$m \in X_k$ (otherwise m is an upper bound for X_k in $\mathbb{N}$, implying X_k is finite). But then $\mathbb{N} \setminus X_k = \{1, \dots, n-1\}$ is a finite set, and so also $\mathbb{N} \setminus X$ must be finite, a contradiction. Thus for each finite X_k there are $n_k, m_k \in \mathbb{N}$ such that X_k can be written $\{n_k, \dots, n_k + m_k\}$.

It also follows that there must be infinitely many X_k , otherwise we could derive a contradiction from being able to write X as the union of finitely many finite sets.

Then, the map $\pi : \{X_k \mid k \in \mathbb{N}\} \rightarrow O$ defined $X_k \mapsto 2k-1$ is a bijection between odd numbers and maximal sets X_k of consecutive elements of X .¹⁴

($\Rightarrow$) Suppose $(g_j : N \rightarrow N \mid j \in O)$ is a $\mathcal{P}$ -winning strategy for O , satisfying:

- $Q \subseteq g_j(Q)$ for all $j \in O$ and all $Q \in N$
- For every N -construction sequence $\overline{Q}$ with $Q_{j+1} = g_j(Q_j)$ for all $j \in O$, its union $\bigcup \overline{Q}$ satisfies $\mathcal{P}$

We construct a $\mathcal{P}$ winning strategy $(f_i : N \rightarrow N \mid i \in X)$ for X . For each $i \in X$, take the unique $X_k = \{n_k, \dots, n_k + m_k\} \subset X$ (as defined above) such that $i \in X_k$. Furthermore, let $j = \pi(X_k) \in O$. For each $i \in X$, we define:

$$f_i : N \rightarrow N = \begin{cases} \text{Id}_N : N \rightarrow N & \text{if } i < n_k + m_k \\ g_j : N \rightarrow N & \text{if } i = n_k + m_k \end{cases}$$

Now, suppose we have an N -construction sequence $\overline{P}$ with $P_{i+1} = f_i(P_i)$ for all $i \in X$. We show $\bigcup \overline{P}$ satisfies $\mathcal{P}$, and thus $(f_i : N \rightarrow N \mid i \in X)$ is $\mathcal{P}$ -winning. Recall that:

$$X = \{n_1, \dots, n_1 + m_1\} \cup \{n_2, \dots, n_2 + m_2\} \cup \{n_3, \dots, n_3 + m_3\} \cup \dots$$

We let $\overline{Q}$ be the N -construction sequence:

$$P_{n_1} \subseteq P_{n_1+m_1+1} \subseteq P_{n_2} \subseteq P_{n_2+m_2+1} \subseteq \dots \subseteq P_{n_k} \subseteq P_{n_k+m_k+1} \subseteq \dots$$

¹⁴The reader might notice that we instead used the sets of ‘consecutive odd numbers’, defined similarly to the X_k , then these are just singleton sets $O_j = \{j\}$ for $j \in O$. Then a bijection between the X_k and the odd numbers j could be just as easily be presented as a bijection between the X_k and the O_j . We have used the odd numbers specifically to make the proof simpler, and for their intuitive correspondence with a game between two players. However, this note should give the reader a good idea of how we could completely avoid using reference to any specific set such as O , and could instead show the equivalence holds between any two arbitrary infinite and coinfinite sets X and Y .

Explicitly, we define $\overline{Q}$ for by setting $Q_1 = P_{n_1}$. For $j \in \mathbb{N}$ we take $k \in \mathbb{N}$ such that $\pi(X_k) - 1 \leq j \leq \pi(X_k)$ and define:

$$Q_{j+1} = \begin{cases} P_{n_k} & \text{if } j \notin O \text{ so } j = \pi(X_k) - 1 \\ P_{n_k+m_k+1} & \text{if } j \in O \text{ so } j = \pi(X_k) \end{cases}$$

Then we claim $\overline{Q}$ satisfies:

(a) $\bigcup \overline{Q} = \bigcup \overline{P}$

(b) $Q_{j+1} = g_j(Q_j)$ for all $j \in O$

For (a), this holds as every component of $\overline{Q}$ occurs in $\overline{P}$ as well so $\bigcup \overline{Q} \subseteq \bigcup \overline{P}$. Furthermore, for any condition $P_i \in \overline{P}$ there is $k \in \mathbb{N}$ such that $i \leq n_k$. Let $j = \pi(X_k) - 1$. Then $P_i \subseteq P_{n_k} = Q_{j+1}$. Thus $\bigcup \overline{P} \subseteq \bigcup \overline{Q}$ and equality follows.

For (b), we let $j \in O$ and show $g_j(Q_j) = Q_{j+1}$. We know for $j - 1 \notin O$ we have $Q_{(j-1)+1} = P_{n_k}$ where $j - 1 = \pi(X_k) - 1$. Furthermore, for $j \in O$ we have defined $Q_{j+1} = P_{n_k+m_k+1}$. Furthermore by construction of the f_i , for $n_k \leq i < n_k + m_k$ we have $P_{i+1} = f_i(P_i) = P_i$. Thus:

$$P_{n_k} = P_{n_k+1} = \dots = P_{n_k+m_k}$$

Putting this together, we obtain:

$$Q_{j+1} = P_{n_k+m_k+1} = f_{n_k+m_k}(P_{n_k+m_k}) = g_j(P_{n_k+m_k}) = g_j(P_{n_k}) = g_j(Q_j)$$

Thus, $\overline{Q}$ has $Q_{j+1} = g_j(Q_j)$ for all $j \in O$, and since $(g_j : N \rightarrow N \mid j \in O)$ is a $\mathcal{P}$ winning strategy for O it follows that $\bigcup \overline{Q}$ satisfies $\mathcal{P}$, and since the unions are exactly equal, also $\bigcup \overline{P}$ satisfies $\mathcal{P}$. Thus, $(f_i : N \rightarrow N \mid i \in X)$ is a $\mathcal{P}$ -winning strategy for X .

($\Leftarrow$) We suppose $(f_i : N \rightarrow N \mid i \in X)$ is a $\mathcal{P}$ -winning strategy for X and construct a $\mathcal{P}$ -winning strategy for O .

For $j \in O$, we have $X_k = \{n_k, \dots, n_k + m_k\}$ such that $\pi(X_k) = j$. We also have maps $f_{n_k}, \dots, f_{n_k+m_k}$. We define:

$$g_j = f_{n_k+m_k} \circ f_{n_k+m_k-1} \circ \dots \circ f_{n_k+1} \circ f_{n_k} \text{ where } j = \pi(X_k)$$

Suppose we have $\overline{Q}$ satisfying $Q_{j+1} = g_j(Q_j)$ for all $j \in O$. We construct $\overline{P}$ such that $\bigcup \overline{P} = \bigcup \overline{Q}$ and $P_{i+1} = f_i(P)$ for all $i \in X$.

Set $P_1 = Q_1$. For each $j \in \mathbb{N}$ we define a finite sequence of conditions for $\overline{P}$, each corresponding to a single step $Q_j \subseteq Q_{j+1}$ in $\overline{Q}$. The j 'th sequence will always begin with Q_j and will be defined so as to ensure that the final component is equal to Q_{j+1} , but may contain additional intermediate conditions. We take all of these sequences together, identify the last component of the j 'th sequence with the first component of the subsequent $(j+1)$ 'th sequence, to form the single chain $\overline{P}$, which will contain $\overline{Q}$ as a subsequence and be of the form:

$$Q_1 \subseteq \dots \subseteq Q_2 \subseteq \dots \subseteq Q_3 \subseteq \dots$$

How the j 'th sequence is defined depends on whether $j \in O$ or $j \notin O$.

If $j \in O$ with $j = \pi(X_k)$, we define P_i for $n_k \leq i \leq n_k + m_k + 1$ to get the sequence $P_{n_k} \subseteq \dots \subseteq P_{n_k+m_k+1}$.

If $j \notin O$ with $j - 1 = \pi(X_k)$ we define P_i for $n_k + m_k + 1 \leq i \leq n_{k+1}$ ¹⁵ and obtain the sequence $P_{n_k+m_k+1} \subseteq \dots \subseteq P_{n_{k+1}}$.

Thus, for the two cases $j \in O$ and $j \notin O$ we define:

- If $j \in O$ with $j = \pi(X_k)$, the j 'th sequence of $\overline{P}$ is the sequence:

$$P_{n_k} \subseteq P_{n_k+1} \subseteq \dots \subseteq P_{n_k+m_k} \subseteq P_{n_k+m_k+1}$$

We set this equal to:

$$Q_j \subseteq (f_{n_k})(Q_j) \subseteq \dots \subseteq (f_{n_k+m_k} \circ \dots \circ f_{n_k})(Q_j)$$

Then, since $f_{n_k+m_k} \circ \dots \circ f_{n_k} = g_j$, we have that:

$$P_{n_k+m_k+1} = (f_{n_k+m_k} \circ \dots \circ f_{n_k})(Q_j) = g_j(Q_j) = Q_{j+1}$$

Furthermore, for each $n_k \leq i \leq n_k + m_k$ (that is, each $i \in X_k$) we have $P_{i+1} = f_i(P_i)$.

¹⁵We choose $j - 1 = \pi(X_k)$ so that, $n_k + m_k + 1, \dots, n_{k+1} - 1$ (but not n_{k+1}) are consecutive elements of $\mathbb{N} \setminus X$ which immediately succeed the elements of X_k .

- If $j \notin O$ with $j - 1 = \pi(X_k)$, the j 'th sequence of $\overline{P}$ is the sequence:

$$P_{n_k+m_k+1} \subseteq P_{n_k+m_k+2} \subseteq \dots \subseteq P_{n_{k+1}-1} \subseteq P_{n_{k+1}}$$

We set this equal to:

$$Q_j \subseteq Q_j \subseteq \dots \subseteq Q_j \subseteq Q_{j+1}$$

That is, $P_{n_k+m_k+1} = \dots = P_{n_{k+1}-1} = Q_j$ and $P_{n_{k+1}} = Q_{j+1}$.

Thus the two definitions combine to give a well defined sequence $\overline{P}$ satisfying $P_{i+1} = f_i(P_i)$ for all $i \in X$. By the same reasoning as earlier, we get that $\bigcup \overline{P}$ and $\bigcup \overline{Q}$ both satisfy $\mathcal{P}$ and we have shown that $(g_j : N \rightarrow N \mid j \in O)$ is a $\mathcal{P}$ -winning strategy for O . $\square$

4.2.7 Corollary. *Let N be a forcing structure in $\mathcal{L}$ and let $\mathcal{P}$ be a property of sets of $\mathcal{L}$ -sentences. Let $X \subset \mathbb{N}$ be an infinite and coinfinite set. Then:*

- (i) *If $Y \subset \mathbb{N}$ is a infinite and coinfinite subset then:*

$$X \text{ has a } \mathcal{P}\text{-winning strategy} \iff Y \text{ has a } \mathcal{P}\text{-winning strategy}$$

- (ii) *If $Z \subseteq \mathbb{N}$ is any infinite subset then:*

$$X \text{ has a } \mathcal{P}\text{-winning strategy} \implies Z \text{ has a } \mathcal{P}\text{-winning strategy}$$

Proof.

- (i) This follows from 4.2.6.
- (ii) This does not follow directly from 4.2.6 but can be proved by a very similar argument, adapting the proof of $(\Leftarrow)$ above to show that a winning strategy for O can be used to construct one for Z .

The key difference is that if Z is *not* coinfinite, we cannot use the same method to partition Z into infinitely many finite sets of consecutive elements. Instead, there will be some finite number (perhaps zero) of finite sets of consecutive elements $Z_1, \dots, Z_{k-1}$ and then a single infinite $Z_k = \{n_k, n_k + 1, \dots\}$.

To get around this, we instead partition Z as follows. The partition consists initially of $Z_1, \dots, Z_{k-1}$ as before, and then consequently contains the singleton

sets $\{n_k\}, \{n_k+1\}, \{n_k+2\}, \dots$. This gives a partition of Z into infinitely many finite sets of consecutive elements which can be put into bijection with O , and the proof proceeds similarly from there.

□

4.2.8 Theorem. *Let N be a forcing structure in $\mathcal{L}$ and let $\mathcal{P}$ be a property of sets of $\mathcal{L}$ -sentences. Suppose there is an infinite and coinfinite subset $X \subset \mathbb{N}$ such that there is a $\mathcal{P}$ -winning strategy for X . Then $\mathcal{P}$ is N -enforceable.*

Proof. By the Lemma 4.2.6 and its Corollary 4.2.7, if one infinite and coinfinite set $X \subset \mathbb{N}$ has a $\mathcal{P}$ -winning strategy, then so does every other infinite subset of $\mathbb{N}$, and thus $\mathcal{P}$ is N -enforceable. □

4.2.9 Remark. It is common in mathematical formalisations of games to formalise a strategy as a family of functions. Hodges takes a more informal approach to strategies, usually informally describing a procedure which tells a player how to make their next move in the game, as in ‘given P_i , choose P_{i+1} as follows: ...’. Hodges does develop some additional formalism for games themselves, which we do not need to include here - this includes ideas such as when a game has been ‘determined’ at a certain point (i.e. one of the players has a winning strategy from that point forward), and other ideas such as ‘winning positions’. To see Hodges’ discussion of this, see [Hod85, pp. 24-25]. These are not included in this dissertation as we prefer to just use games to aid intuition, rather than when actually defining enforceable properties.

Later in this dissertation, we will relax our approach and describe strategies in a similarly informal way. In fact, it is helpful to notice the cumbersome level of detail that we have needed to use in the proofs above, as this will be contrasted against the simpler, more informal descriptions we will be able to use later. This shows just how useful the games framework is for understanding what is going on in forcing. We have presented the formal approach first, not only so that we can be certain that an informal description of a strategy stands on rigorous foundations, but also because there will be occasions where the formal approach is easier to work with.

4.2.10 Remark. Forcing, games, and strategies are formalised similarly, though very briefly, in [Zie80, pp. 544-545], where Ziegler applies a games approach to use forcing in group theory. Hodges credits this work by Ziegler as the first to identify that

Robinson's weak forcing (see section 5) could be studied within games ([Hod85, p. 34]).

One further difference between Hodges' and Ziegler's approaches is exactly how they define what suffices for a game to be 'won' by $\exists$. Our definition follows Hodges approach: $\mathcal{P}$ is a property of theories, and a game is won (for $\mathcal{P}$) if the compiled theory T^+ has $\mathcal{P}$. Meanwhile, Ziegler defines a class Γ ¹⁶ of $\mathcal{L}$ -theories and defines a game as 'won for Γ ' if the compiled theory has $T^+ \in \Gamma$. Ziegler's approach sidesteps all the complications associated with giving a precise mathematical definition of a property, but Hodges' approach simplifies our terminology.

In fact, these approaches are very similar. We can understand Hodges' use of properties in terms of classes: a property (of theories) is a predicate - a piece of language - which defines a class (of theories). In this way, predicates which do not define a class can be excluded from our use of the term 'property' in forcing, such as those which result in paradox.¹⁷ The relationship between properties and classes is therefore very similar to the relationship between formulas and definable sets in first order model theory - but unlike definable sets, which can only be defined by a first-order formula, we allow classes to be defined by a non-first order property.

We now present the most important result of this section, which will be needed for almost all of the following material.

4.2.11. Conjunction Lemma. ([Hod85, Lemma 2.3.3 (e)]) *Let N be a forcing structure in $\mathcal{L}$. Suppose we have a countable family of properties $(\mathcal{P}_k \mid k \in \mathbb{N})$ of a theory. Let the property $\mathcal{P}$ be the conjunction of all the properties $\mathcal{P}_k$, so a theory T^+ has $\mathcal{P}$ if and only if T^+ has $\mathcal{P}_k$ for all $k \in \mathbb{N}$.*

Then $\mathcal{P}$ is N -enforceable if and only if $\mathcal{P}_k$ is N -enforceable for every $k \in \mathbb{N}$.

Proof. ($\Rightarrow$) Suppose $\mathcal{P}$ is N -enforceable. Then for any $k \in \mathbb{N}$, any $\mathcal{P}$ -winning strategy is also a $\mathcal{P}_k$ -winning strategy as well. This is because whenever the compiled theory T^+ has $\mathcal{P}$ then also T^+ has $\mathcal{P}_k$.

¹⁶This notation differs from Ziegler's, who uses $\mathcal{A}$ for a class of $\mathcal{L}$ -theories, which we are already using for term algebras (and later the compiled structure). Also, note that our use of the term 'class' is somewhat unnecessary, as in this case Γ will be a subset of the set of all $\mathcal{L}$ -theories. However, it is more common to speak of classes than sets when discussing properties.

¹⁷The classic example of a predicate which cannot be a property on pain of contradiction is the property form of Russell's paradox: *is a property which is false of itself* - which is true of itself iff it is false of itself. This is therefore a meaningless piece of language rather than a property which defines a class.

($\Leftarrow$) Suppose $\mathcal{P}_k$ is N -enforceable for every $k \in \mathbb{N}$ and let $X \subset \mathbb{N}$ be an infinite and coinfinite set.

We partition¹⁸ X into infinitely many disjoint, infinite sets $\{X_k \mid k \in \mathbb{N}\}$. An explicit example of such a partition is as follows: because both X and $\mathbb{N} \times \mathbb{N}$ are countably infinite sets, there is a bijection $\pi : \mathbb{N} \times \mathbb{N} \rightarrow X$. Then, for each $k \in \mathbb{N}$ we let $X_k = \pi(\{k\} \times \mathbb{N})$. As there are infinitely many $k \in \mathbb{N}$, and each $\{k\} \times \mathbb{N}$ is infinite, it follows that there are infinitely many infinite X_k .

Then, for each $k \in \mathbb{N}$, since the property $\mathcal{P}_k$ is N -enforceable, there is a $\mathcal{P}_k$ -winning strategy $(f_i : N \rightarrow N \mid i \in X_k)$.

Take as a strategy for $\mathcal{P}$ the family $(f_i : N \rightarrow N \mid i \in X)$ and suppose $\bar{P}$ is an N -construction sequence satisfying $P_{i+1} = f_i(P_i)$ for all $i \in X$. Then certainly also $P_{i+1} = f_i(P_i)$ for all $i \in X_k$ and all $k \in \mathbb{N}$. Thus $\bigcup \bar{P}$ satisfies $\mathcal{P}_k$ for all $k \in \mathbb{N}$, and so also satisfies $\mathcal{P}$.

It follows by 4.2.8 therefore, that since there is a $\mathcal{P}$ -winning strategy for some infinite and coinfinite $X \subset \mathbb{N}$, that $\mathcal{P}$ is N -enforceable. $\square$

4.2.12 Corollary. *Let N be a forcing structure in $\mathcal{L}$ and suppose $\mathcal{P}_1, \mathcal{P}_2, \dots$ is a possibly infinite but at most countable list of properties of theories, all of which are N -enforceable of the compiled theory. Then there exists an $\mathcal{L}$ -theory which has all of the properties $\mathcal{P}_k$ for $k \in \mathbb{N}$.*

Proof. This follows from 4.2.5 and the conjunction lemma. $\square$

4.2.13 Corollary. *Let N be a forcing structure in $\mathcal{L}$ and let $\mathcal{P}$ be a property of an $\mathcal{L}$ -theory. Then, if $\mathcal{P}$ is N -enforceable then it is not the case that the negation of $\mathcal{P}$ is N -enforceable.*

Proof. If this were not the case then, by 4.2.12 there would exist an $\mathcal{L}$ -theory T^+ with both $\mathcal{P}$ and its negation - a contradiction as such a theory cannot exist. To further justify this point, recall from 4.2.10 that, in terms of enforceable properties, a property of theories $\mathcal{P}$ can also be thought of as the class Γ it defines. Then, to say that the compiled theory T^+ has both $\mathcal{P}$ and its negation is to say that both $T^+ \in \Gamma$ and $T^+ \notin \Gamma$, a clear contradiction. $\square$

¹⁸In fact, we do not always need a partition of X . It is necessary that we obtain an infinite collection of disjoint infinite subsets of X , but it is not necessary for their union to equal X .

4.2.14 Remark. The Conjunction Lemma shows exactly why forcing is such a powerful technique. It tells us that we can enforce countably many different properties simultaneously, and we can show this to be true just by showing each property is separately enforceable. Conversely, if we want to show a particular property is enforceable, it may be easier to show that it is enforceable in conjunction with several other properties. We will show that several particularly convenient properties are enforceable, and will then be able to assume that they are being enforced ‘in the background’ while we focus on enforcing the properties we are interested in. One such example will be Hintikka theories. If we want to prove that a property $\mathcal{P}$ is enforceable then by this Lemma, it suffices to show that ‘ $\bigcup \overline{P}$ is a Hintikka set **and** satisfies $\mathcal{P}$ ’ is enforceable.

We remark on one area we need to take care around. If the conjunction of $\mathcal{P}_1$ and $\mathcal{P}_2$ are enforceable together then the conjunction lemma tells us that $\mathcal{P}_1$ and $\mathcal{P}_2$ are also separately enforceable. However, it may be that $\mathcal{P}_1$ and $\mathcal{P}_2$ are inseparable - to enforce one is to enforce the other. For example, one cannot enforce that T^+ is a Hintikka theory without also enforcing that it is $=$ -closed, though we can still consider both properties to be enforceable individually. Furthermore, just because we say that the properties are ‘separately’ enforceable does not mean that we can enforce one alongside the negation of the other. In fact, if both $\mathcal{P}_1$ and $\mathcal{P}_2$ are enforceable, neither of the negations of $\mathcal{P}_1$ or $\mathcal{P}_2$ can be enforceable.

We now look at a further feature of forcing structures and enforceable properties. We have explored how enforceable properties can be seen within the framework of a game between two players, who take turns to choose successive elements of a forcing structure to form a chain. We have described how a property is enforceable in the game if, no matter how $\forall$ chooses (and assuming $\forall$ is allowed to choose first), there is always a winning strategy for $\exists$.

However, it is also helpful to be able to think of properties which may not be N -enforceable, but become enforceable as soon as the chain contains a certain condition. We can think of these properties as enforceable within the game in a slightly weaker sense: where $\exists$ may not have a winning strategy for every game, but she does have a winning strategy for any game where she is allowed to choose first.

4.2.15 Definition. Let $(X, \leq)$ be a partially ordered set. A subset $S \subseteq X$ is called an **upper set** of X if it satisfies upward closure: for all $x \in X$ and all $y \in S$, if $x \geq y$ then $x \in S$.

Furthermore, if S is an upper set which contains a smallest element $s \in S$ then we

call S the **upper set of X generated by s** , and we have:

$$S = \{x \in X \mid x \geq s\}$$

4.2.16 Proposition. *Let N be a forcing structure in $\mathcal{L}$ and let $P \in N$. Then the upper set of N generated by P is a forcing structure and denoted:*

$$N/P = \{Q \in N \mid Q \supseteq P\}$$

Proof. It is simple to check that, as N is a forcing structure which satisfies every one of (C1)-(C8) and (F1)-(F2), they are also satisfied by the forcing structure N/P precisely because of its upward closure property. $\square$

4.2.17 Observation. Given this definition, we now have a formal way to describe a property $\mathcal{P}$ which is enforceable, but only once a certain condition P has been added to the chain. Equivalently, a property which is enforceable provided P is chosen (by either player) in the opening move of the game. Such a property is N/P -enforceable.

The following Lemma states what we call the **monotonicity** property of enforceable properties. This will connect with some of our results in section 5.

4.2.18 Lemma. *Let N be a forcing structure and let $P, Q \in N$. Then for any property $\mathcal{P}$ of theories, we have:*

- (i) *If $\mathcal{P}$ is N/P enforceable and $Q \supseteq P$ then $\mathcal{P}$ is N/Q -enforceable*
- (ii) *If $\mathcal{P}$ is not N/P enforceable and $Q \subseteq P$ then $\mathcal{P}$ is not N/Q -enforceable*

Proof. The statements are equivalent, so we only need to prove one. Suppose $\mathcal{P}$ is N/P enforceable and $P \subseteq Q$. First, we note that not only do we have $N/Q \subseteq N/P$ as every $R \in N$ with $R \supseteq Q$ also has $R \supseteq P$, but also we have $N/Q = (N/P)/Q$. That is, N/Q is upward closed in N/P with smallest element Q .

Then for any infinite and coinfinite $X \subset \mathbb{N}$ there is a $\mathcal{P}$ -winning strategy $(f_i : N/P \rightarrow N/P \mid i \in X)$. This satisfies for all $R \in N/P$ and all $i \in X$ that $R \subseteq f_i(R)$. Therefore, if $R \in N/Q$ then also $f_i(R) \in N/Q$. It follows that if we take $g_i = f_i|_{N/Q}$ for each $i \in X$ we get a family of functions $(g_i : N/Q \rightarrow N/Q \mid i \in X)$. Furthermore, every N/Q -construction sequence chosen according to the g_i is an N/P -construction sequence chosen according to the f_i . It follows that as the f_i

form a $\mathcal{P}$ -winning strategy in N/P , therefore the g_i are a $\mathcal{P}$ -winning strategy in N/Q . $\square$

4.2.19 Corollary. *Let N be a forcing structure and $\mathcal{P}$ a property. Then $\mathcal{P}$ is N -enforceable iff for every $P \in N$, the property $\mathcal{P}$ is N/P -enforceable.*

4.2.20 Lemma ([Hod85, Lemma 2.3.3 (d)]). *Let N be a forcing structure in $\mathcal{L}$ and let $\mathcal{P}$ be a property. Then, if for every $P \in N$ there is $Q \supseteq P$ such that $\mathcal{P}$ is N/Q -enforceable, then $\mathcal{P}$ is N -enforceable.*

Proof. We show there is a winning strategy for $\mathcal{P}$. Suppose we are given infinite and coinfinite $X \subset \mathbb{N}$ and P_i with $i \in X$. Then as $P_i \in N$, we can choose P_{i+1} so that $\mathcal{P}$ is N/P_{i+1} -enforceable, and thus has a winning strategy from this point on. We can now use this strategy to win for $\mathcal{P}$. $\square$

4.3 The compiled structure

Ideally, we would like to use forcing to prove things about not just sets of sentences, but structures as well. This section will develop a way to associate (with some caveats) each N -construction sequence with a uniquely determined structure, using the machinery of term algebras developed in section 3.

4.3.1 Remark. So far, we have generally given very specific, complete, and explicit constructions of strategy functions when discussing enforceable properties. We have seen that the exact construction of a strategy to enforce a property is often not particularly deep, interesting, or difficult, but can still be quite a technical task. Therefore, from this point on, we will describe strategies a little bit more informally. We will aim to give a briefer description which allows us to focus on more interesting topics, but will always aim to make sure that strategies are described in sufficient detail so that a reader could write down the explicit strategy functions without difficulty.

4.3.2 Proposition. *Let $\mathcal{L}$ be a language with witnesses C and let N be a forcing structure in $\mathcal{L}$. Then it is N -enforceable that the compiled theory is a Hintikka theory.*

Proof. Let $X \subset \mathbb{N}$ be infinite and coinfinite. Partition X into one infinite subset for each (H2)-(H8) which we call $X_2, \dots, X_8$. That the X_i are infinite ensures that for any X_i, X_j , whenever we have $x \in X_i$ there are always infinitely many $y \in X_j$ with $y > x$.

We will use the following enumerations:

- $\mathcal{L}$ -sentences: $\Sigma = \{\sigma_1, \sigma_2, \dots\}$
- Atomic $\mathcal{L}$ -formulas in one variable: $\Theta = \{\theta_1(x), \theta_2(x), \dots\}$
- All $\mathcal{L}$ -formulas in one variable: $\Phi = \{\varphi_1(x), \varphi_2(x), \dots\}$
- Closed $\mathcal{L}$ -terms: $\Omega = \{t_1, t_2, \dots\}$

Each of the $X_2, \dots, X_8$ will be used to make sure that the compiled theory satisfies the corresponding (H2)-(H8).

Given X_2 we put X_2 in bijection with Ω so for each $i \in X_2$ we have $t_i \in \Omega$. Then, given P_i with $i \in X_2$ we put $P_{i+1} = P \cup \{t_i = t_i\}$.

Given X_3 we partition X_3 into infinitely many infinite sets, and assign each distinct triple $(\theta(x), s, t) \in \Theta \times \Omega \times \Omega$ to an infinite subset $Y_{\theta(x), s, t} \subset X_3$. Then for P_i with $i \in Y_{\theta(x), s, t}$, we set:

$$P_{i+1} = \begin{cases} P_i \cup \{\theta(t)\} & \text{if } \theta(s) \in P_i \text{ and at least one of } s = t \in P_i \text{ or } t = s \in P_i \\ P_i & \text{otherwise} \end{cases}$$

Given X_4 we partition X_4 into infinitely many infinite sets and assign to each $\sigma \in \Sigma$ an infinite subset $Y_\sigma \subset X_4$. Then given P_i with $i \in Y_\sigma$ we put:

$$P_{i+1} = \begin{cases} P_i \cup \{\sigma\} & \text{if } \neg\neg\sigma \in P_i \\ P_i & \text{otherwise} \end{cases}$$

Given X_5 we partition X_5 into infinitely many infinite sets and assign to each pair $(\sigma, \tau) \in \Sigma \times \Sigma$, an infinite subset $Y_{\sigma, \tau} \subset X_5$. Then given P_i with $i \in Y_{\sigma, \tau}$ we put:

$$P_{i+1} = \begin{cases} P_i \cup \{\sigma, \tau\} & \text{if } \sigma \wedge \tau \in P_i \\ P_i \cup \{\neg\sigma\} & \text{if } \neg(\sigma \wedge \tau) \in P_i \text{ and } P_i \cup \{\neg\sigma\} \text{ is consistent} \\ P_i \cup \{\neg\tau\} & \text{if } \neg(\sigma \wedge \tau) \in P_i \text{ and } P_i \cup \{\neg\sigma\} \text{ is inconsistent} \\ P_i & \text{otherwise} \end{cases}$$

Given X_6 we partition X_6 into infinitely many infinite sets and assign to each pair

$(\sigma, \tau) \in \Sigma \times \Sigma$, an infinite subset $Y_{\sigma, \tau} \subset X_6$. Then given P_i with $i \in Y_{\sigma, \tau}$ we put:

$$P_{i+1} = \begin{cases} P_i \cup \{\sigma\} & \text{if } \sigma \vee \tau \in P_i \text{ and } P_i \cup \{\sigma\} \text{ is consistent} \\ P_i \cup \{\tau\} & \text{if } \sigma \vee \tau \in P_i \text{ and } P_i \cup \{\sigma\} \text{ is inconsistent} \\ P_i \cup \{\neg\sigma, \neg\tau\} & \text{if } \neg(\sigma \vee \tau) \in P_i \\ P_i & \text{otherwise} \end{cases}$$

Given X_7 we partition X_7 into infinitely many infinite sets and assign to each pair $(\varphi(x), t) \in \Phi \times \Omega$ an infinite subset $Y_{\varphi(x), t} \subset X_7$. Then given P_i with $i \in Y_{\varphi(x), t}$ we put:

$$P_{i+1} = \begin{cases} P_i \cup \{\varphi(t)\} & \text{if } \forall x \varphi(x) \in P_i \\ P_i \cup \{\neg\varphi(t)\} & \text{if } \neg\forall x \varphi(x) \in P_i \text{ and } t \text{ does not already appear in } P_i \\ P_i & \text{otherwise} \end{cases}$$

Given X_8 we partition X_8 into infinitely many infinite sets and assign to each pair $(\varphi(x), t) \in \Phi \times \Omega$ an infinite subset $Y_{\varphi(x), t} \subset X_8$. Then given P_i with $i \in Y_{\varphi(x), t}$ we put:

$$P_{i+1} = \begin{cases} P_i \cup \{\varphi(t)\} & \text{if } \exists x \varphi(x) \in P_i \text{ and } t \text{ does not already appear in } P_i \\ P_i \cup \{\neg\varphi(t)\} & \text{if } \neg\exists x \varphi(x) \in P_i \\ P_i & \text{otherwise} \end{cases}$$

The description above should provide sufficient detail to infer how we would construct an explicit strategy.

Now, suppose $\overline{P}$ has been constructed according to all of the above. We claim $\bigcup \overline{P}$ is a Hintikka theory. That $\bigcup \overline{P}$ satisfies each of (H1)-(H8) can be checked without much difficulty. We do not check them all individually, but will for an example show the case of (H4).

Since each σ is assigned its own infinite subset $Y_\sigma \subset X_4$, if $\neg\neg\sigma \in \bigcup \overline{P}$ then there must be some finite $j \in \mathbb{N}$ with $\neg\neg\sigma \in P_j$. But then there must still be infinitely many $i \in Y_\sigma$ such that $i > j$, so at some point we will have $\sigma \in P_i$. This illustrates the key step in ensuring that each unique task is repeated infinitely many times, so that if some prerequisite for the task is satisfied at any point in the chain, there will always be subsequent iterations of that exact task. Thus each task must be

fulfilled by the end of the chain.

It follows that for any forcing structure N , it is N -enforceable that the compiled theory is a Hintikka theory.

□

4.3.3 Definition. Let $\mathcal{L}$ be a language and $\mathcal{L}^+$ an extension by the addition of witnesses C . Let N be a forcing structure in $\mathcal{L}^+$. Let $\bar{P}$ be an N -construction sequence such that $\bigcup \bar{P}$ is a Hintikka theory. Then we call the term algebra $\mathcal{A}^+$ of $\bigcup \bar{P}$ the **compiled structure of $\bar{P}$** , or just the **compiled structure**, and we write $\mathcal{A}$ for the reduct of $\mathcal{A}^+$ to $\mathcal{L}$.

4.3.4 Theorem. *Let N be a forcing structure in a countable language $\mathcal{L}^+$, an extension of $\mathcal{L}$ by the addition of an infinite set of witnesses C . Then it is N -enforceable that the compiled structure $\mathcal{A}^+$ of T^+ satisfies both:*

- (i) $\mathcal{A}^+ \models T^+$
- (ii) *Each element of $\mathcal{A}^+$ is of the form c^{A^+} for infinitely many witnesses $c \in C$.*

Proof. (i) By 4.3.2, it is N -enforceable that T^+ is a Hintikka theory. Thus by 3.3.11, we have $\mathcal{A}^+ \models T^+$. It follows that it is enforceable that the compiled structure has $\mathcal{A}^+ \models T^+$.

- (ii) Let $X \subset \mathbb{N}$ be infinite and coinfinite. Partition X into infinitely many infinite sets, and assign to each closed $\mathcal{L}^+$ -term t an infinite set $X_t \subset X$. Given P_i with $i \in X_t$, we take some witness c which does not appear in P_i (such a witness must always exist by (F2))¹⁹ and set:

$$P_{i+1} = P_i \cup \{t = c\}$$

We can always do this by (F1). Now, since every element of $\mathcal{A}^+$ is of the form t^{A^+} for some closed term t , and for each such t this process is repeated infinitely many times and each time we add $t = c$ for some new witness $c \in C$, we conclude that every element of $\mathcal{A}^+$ is named by infinitely many witnesses.

□

¹⁹In order to write down an explicit strategy function for this process, we require a way to uniquely determine a particular witness which does not appear in P_i . To achieve this, we enumerate the set of witnesses C , and each time take the witness c_j for the smallest $j \in \mathbb{N}$ such that c_j does not appear in P_i .

4.3.5 Remark. From this point forward, for simplicity we will not unnecessarily differentiate between properties of structures and properties of theories. To be completely precise, in this dissertation it would suffice to restrict the scope of the term ‘property’ to refer to only properties of theories. This is because a theory uniquely determines its term algebra, but this is not a 1-1 correspondence - there are usually several distinct theories which determine the same term algebra. For example, as a consequence of our work in 3.3, if T is a Hintikka theory and U is the set of its atomic sentences, then $\text{TmAlg}(T) = \text{TmAlg}(U)$ as structures.

Thus, if we want to ensure that every property is technically a property of theories, rather than structures, how do we understand the claim that a property of structures is enforceable? If $\mathcal{P}$ is a property of structures, such as *is an existentially closed model of a theory T* , we rephrase this as a property $\mathcal{P}'$ of theories: *is a theory which determines the term algebra $\mathcal{A}^+$ which has $\mathcal{P}$* . The crucial observation is that the compiled theory T^+ will have $\mathcal{P}'$ if and only if the compiled structure $\mathcal{A}^+$ has $\mathcal{P}$.

If we attempted the opposite approach, and defined all enforceable properties in terms of properties of structures, it is less obvious how we would interpret claims about enforceable properties of theories. If $\mathcal{P}$ is a property of theories, the obvious way to define whether $\mathcal{P}$ is enforceable in terms of enforceable properties of structures is to consider whether if $\mathcal{P}'$ is the property *is the term algebra of a theory T^+ which has $\mathcal{P}$* is enforceable. However, in this case, it is not always true that if the compiled structure $\mathcal{A}^+$ has $\mathcal{P}$ then the compiled theory will have $\mathcal{P}$. This is because $\mathcal{A}^+$ could also be the term algebra of a different theory with $\mathcal{P}'$, even while the compiled theory does not itself have $\mathcal{P}'$, as several distinct theories can determine the same term algebra.

4.3.6 Corollary. *Let N be a forcing structure in $\mathcal{L}$ and let $\mathcal{P}_1, \mathcal{P}_2, \dots$ be a possibly infinite, but at most countable list of N -enforceable properties of $\mathcal{L}$ -structures. Then there exists an $\mathcal{L}$ -structure $\mathcal{A}^+$ in which each of the $\mathcal{P}_k$ are true.*

Proof. By the remark above, we can think of each of the properties $\mathcal{P}_k$ which could be true of an $\mathcal{L}$ -structure instead as a property of theories: *determines the term algebra $\mathcal{A}^+$ which has the property $\mathcal{P}_k$* . Then, by 4.2.12, there is an $\mathcal{L}$ -theory T^+ which has all of these properties. That is, the term algebra of T^+ has all of the properties $\mathcal{P}_k$ for $k \in \mathbb{N}$. Thus, we have found an $\mathcal{L}$ -structure $\mathcal{A}^+$ in which all of the $\mathcal{P}_k$ are true. $\square$

4.3.7 Remark. In the following section, we will introduce the weak forcing relation,

which holds between conditions and sentences. In fact, model-theoretic forcing was originally defined in [Rob71] using the weak forcing relation, alongside a strong forcing relation as well. We will define syntactically in 5.1.1 and 5.2.1 the meaning of the statement ‘ P weakly forces σ ’. The following section will conclude that, according to this syntactic definition, P weakly forces σ if and only if it is N/P -enforceable that $\mathcal{A}^+ \models \sigma$.

In order to show this equivalence between the semantic presentation of forcing with games and enforceable properties, and the syntactic definition of the forcing relations, we will conclude this section with some results which will be important in the following section. These will help us to explore, given a condition P and a sentence σ , exactly when it is N/P -enforceable that $\mathcal{A}^+ \models \sigma$.

In some cases, this may be a simple question of whether it is N/P -enforceable that $\sigma \in T^+$, since we know we will have $\mathcal{A}^+ \models \sigma$. In some cases, there may be no $P \in N$ with $\sigma \in P$, but there is still a way to enforce $T^+ \models \sigma$. The more interesting question is in the case where neither $T^+ \models \sigma$ nor $T^+ \models \neg\sigma$ are enforceable. In this case, it is possible to have $T^+ \not\models \sigma$ but still $\mathcal{A}^+ \models \sigma$.

For example, in section 6, we will look at **finite forcing structures**, in which the conditions contain only atomic and negated atomic sentences. Then, if $\varphi(x)$ is an atomic formula, it is usually not possible to enforce $T^+ \models \forall x \varphi(x)$. However, if it is enforceable that $\varphi(t) \in T^+$ for every closed term t , then we will have $\mathcal{A}^+ \models \varphi(t)$ for every closed term t . Since every element of $\mathcal{A}^+$ is named by a closed term, as a term algebra, this would entail that it is enforceable that $\mathcal{A}^+ \models \forall x \varphi(x)$, even if $T^+ \models \forall x \varphi(x)$ is not enforceable.

4.3.8 Lemma ([Hod85, Theorem 2.3.4 (c) and (f), pp. 30-32]). *Let N be a forcing structure in $\mathcal{L}$.*

- (i) *Let σ be an atomic $\mathcal{L}$ -sentence. Then it is N -enforceable that $\mathcal{A}^+ \models \neg\sigma$ if and only if every $P \in N$ has $\sigma \notin P$.*
- (ii) *Let σ be an atomic $\mathcal{L}$ -sentence. Then it is N -enforceable that $\mathcal{A}^+ \models \sigma$ if and only if for every $P \in N$ there is $Q \supseteq P$ with $\sigma \in Q$.*
- (iii) *Let σ be an $\mathcal{L}$ -sentence. For all $P \in N$, if for all $Q \supseteq P$ it is not N/Q -enforceable that $\mathcal{A}^+ \models \neg\sigma$ then there is $R \supseteq P$ such that $\mathcal{A}^+ \models \sigma$ is N/R -enforceable.*

Proof.

- (i) Suppose first that there is $P \in N$ with $\sigma \in P$. Then there is $P \in N$ such that it is not N/P -enforceable that $\mathcal{A}^+ \models \neg\sigma$. Thus by 4.2.18, it is also not N -enforceable that $\mathcal{A}^+ \models \neg\sigma$.

Conversely, suppose that every $P \in N$ has $\sigma \notin P$. Thus any compiled theory will have $\sigma \notin T^+$. Then, as it is N -enforceable that T^+ is a Hintikka theory with $\sigma \notin T^+$, it is N -enforceable that $\mathcal{A}^+ \not\models \sigma$, that is, $\mathcal{A}^+ \models \neg\sigma$.

- (ii) If for every $P \in N$ there is $Q \supseteq P$ with $\sigma \in Q$, then there is always a strategy to ensure $\mathcal{A}^+ \models \sigma$. Given P_i in N , choose P_{i+1} satisfying $\sigma \in P_{i+1}$. Therefore, it is N -enforceable that $\mathcal{A}^+ \models \sigma$.

Conversely, if there is $P \in N$ such that every $Q \supseteq P$ has $\sigma \notin Q$, then by (i), for such a P it is N/P -enforceable that $\mathcal{A}^+ \models \neg\sigma$. Thus by 4.2.18, it is not N -enforceable that $\mathcal{A}^+ \models \neg\sigma$.

- (iii) We prove this by induction on the complexity of sentences. For the base case, let $P \in N$ and we have:

- Suppose σ is atomic, and for every $Q \supseteq P$ it is not N/Q -enforceable that $\mathcal{A}^+ \models \neg\sigma$. Then by (i) for every $Q \supseteq P$ there is $Q \supseteq P$ with $\sigma \in Q$. Thus by (ii), it is N -enforceable that $\mathcal{A}^+ \models \sigma$ (and so also N/Q -enforceable).

For the inductive step, suppose σ, τ are $\mathcal{L}$ -sentences satisfying that, for all $P \in N$, if for all $Q \supseteq P$ it is not N/Q -enforceable that $\mathcal{A}^+ \models (-)$ then there is $R \supseteq P$ such that $\mathcal{A}^+ \models \neg(-)$ is N/R -enforceable.

- For $\neg\sigma$, if there is no $Q \supseteq P$ such that $\mathcal{A}^+ \models \neg\neg\sigma$ is N/Q -enforceable, then equivalently, no $Q \supseteq P$ satisfies that $\mathcal{A}^+ \models \sigma$ is N/Q -enforceable. Then, by the inductive hypothesis there is $R \supseteq P$ such that $\mathcal{A}^+ \models \neg\sigma$ is N/R -enforceable.
- For $\sigma \wedge \tau$, suppose there is no $Q \supseteq P$ such that $\mathcal{A}^+ \models \neg(\sigma \wedge \tau)$, equivalently $\mathcal{A}^+ \models \neg\sigma$ or $\mathcal{A}^+ \models \neg\tau$, is N/Q -enforceable. Then, by 4.2.11, for all $Q \supseteq P$, it must not be N/Q -enforceable that $\mathcal{A}^+ \models \neg\sigma$, and not N/Q -enforceable that $\mathcal{A}^+ \models \neg\tau$.

By the inductive hypothesis, it follows that there is $R \supseteq P$ such that it is N/R -enforceable that $\mathcal{A}^+ \models \sigma$. But then, it is also the case that for all $Q \supseteq P/R$, it is not $(N/R)/P = N/Q$ -enforceable that $\mathcal{A}^+ \models \tau$. Again, by the induction hypothesis, there is $R \in N/R \subseteq N$ such that it is N/R -enforceable that $\mathcal{A}^+ \models \tau$. By 4.2.18, it is also N/R -enforceable

that $\mathcal{A}^+ \models \sigma$.

Thus, we have found $R \in N$ such that it is N/R -enforceable that $\mathcal{A}^+ \models \sigma \wedge \tau$.

- For $\sigma \vee \tau$, suppose there is no $Q \supseteq P$ such that $\mathcal{A}^+ \models \neg(\sigma \vee \tau)$, equivalently $\mathcal{A}^+ \models \neg\sigma \wedge \neg\tau$, is N/Q -enforceable. This case then follows from the $\neg$ and $\wedge$ steps of the induction.

Finally, suppose $\varphi(x)$ is an $\mathcal{L}$ -formula satisfying for all $P \in N$ and all closed terms t that if for all $Q \supseteq P$ it is not N/Q -enforceable that $\mathcal{A}^+ \models \neg\varphi(t)$ then there is $R \supseteq P$ such that $\mathcal{A}^+ \models \varphi(t)$ is N/R -enforceable.

- For $\forall x \varphi(x)$, suppose there is no $Q \supseteq P$ such that it is N/Q -enforceable that $\mathcal{A}^+ \models \neg\forall x \varphi(x)$, equivalently $\mathcal{A}^+ \models \exists x \neg\varphi(x)$. We know that, as $\mathcal{A}^+$ is a term algebra in which every element is named by a closed term, that we will have $\mathcal{A}^+ \models \exists x \neg\varphi(x)$ iff $\mathcal{A}^+ \models \neg\varphi(t)$ for some closed term t .

Thus, for all $Q \supseteq P$, it is not N/Q -enforceable that there is a closed term t such that $\mathcal{A}^+ \models \neg\varphi(t)$. Then, for all $Q \supseteq P$ and all closed terms t , it is not N/Q -enforceable that $\mathcal{A}^+ \models \neg\varphi(t)$.

Let t be a closed term and let $Q \supseteq P$. Then for all $R \supseteq Q$ we also have $R \supseteq P$, and so it is not N/R -enforceable that $\mathcal{A}^+ \models \neg\varphi(t)$. Then, by the inductive hypothesis, there is $T \supseteq Q$ such that it is N/T -enforceable that $\mathcal{A}^+ \models \varphi(t)$.

Thus, for every $Q \supseteq P$ there is $T \supseteq Q$ such that it is N/T -enforceable that $\mathcal{A}^+ \models \varphi(t)$. From 4.2.20, it follows that it is N/P -enforceable that $\mathcal{A}^+ \models \varphi(t)$. Since t was arbitrary, this holds for all closed terms.

Now, as there are countably many closed $\mathcal{L}$ -terms, we can apply the conjunction lemma (4.2.11) and obtain that it is N/P -enforceable that $\mathcal{A}^+ \models \varphi(t)$ for all closed terms t . Thus, it is N/P -enforceable that $\mathcal{A}^+ \models \forall x \varphi(x)$.

- For $\exists x \varphi(x)$, we suppose there is no $Q \supseteq P$ such that it is N/Q -enforceable that $\mathcal{A}^+ \models \neg\exists x \varphi(x)$, equivalently $\mathcal{A}^+ \models \forall x \neg\varphi(x)$. Then, this case follows from the $\forall$ and $\neg$ steps of the induction.

□

4.3.9 Corollary. *Let N be a forcing structure in $\mathcal{L}$. Then for all $P \in N$ and all $\sigma \in \text{Sen}(\mathcal{L})$, the following are equivalent:*

- (i) *It is N/P -enforceable that $\mathcal{A}^+ \models \sigma$.*
- (ii) *For all $Q \supseteq P$, it is not N/Q -enforceable that $\mathcal{A}^+ \models \neg\sigma$*

Proof. (i) $\Rightarrow$ (ii) Let $Q \supseteq P$. Then by 4.2.18, it is N/Q -enforceable that $\mathcal{A}^+ \models \sigma$. By 4.2.13, $\mathcal{A}^+ \models \neg\sigma$ cannot also be N/Q -enforceable.

(ii) $\Rightarrow$ (i) Let $Q \supseteq P$. Then all $R \supseteq Q$ also have $R \supseteq P$, and so satisfy that it is not N/R -enforceable that $\mathcal{A}^+ \models \neg\sigma$. By 4.3.8 (iii), there is $T \supseteq Q$ such that it is N/T -enforceable that $\mathcal{A}^+ \models \sigma$.

Then we have shown that for all $Q \supseteq P$ there is $R \supseteq Q$ such that $\mathcal{A}^+ \models \sigma$ is N/R -enforceable. It follows by 4.2.20 that also $\mathcal{A}^+ \models \sigma$ is N/P -enforceable. $\square$

5 The Forcing Relation

In [Rob71], Robinson built on Cohen's work in set-theoretic forcing, developing a theory of forcing in the more general setting of model theory. The foundation of Robinson's model theoretic forcing is the **forcing relation**, which holds between conditions (in a particular forcing structure) and sentences. Both Robinson and Cohen introduced forcing in a different form to ours: our forcing, as in enforceable properties with games, is *weak* (as in [Hod85]). On the other hand, Robinson first introduces *strong* forcing and then defines a secondary weak forcing relation in terms of his strong forcing.

Only later did weak forcing gained prominence, particularly in [Hod85] through the games framework presented in the previous section. The games approach provides a much more intuitive and semantic way to understand weak forcing. In fact, in [Hod85] the weak forcing relation ($\Vdash^w$) which holds between conditions and sentences (as in, $P \Vdash^w \sigma$) is defined entirely semantically, in terms of enforceable properties of the compiled structure, as we discussed in remark 4.3.7.

However, Robinson's definitions of the forcing relations (both weak and strong) were originally entirely syntactic. Strong forcing was defined by induction on the complexity of a sentence σ , and weak forcing is defined in terms of strong forcing. The (strong) forcing relation is defined for sentences involving primitive symbols ($\neg, \wedge, \vee, \exists$), and from this we derive the definition of forcing for sentences involving the non-primitive symbols ($\rightarrow, \leftarrow, \leftrightarrow, \forall$).

The goal of this section is to show how the games based, more semantic approach to weak forcing is indeed equivalent to Robinson's original, syntactic definition of weak forcing. This culminates in the following statement, which we will state and prove in 5.2.14:

For a forcing structure N , an N -condition P and a sentence σ , it is N/P -enforceable that the compiled structure is a model of σ if and only if P weakly forces σ .

With this equivalence in mind, a reader should be able to see several similarities between the results proved in this section about the weak forcing relation and those proved about enforceable properties in the previous section. Note that Hodges does not prove the connection between his semantic approach to forcing and Robinson's syntactic forcing relations, but instead leaves it as an exercise to the reader in [Hod85, Exercise 7, pp. 33-34].

5.0.1. Notation. In this part of the dissertation, where both strong and weak forcing are in play, we will be careful to distinguish the two relations with $\Vdash^s$ and $\Vdash^w$ for the strong and weak forcing relations respectively. However, because the forcing that we use in this dissertation is weak, and strong forcing will not be needed after the following syntactic exploration of forcing, we will eventually be able to use $\Vdash$ to unambiguously refer to *weak* forcing, and we will say ‘ P forces σ ’ to mean ‘ P weakly forces σ ’.

5.1 Strong forcing

The following definition of strong forcing is given in [Rob71, p. 70], though this is in the specific context of finite forcing. The definition is also given in full generality in [Hod85, Exercise 7, pp. 33-34].

5.1.1 Definition (Strong forcing). Let N be a forcing structure in $\mathcal{L}$ and let $P \in N$. Let σ be an $\mathcal{L}$ -sentence. We define by induction on the complexity of σ the relation $P \Vdash^s \sigma$ (‘ P **strongly** forces σ ’).

- If σ is atomic, then:

$$P \Vdash^s \sigma \iff P \models \sigma$$

- If σ is of the form $\neg\tau$, then:

$$P \Vdash^s \neg\tau \iff \text{for all } N\text{-conditions } Q \supseteq P \text{ we have } Q \nVdash^s \tau$$

- If σ is of the form $\tau_1 \vee \tau_2$, then:

$$P \Vdash^s \tau_1 \vee \tau_2 \iff P \Vdash^s \tau_1 \text{ or } P \Vdash^s \tau_2$$

- If σ is of the form $\tau_1 \wedge \tau_2$, then:

$$P \Vdash^s \tau_1 \wedge \tau_2 \iff P \Vdash^s \tau_1 \text{ and } P \Vdash^s \tau_2$$

- If σ is of the form $\exists x \varphi(x)$, then:

$$P \Vdash^s \exists x \varphi(x) \iff \text{there is a constant } c \in \mathcal{L} \text{ with } P \Vdash^s \varphi(c)$$

The trickiest part of the forcing relation is dealing with negation. Importantly, it

is possible for both $P \Vdash^s \sigma$ and $P \nVdash^s \neg\sigma$ to be true (though we will soon show that it is **not** possible to have both $P \Vdash^s \sigma$ and $P \Vdash^s \neg\sigma$). The following observation will help us to unwind (strong) forcing statements involving negation.

5.1.2 Observation. Let N be a forcing structure in $\mathcal{L}$ and let $P \in N$ and $\sigma \in \text{Sen}(\mathcal{L})$. Then, just from the definition of strong forcing and unwinding negations, we have:

$$\begin{aligned} P \Vdash^s \neg\sigma &\iff (\forall Q \supseteq P) : Q \nVdash^s \sigma \\ P \nVdash^s \neg\sigma &\iff (\exists Q \supseteq P) : Q \Vdash^s \sigma \end{aligned}$$

Moving on, we have the following property possessed by all forcing structures and known as the **monotonicity** property of forcing. It resembles the usual monotonicity property of logical entailment, where if $\Sigma \models \sigma$ and $\Sigma' \supseteq \Sigma$ then $\Sigma' \models \sigma$. This corresponds to an intuitive idea that if a sentence is already forced by a condition, it cannot be ‘unforced’ by adding new information (within the limits of what information can permissably be added, determined by the forcing structure). We have already shown an analogue of this for enforceable properties in 4.2.18, and we will show an analogue for weak forcing in 5.2.6.

5.1.3. Lemma (Monotonicity of strong forcing). *Let N be a forcing structure in $\mathcal{L}$. Let $P, Q \in N$ and let $\sigma \in \text{Sen}(\mathcal{L})$.*

- (i) *If $P \Vdash^s \sigma$ and $Q \supseteq P$ then $Q \Vdash^s \sigma$.*
- (ii) *If $P \nVdash^s \sigma$ and $Q \subseteq P$ then $Q \nVdash^s \neg\sigma$.*

Proof. Since the two statements are equivalent, we only need to prove one. Let $Q \supseteq P$ and let $\sigma \in \text{Sen}(\mathcal{L})$. We prove (i) by induction on the complexity of σ .

For the base case:

- Suppose σ is atomic and $P \Vdash^s \sigma$. Then $P \models \sigma$, and thus by the monotonicity of entailment, all $Q \supseteq P$ have $Q \models \sigma$. Thus $Q \Vdash^s \sigma$.

For the inductive step, suppose the sentences $\tau, \tau_1, \tau_2 \in \text{Sen}(\mathcal{L})$ satisfy (i) for all $P, Q \in N$.

- Suppose σ is of the form $\neg\tau$ and $P \Vdash^s \neg\tau$. Then every $R \supseteq P$ has $R \not\models \tau$ and so has $R \Vdash^s \neg\tau$. Therefore $Q \Vdash^s \neg\tau$.
- Suppose σ is of the form $\tau_1 \vee \tau_2$ and $P \Vdash^s \tau_1 \vee \tau_2$. Then $P \Vdash^s \tau_1$ or $P \Vdash^s \tau_2$.

Without loss of generality, suppose $P \Vdash^s \tau_1$. Then by the inductive hypothesis, since $Q \supseteq P$ we have $Q \Vdash^s \tau_1$. Therefore $Q \Vdash^s \tau_1 \vee \tau_2$.

- Suppose σ is of the form $\tau_1 \wedge \tau_2$ and $P \Vdash^s \tau_1 \wedge \tau_2$. Then $P \Vdash^s \tau_1$ and $P \Vdash^s \tau_2$. Then for $Q \supseteq P$, also $Q \Vdash^s \tau_1$ and $Q \Vdash^s \tau_2$. Thus $Q \Vdash^s \tau_1 \wedge \tau_2$.

Additionally, suppose $\varphi(x) \in \text{Fml}(\mathcal{L})$ such that for all witnesses c , the sentence $\varphi(c)$ satisfies (i) for all $P, Q \in N$.

- Suppose σ is of the form $\exists x \varphi(x)$ and $P \Vdash^s \exists x \varphi(x)$. Then there is a witness c such that $P \Vdash^s \varphi(c)$. Therefore, by the inductive hypothesis, also $Q \Vdash^s \varphi(c)$, and so $Q \Vdash^s \exists x \varphi(x)$.

□

5.1.4 Proposition. *Let N be a forcing struture in $\mathcal{L}$. For all $\sigma \in \text{Sen}(\mathcal{L})$, a condition $P \in N$ can satisfy at most one of $P \Vdash^s \sigma$ and $P \Vdash^s \neg\sigma$.*

Proof. If $P \Vdash^s \neg\sigma$ then by 5.1.3, all $Q \supseteq P$ have $Q \nVdash^s \sigma$, including P itself, so $P \nVdash^s \sigma$. Similarly we have that if $P \Vdash^s \sigma$ then $P \nVdash^s \neg\sigma$. □

5.1.5 Theorem. *Let N be a forcing struture in $\mathcal{L}$. Let $P \in N$ and let $\forall x \varphi(x)$ be an $\mathcal{L}$ -sentence. Then:*

$$P \Vdash^s \forall x \varphi(x) \quad \Longleftrightarrow \quad \text{for all } Q \supseteq P \text{ and closed terms } t \text{ we have } Q \nVdash^s \neg\varphi(t)$$

Proof. We consider $\forall x \varphi(x)$ to be an abbreviation for $\neg\exists x \neg\varphi(x)$. Then we have $P \Vdash^s \neg\exists x \neg\varphi(x)$ iff every $Q \supseteq P$ has $Q \nVdash^s \exists x \neg\varphi(x)$. Equivalently, for any $Q \supseteq P$, there is no closed term t with $Q \Vdash^s \neg\varphi(t)$ (otherwise $Q \Vdash^s \exists x \neg\varphi(x)$). This is the case if and only if for all $Q \supseteq P$ and closed terms t we have $Q \nVdash^s \neg\varphi(t)$. □

5.2 Weak forcing

We now define the weak forcing relation in terms of strong forcing. Recall that the aim of this section is to show that, for a condition P in a forcing structure N , the statement ‘ P weakly forces σ ’ is equivalent to the property ‘the compiled structure is a model of σ ’ being N/P -enforceable. As a result, to a greater extent than in our previous discussion of the strong forcing relation, the following results about the weak forcing relation will greatly resemble some of the work we have done on enforceable properties, and the syntactic motivation of the games framework for

forcing should become apparent. The weak forcing relation is defined alongside the strong forcing relation in both [Rob71, p. 70] and [Hod85, Exercise 7, pp. 33-34].

5.2.1 Definition. Let N be a forcing structure in $\mathcal{L}$. For all $P \in N$ and $\sigma \in \text{Sen}(\mathcal{L})$, we define the weak forcing relation $\Vdash^w$ as follows:

$$P \Vdash^w \sigma \iff P \Vdash^s \neg\neg\sigma$$

5.2.2 Remark. The difference between strong forcing and weak forcing is can be compared to the difference between intuitionistic and classical logic. In classical logic, $\sigma \vee \neg\sigma$ is always a tautology, as is $\sigma \leftrightarrow \neg\neg\sigma$. On the other hand, intuitionistic logics do not assume the law of excluded middle, and thus in intuitionistic logic $\sigma \vee \neg\sigma$ is not always a tautology, and similarly $\neg\neg\sigma$ is not logically equivalent to σ . Similarly, while we have $P \Vdash^w \sigma$ iff $P \Vdash^w \neg\neg\sigma$ (we will show this in 5.2.10) and all classical tautologies are weakly forced by every condition (shown in 5.2.16), only the tautologies of intuitionistic logic are always strongly forced, so there may be some condition P with $P \not\Vdash^s \sigma \vee \neg\sigma$. Furthermore $P \Vdash^s \neg\neg\sigma$ does not entail $P \Vdash^s \sigma$. This idea is discussed in [Hod85, p. 34].

5.2.3 Observation. Let N be a forcing structure in $\mathcal{L}$. Let $P \in N$ and let $\sigma \in \text{Sen}(\mathcal{L})$. Then unwinding how strong forcing is defined for negation we have:

$$P \Vdash^s \neg\neg\sigma \iff (\forall Q \supseteq P) : Q \Vdash^s \neg\sigma \iff (\forall Q \supseteq P)(\exists R \supseteq Q) : R \Vdash^s \sigma$$

Thus it follows that:

$$P \Vdash^w \sigma \iff (\forall Q \supseteq P)(\exists R \supseteq Q) : R \Vdash^s \sigma$$

We will apply this equivalence many times in the proofs that follow.

In the following proposition, we can see some ways in which the two forcing relations relate to each other. This can be found in [Rob71, p. 73].

5.2.4 Proposition. Let N be a forcing structure in $\mathcal{L}$. For all $P \in N$ and $\sigma \in \text{Sen}(\mathcal{L})$ we have:

- (i) If $P \Vdash^s \sigma$ then $P \Vdash^w \sigma$
- (ii) If $P \Vdash^w \neg\sigma$ then $P \Vdash^s \neg\sigma$

The converse of (ii) holds as an incidence of (i), but the converse of (i) is false.

Proof.

- (i) Suppose $P \Vdash^s \sigma$. We show $P \Vdash^s \neg\neg\sigma$, which holds iff for all $Q \supseteq P$ we have $Q \nVdash^s \neg\sigma$. Let $Q \supseteq P$ and suppose for contradiction that $Q \Vdash^s \neg\sigma$. Then, because $P \Vdash^s \sigma$ and $Q \supseteq P$ we get $Q \Vdash^s \sigma$. But now we have both $Q \Vdash^s \sigma$ and $Q \Vdash^s \neg\sigma$. It follows that we must have $Q \Vdash^s \neg\sigma$, and thus $P \Vdash^s \neg\neg\sigma$. Therefore, $P \Vdash^w \sigma$.
- (ii) Suppose $P \Vdash^w \neg\sigma$. Then $P \Vdash^s \neg\neg\neg\sigma$. If we did not have $P \Vdash^s \neg\sigma$ then there would be $Q \supseteq P$ with $Q \Vdash^s \neg\neg\sigma$, a contradiction to the fact that also $Q \Vdash^s \neg\neg\neg\sigma$ from $Q \supseteq P$.

□

5.2.5 Corollary. *Let N be a forcing structure in $\mathcal{L}$. Let $P \in N$ and let $\sigma \in \text{Sen}(\mathcal{L})$. Then if $P \Vdash^w \sigma$ then $P \nVdash^w \neg\sigma$ (cf. 4.2.13).*

Proof. If $P \Vdash^w \sigma$ then $P \Vdash^s \neg\neg\sigma$. Then by 5.1.4 we have $P \nVdash^s \neg\neg\neg\sigma$, and therefore by the contrapositive of 5.2.4, $P \nVdash^w \neg\sigma$. □

Recalling the analogous results for strong forcing (5.1.3) and enforceable properties (4.2.18), the monotonicity property holds for weak forcing as well:

5.2.6. Lemma (Monotonicity of weak forcing). *Let N be a forcing structure in $\mathcal{L}$. Let $P, Q \in N$ and let $\sigma \in \text{Sen}(\mathcal{L})$. Then:*

- (i) *If $P \Vdash^w \sigma$ and $Q \supseteq P$ then $Q \Vdash^w \sigma$*
- (ii) *If $P \nVdash^w \sigma$ and $Q \subseteq P$ then $Q \nVdash^w \sigma$*

Proof. Since the statements are equivalent, we only need to prove one. Then, note that if $P \Vdash^w \sigma$ then $P \Vdash^s \neg\neg\sigma$. Then by monotonicity of strong forcing (5.1.3), if $Q \supseteq P$ then $Q \Vdash^s \neg\neg\sigma$. Thus $Q \Vdash^w \sigma$. □

5.2.7 Remark. The next theorem is a crucial property of the weak forcing relation, called the **denseness** property of weak forcing (not to be confused with the denseness of ordered sets). In forcing, given two forcing structures $N' \subseteq N$, we say that N' is **dense** in N if for every $P \in N$ there is $Q \in N'$ with $Q \supseteq P$ (See [Hod85, Exercise 6, p. 33])

Intuitively, if N' is the forcing structure consisting of all $P \in N$ with $P \Vdash^w \sigma$ (one can easily show this is a forcing structure by the monotonicity of weak forcing) then the following theorem is exactly the claim that $P \Vdash^w \sigma$ iff N' is dense in N .

5.2.8 Theorem. *Let N be a forcing structure in $\mathcal{L}$. Let $P \in N$ and let $\sigma \in \text{Sen}(\mathcal{L})$. Then $P \Vdash^w \sigma$ if and only if for all $Q \supseteq P$ there is $R \supseteq Q$ with $R \Vdash^s \sigma$.*

Proof. Using 5.2.1 and 5.2.3, we have:

$$\begin{aligned}
P \Vdash^w \sigma & \\
\iff (\forall Q \supseteq P) (\exists R \supseteq Q) : R \Vdash^s \sigma & \\
\iff (\forall Q \supseteq P) (\exists T \supseteq Q) (\forall S \supseteq T) (\exists R \supseteq S) : R \Vdash^s \sigma & \quad (*) \\
\iff (\forall Q \supseteq P) (\exists R \supseteq Q) : R \Vdash^s \neg\neg\sigma & \\
\iff (\forall Q \supseteq P) (\exists R \supseteq Q) : R \Vdash^w \sigma &
\end{aligned}$$

The one step which a reader may want extra justification for, indicated by (*), is the equivalence:

$$(\forall Q \supseteq P)(\exists R \supseteq Q) : R \Vdash^s \sigma \iff (\forall Q \supseteq P)(\exists T \supseteq Q)(\forall S \supseteq T)(\exists R \supseteq S) : R \Vdash^s \sigma$$

Subproof. ($\Leftarrow$) Let $Q \supseteq P$ be arbitrary. By assumption, there is $T \supseteq Q$ such that for all $S \supseteq T$ there is $R \supseteq S$ with $R \Vdash^s \sigma$. This holds for any S extending T , so in particular it holds for T itself. Thus, we get that there is $R \supseteq S = T \supseteq Q$ satisfying $R \Vdash^s \sigma$. Thus, as such an $R \supseteq Q$ always exists for arbitrary $Q \supseteq P$, the implication holds.

($\Rightarrow$) Let $Q \supseteq P$. We claim the statement on the right of the equivalence holds if we take $T = Q$. To show this, we let $S \supseteq Q$. Then, as $S \supseteq P$ as well, as we have assumed the left hand statement, we get that there is $R \supseteq S$ with $R \Vdash^s \sigma$. $\square$

5.2.9 Corollary. *Let N be a forcing structure in $\mathcal{L}$. Let $P \in N$ and $\sigma \in \text{Sen}(\mathcal{L})$. Then:*

$$(\forall Q \supseteq P)(\exists R \supseteq Q) : R \Vdash^w \sigma \iff (\forall Q \supseteq P)(\exists R \supseteq Q) : R \Vdash^s \sigma$$

Proof. This follows from combining 5.2.3 with 5.2.8. $\square$

Using these results, we can now prove some key features about the weak forcing relation. This theorem shows that the weak and strong forcing relations behave

similarly for (single) **negations** and **conjunctions**. However, weak forcing has elimination of **double negations**, unlike strong forcing. Furthermore, weak forcing has a simple characterisation of **universal quantification** which does not hold for strong forcing.

The following theorem notably does not make any statements about **disjunction** or **existential quantification** in weak forcing, even though these can be characterised simply in strong forcing. Disjunction and existential quantification is a crucial way in which the strong and forcing relations differ; this will be discussed further in remark 5.2.12.

5.2.10 Theorem. *Let N be a forcing structure in $\mathcal{L}$. Let $P \in N$ and let $\sigma, \tau \in \text{Sen}(\mathcal{L})$. Then:*

- (i) $P \Vdash^w \neg\sigma$ if and only if for all $Q \supseteq P$ we have $Q \nVdash^w \sigma$
- (ii) $P \Vdash^w \sigma$ if and only if $P \Vdash^w \neg\neg\sigma$
- (iii) $P \Vdash^w \sigma \wedge \tau$ if and only if $P \Vdash^w \sigma$ and $P \Vdash^w \tau$
- (iv) $P \Vdash^w \forall x \varphi(x)$ if and only if $P \Vdash^w \varphi(t)$ for every closed $\mathcal{L}$ -term t

Proof.

- (i) Suppose $P \Vdash^w \neg\sigma$. Then for all $Q \supseteq P$ there is $R \supseteq Q$ with $R \Vdash^w \neg\sigma$. Therefore, for all $Q \supseteq P$ there is $R \supseteq Q$ with $R \nVdash^w \sigma$. By monotonicity of weak forcing (5.2.6), since $R \supseteq P$ with $R \nVdash^w \sigma$ we have $Q \nVdash^w \sigma$ for all $Q \supseteq P$.

Conversely, suppose for all $Q \supseteq P$ we have $Q \nVdash^w \sigma$. Then also for all $Q \supseteq P$ we have $Q \nVdash^s \neg\neg\sigma$. This is the definition of $P \Vdash^s \neg\neg(\neg\sigma)$, equivalently $P \Vdash^w \neg\sigma$.

- (ii) We have:

$$\begin{aligned}
& P \Vdash^w \neg\neg\sigma \\
& \iff (\forall Q \supseteq P) : Q \nVdash^w \neg\sigma && \text{(By (i) for } P \Vdash^w \neg(\neg\sigma)\text{)} \\
& \iff (\forall Q \supseteq P)(\exists R \supseteq Q) : R \Vdash^w \sigma && \text{(By (i) for } Q \nVdash^w \neg\sigma \text{ (cf. 5.1.2))} \\
& \iff P \Vdash^w \sigma && \text{(By 5.2.8)}
\end{aligned}$$

- (iii) By 5.2.3, we know that $P \Vdash^w \sigma$ iff $(\forall Q \supseteq P)(\exists R \supseteq Q) : R \Vdash^s \sigma$

We have:

$$\begin{aligned}
& P \Vdash^w \sigma \wedge \tau \\
& \iff (\forall Q \supseteq P)(\exists R \supseteq Q) : R \Vdash^s \sigma \wedge \tau \\
& \iff (\forall Q \supseteq P)(\exists R \supseteq Q) : (R \Vdash^s \sigma \text{ and } R \Vdash^s \tau) \quad (*)
\end{aligned}$$

And we have:

$$\begin{aligned}
& P \Vdash^w \sigma \text{ and } P \Vdash^w \tau \\
& \iff (\forall Q \supseteq P)(\exists R \supseteq Q) : R \Vdash^w \sigma \text{ and } (\forall Q \supseteq P)(\exists R \supseteq Q) : R \Vdash^w \tau \\
& \iff (\forall Q \supseteq P) \left[((\exists R_1 \supseteq Q) : R_1 \Vdash^s \sigma) \text{ and } ((\exists R_2 \supseteq Q) : R_2 \Vdash^s \tau) \right] \quad (\dagger)
\end{aligned}$$

Therefore, we want to show that the following statements are equivalent:

$$\begin{aligned}
(*) & (\forall Q \supseteq P)(\exists R \supseteq Q) : (R \Vdash^s \sigma \text{ and } R \Vdash^s \tau) \\
(\dagger) & (\forall Q \supseteq P) \left[((\exists R_1 \supseteq Q) : R_1 \Vdash^s \sigma) \text{ and } ((\exists R_2 \supseteq Q) : R_2 \Vdash^s \tau) \right]
\end{aligned}$$

Suproof.

$(*) \Rightarrow (\dagger)$ This is the easier direction, as we can simply take $R_1 = R_2$ to both be some $R \supseteq Q$ with $R \Vdash^s \sigma$ and $R \Vdash^s \tau$, which exists by $(*)$.

$(\dagger) \Rightarrow (*)$ This direction is harder to see. Let $Q \supseteq P$. Then there is some $R_1 \supseteq Q$ with $R_1 \Vdash^s \sigma$. We then note that $R_1 \supseteq Q \supseteq P$ and apply $(\dagger)$, to obtain from $R_1 \supseteq P$ that there is $R_2 \supseteq R_1$ with $R_2 \Vdash^s \tau$.

Then, as $R_2 \supseteq R_1$ also $R_2 \Vdash^s \sigma$. Thus we have $R_2 \supseteq Q$ with $R_2 \Vdash^s \sigma$ and $R_2 \Vdash^s \tau$. $\diamond$

This concludes our subproof, and thus also proves $P \Vdash^w \sigma \wedge \tau$ iff $P \Vdash^w \sigma$ and $P \Vdash^w \tau$.

- (iv) Here, recall that we consider $\forall x \varphi(x)$ to be an abbreviation for $\neg \exists x \neg \varphi(x)$. We have that:

$$P \Vdash^w \neg \exists x \neg \varphi(x) \iff P \Vdash^s \neg \neg \neg \exists x \neg \varphi(x) \iff P \Vdash^s \neg \exists x \neg \varphi(x)$$

By 5.1.5, we know: $P \Vdash^s \neg \exists x \neg \varphi(x)$ iff for all $Q \supseteq P$ and all closed terms t we have $Q \nVdash^s \neg \varphi(t)$.

Now, for any term t , if we let $Q \supseteq P$ then we have $Q \not\Vdash^s \neg\varphi(t)$. This is the definition, for any term t , of $P \Vdash^s \neg\neg\varphi(t)$ - i.e. $P \Vdash^w \varphi(t)$.

This argument works backwards as well, so we get that $P \Vdash^w \forall x \varphi(x)$ iff $P \Vdash^w \varphi(t)$ for all closed $\mathcal{L}$ -terms t .

□

In the following corollary, we now summarise some of the key features which characterise the weak forcing relation.

5.2.11. Let N be a forcing structure in $\mathcal{L}$. Let $P \in N$ and let $\sigma \in \text{Sen}(\mathcal{L})$. Then:

- (i) $P \Vdash^w \sigma$ if and only if for all $Q \supseteq P$ there is $R \supseteq P$ with $R \Vdash^w \sigma$
- (ii) $P \not\Vdash^w \neg\sigma$ if and only if there is some $Q \supseteq P$ with $Q \Vdash^w \sigma$
- (iii) $P \not\Vdash^w \sigma$ if and only if there is some $Q \supseteq P$ with $Q \Vdash^w \neg\sigma$

Proof.

- (i) This is 5.2.8.
- (ii) This is 5.2.10 (i) after taking negations.
- (iii) By 5.2.10 (ii), we have $P \not\Vdash^w \sigma$ iff $P \not\Vdash^w \neg\neg\sigma$. Then, by 5.2.10 (i), we have $P \not\Vdash^w \neg(\neg\sigma)$ iff there is $Q \supseteq P$ with $Q \Vdash^w \neg\sigma$.

□

5.2.12 Remark. We now consider the different behaviour of the weak and strong forcing relations around disjunction and both universal and existential quantification. Notice that we know the following about the strong forcing relation, by definition:

$$P \Vdash^s \exists x \varphi(x) \text{ iff there is a closed term } t \text{ with } P \Vdash^s \varphi(t)$$

And we have proved the following fact about the weak forcing relation:

$$P \Vdash^w \forall x \varphi(x) \text{ iff for all closed terms } t \text{ we have } P \Vdash^w \varphi(t)$$

However, if we want to know about existential quantification in weak forcing, or universal quantification in strong forcing, we get the following, slightly less simple

results:

$$P \Vdash^s \forall x \varphi(x) \text{ iff for every } Q \supseteq P \text{ and all closed terms } t \text{ we have } Q \nVdash^s \neg\varphi(t)$$

And:

$$P \Vdash^w \exists x \varphi(x) \text{ iff for every } Q \supseteq P \text{ there is } R \supseteq Q \text{ and closed } t \text{ with } R \Vdash^w \varphi(t)$$

The idea is that P can only strongly force $\exists x \varphi(x)$ if there is a particular closed term t with $P \Vdash^s \varphi(t)$ - exactly which closed term this is must already be determined by P itself, and cannot be determined by some stronger condition. On the other hand, P can weakly force $\exists x \varphi(x)$ by ensuring that there is always some $Q \supseteq P$ for which there is a closed term t with $Q \Vdash^w \varphi(t)$. Unlike in the strong case, there need not be a particular t with $P \Vdash^w \varphi(t)$ - exactly which closed term ends up having this property could be different depending on which path is chosen.

This is similar to the case of disjunction. We have:

$$P \Vdash^s \sigma \vee \tau \iff P \Vdash^s \sigma \text{ or } P \Vdash^s \tau$$

In fact, the same statement for weak forcing is not true. Instead, we have:

$$P \Vdash^w \sigma \vee \tau \iff \text{for all } Q \supseteq P \text{ there is } R \supseteq Q \text{ with } R \Vdash^w \sigma \text{ or } R \Vdash^w \tau$$

Just like in the case of weakly forced existential statements, a disjunction may be weakly forced by P without it yet being decided which of the disjuncts will be forced. Instead, a disjunction is weakly forced so long as at every point in the path forward, there is a way to enforce at least one of σ or τ .

In other words, a disjunction is weakly forced as long as no path forward closes off the possibility of forcing both of the disjuncts - the option to enforce at least one of them must always remain open. This is similar to how an existential statement is weakly forced as long as there is no path from P which makes every possible closed term unavailable to witness the existential statement.

5.2.13 Remark. Now, we can state the main theorem of this section. This allows us to consider the forcing relations, weak and strong, as they were originally defined by Robinson in [Rob71] and relate them to Hodges' framing of forcing in terms of games in [Hod85]. Recall that given a forcing structure N and a condition $P \in N$, we write N/P for the forcing structure containing exactly the N -conditions which

contain P as a subset. This set N/P is the upper set generated by P in N .

5.2.14 Theorem. *Let N be a forcing structure in $\mathcal{L}$ and let $P \in N$. Then the following are equivalent for $\sigma \in \text{Sen}(\mathcal{L})$:*

- (i) *It is N/P -enforceable that the compiled structure $\mathcal{A}^+$ is a model of σ*
- (ii) *P weakly forces σ*

Proof. We prove by induction on the complexity of σ that for all $P \in N$ we have $P \Vdash^w \sigma$ iff $\mathcal{A}^+ \models \sigma$ is N/P -enforceable.

For the base case, suppose σ is atomic.

- Let $P \in N$. Then we have that $P \Vdash^w \sigma$ iff for all $Q \supseteq P$ there is $R \supseteq Q$ with $R \Vdash^s \sigma$, that is, $R \models \sigma$. Then, by 4.3.8 (ii), this is equivalent to $\mathcal{A}^+ \models \sigma$ being N/P -enforceable.

For the inductive step, suppose we have $\sigma, \tau \in \text{Sen}(\mathcal{L})$ satisfying for all $P \in N$ that $P \Vdash^w (-)$ iff $\mathcal{A}^+ \models (-)$ is N/P -enforceable.

- Let $P \in N$. Then we have $P \Vdash^w \neg\sigma$ iff every $Q \supseteq P$ satisfies $Q \nVdash^w \sigma$. By the inductive hypothesis, this is true iff for every $Q \supseteq P$, it is not N/Q -enforceable that $\mathcal{A}^+ \models \sigma$. Then, by 4.3.9, this is equivalent to the statement that it is N/P -enforceable that $\mathcal{A}^+ \models \sigma$.
- Let $P \in N$. Then $P \Vdash^w \sigma \wedge \tau$ iff $P \Vdash^w \sigma$ and $P \Vdash^w \tau$. By the inductive hypothesis, this is equivalent to both $\mathcal{A}^+ \models \sigma$ and $\mathcal{A}^+ \models \tau$ being N/P -enforceable. By 4.2.11, this is true iff $\mathcal{A}^+ \models \sigma \wedge \tau$ is N/P -enforceable.
- Let $P \in N$. Suppose $P \Vdash^w \sigma \vee \tau$, so for all $Q \supseteq P$ there is $R \supseteq Q$ with $R \Vdash^w \sigma$ or $R \Vdash^w \tau$. By the inductive hypothesis, for all $Q \supseteq P$, there is $R \supseteq Q$ such that either $\mathcal{A}^+ \models \sigma$ or $\mathcal{A}^+ \models \tau$ is N/R -enforceable. But since $\sigma \models \sigma \vee \tau$ and $\tau \models \sigma \vee \tau$, in either case, it follows that it is also N/R -enforceable that $\mathcal{A}^+ \models \sigma \vee \tau$. But then, since for all $Q \supseteq P$, there is such an $R \supseteq Q$, by 4.2.20, we get that it is N/P -enforceable that $\mathcal{A}^+ \models \sigma \vee \tau$.

Conversely, suppose $P \nVdash^w \sigma \vee \tau$. Then there is $Q \supseteq P$ such that for all $R \supseteq Q$ we have $R \nVdash^w \sigma$ and $R \nVdash^w \tau$, including Q itself. Thus we have found $Q \supseteq P$ with both $Q \nVdash^w \sigma$ and $Q \nVdash^w \tau$. Then, by the inductive hypothesis, neither $\mathcal{A}^+ \models \sigma$ nor $\mathcal{A}^+ \models \tau$ are N/Q -enforceable. Thus neither are N/P -enforceable.

either, by 4.2.18. But then also $\mathcal{A}^+ \models \sigma \vee \tau$ must not be N/P -enforceable, as required.

Next, suppose we have the $\mathcal{L}$ -formula $\varphi(x)$ satisfying that for all closed terms t and all $P \in N$, we have $P \Vdash^w \varphi(t)$ iff $\mathcal{A}^+ \models \varphi(t)$.

- Let $P \in N$. Then we have $P \Vdash^w \forall x \varphi(x)$ iff $P \Vdash^w \varphi(t)$ for all closed terms t , by 5.2.10 (iv). By the inductive hypothesis, equivalently we have that for all closed terms t , it is N/P -enforceable that $\mathcal{A}^+ \models \varphi(t)$. Finally, by 4.2.11, this is true iff it is N/P -enforceable that $\mathcal{A}^+ \models \varphi(t)$ for all closed terms t . Since every element of $\mathcal{A}^+$ is named by a closed term, this is equivalent to the statement that it is N/P -enforceable that $\mathcal{A}^+ \models \forall x \varphi(x)$.
- Let $P \in N$. Recall that $P \Vdash^w \exists x \varphi(x)$ iff for all $Q \supseteq P$ there is $R \supseteq Q$ and a closed term t with $R \Vdash^w \varphi(t)$. Let $Q \supseteq P$. Then there is $R \supseteq Q$ and a closed term t with $R \Vdash^w \varphi(t)$. By the inductive hypothesis, we have $R \supseteq Q$ such that $\mathcal{A}^+ \models \varphi(t)$ is N/R -enforceable. Then $\mathcal{A}^+ \models \exists x \varphi(x)$ is N/R -enforceable as well, and for every $Q \supseteq P$ such an $R \supseteq Q$ exists. It follows from 4.2.20 that $\mathcal{A}^+ \models \exists x \varphi(x)$ is N/P -enforceable as well.

Conversely, suppose $P \nVdash^w \exists x \varphi(x)$. Then there is $Q \supseteq P$ with $Q \Vdash^w \neg \exists x \varphi(x)$. That is, $Q \Vdash^w \forall x \neg \varphi(x)$. By the previous step in the induction, it follows that it is N/Q -enforceable that $\mathcal{A}^+ \models \forall x \neg \varphi(x)$. Since a contradiction is never enforceable by 4.2.13, is therefore not N/Q -enforceable that $\mathcal{A}^+ \models \neg \forall x \neg \varphi(x)$, that is $\mathcal{A}^+ \models \exists x \varphi(x)$. By 4.2.18, we get that it is also not N/P -enforceable that $\mathcal{A}^+ \models \exists x \varphi(x)$, as required.

□

5.2.15 Corollary. *Let N be a forcing structure in $\mathcal{L}$. Let $P \in N$ and let $\sigma \in \text{Sen}(\mathcal{L})$. Then we have:*

$$P \models \sigma \implies P \Vdash^w \sigma$$

Proof. We show the contrapositive. We know that if $P \nVdash^w \sigma$ then there is some $Q \supseteq P$ with $Q \Vdash^w \neg \sigma$. But then any compiled structure in N/Q will have $\mathcal{A}^+ \models P \cup \{\neg \sigma\}$. We obtain $P \nVdash \sigma$ as required. □

5.2.16 Corollary. *Let N be a forcing structure in $\mathcal{L}$. Let $P \in N$ and let $\sigma, \tau \in \text{Sen}(\mathcal{L})$. Then:*

- (i) *Every tautology is weakly forced by P*

(ii) If $P \models \sigma$ then $P \Vdash^w \sigma$

(iii) If $P \Vdash^w \sigma$ and $\sigma \models \tau$ then $P \Vdash^w \tau$

Proof. These are all examples of propositions which are difficult to prove directly from the definition of the forcing relation,²⁰ but follow immediately from 5.2.14, which allows us to interpret the forcing relation in a more semantic setting.

- (i) Since every tautology is true in any compiled structure $\mathcal{A}^+$, it is always N/P -enforceable for any $P \in N$. Thus every tautology is weakly forced by P .
- (ii) Since every compiled structure of an N/P -construction sequence is a model of P , and thus also σ , it is always N/P -enforceable that the compiled structure is a model of σ . Thus $P \Vdash^w \sigma$.
- (iii) Since every model of σ is a model of τ , if $P \Vdash^w \sigma$, then as it is therefore N/P -enforceable that $\mathcal{A}^+ \models \sigma$, it must also be N/P -enforceable that $\mathcal{A}^+ \models \tau$, and thus $P \Vdash^w \tau$.

□

5.2.17 Remark. Now that we have seen how the forcing relation can be defined syntactically, we consider how this relates to the more semantic approach taken in [Hod85]. We have defined the weak forcing relation $P \Vdash^w \sigma$ to occur whenever $P \Vdash^s \neg\neg\sigma$, which is itself defined by induction on the complexity of σ . This is the approach in [Rob71].

However, the weak forcing relation can be defined entirely semantically using the equivalence of theorem 5.2.14, which states that, in a given forcing structure N , we have $P \Vdash^w \sigma$ if and only if it is N/P -enforceable that σ is true in the compiled structure. This is the definition of $P \Vdash^w \sigma$ given in [Hod85, p. 29].²¹ For the remainder of this remark, recall that when we speak of ‘forcing’ in the semantic sense, we always mean weak forcing.

²⁰In particular, deriving proofs of (i) and (iii) of this corollary from the syntactic definition of the forcing relation is set as an exercise in [Hod85, pp. 33-34]. Hodges suggests for this exercise to prove these facts using a proof-calculus for first-order logic and to go by induction on the lengths of proofs. The semantic approach helps us avoid these technicalities!

²¹This is a slight oversimplification. In fact, Hodges’ definition of the forcing relation (which holds between a condition P and a property $\mathcal{P}$ - this is explained later in this remark) is defined in terms of the specific game terminology used in [Hod85], which we have chosen to omit in this dissertation. The statement ‘ P forces $\mathcal{P}$ iff $\mathcal{P}$ is N/P -enforceable’ is given as a lemma ‘immediate from the definition’ ([Hod85, Lemma 2.3.3 (a), pp. 29-30]), and indeed is just a rewording of the games-based definition.

The advantage of the more syntactic relation is that we can clearly see the rigorous formal foundations of forcing relation, and with statements such as theorem 5.2.8, the relationship between the weak forcing relation and its characterisation in terms of a game can be more clearly understood: P weakly forces σ if, once P has been added to the chain (so we are playing in N/P), for every possible move by $\forall$ after P (for every $Q \supseteq P$) there is still a move available to $\exists$ which continues her winning strategy to enforce the truth of σ in the compiled structure (there is $R \supseteq Q$ with $R \Vdash^w \sigma$).

We have already commented that the semantic games approach has the great advantage of being much more intuitive, while the details of the syntactic forcing relation can be hard to work with. However, a further advantage of the semantic approach taken by Hodges is that the forcing relation is able to be defined in slightly greater generality. Instead of restricting ‘what can be forced’ by a condition P to just sentences, Hodges expands this to include properties, so a condition P forces a property $\mathcal{P}$ if and only if it is N/P -enforceable that the compiled structure has the property $\mathcal{P}$. Then, we say P forces σ to mean that P forces the property ‘the compiled structure is a model of σ ’.

The advantage of this is that tools such as the conjunction lemma (4.2.11) become available to us when considering the forcing relation. For example, if Σ is a countable set of sentences such that $P \Vdash^w \sigma$ for all $\sigma \in \Sigma$, then the conjunction lemma tells us that it is N/P -enforceable that the compiled structure is a model of Σ - and this applies even if there are infinitely many sentences in Σ . This is a simple argument which would be rather more complicated using only the syntactic definition, and could require going beyond first-order logic to consider how the forcing relation is defined in infinitary logics.

6 Finite Forcing

In this section, we will look at one specific kind of forcing called **finite** forcing, also known as Robinson forcing (or Robinson finite forcing, to distinguish it from Robinson infinite forcing). In finite forcing, we always start from a given theory T which uniquely determines its finite forcing structure. Finite forcing is particularly interesting when working with $\forall\exists$ -theories, and we will see that finite forcing produces existentially closed models of $\forall\exists$ -theories.

Finite forcing is the form in which it was first introduced to model theory by Robinson in [Rob71]. Now, there are many different kinds of model-theoretic forcing. Finite forcing is explored within the games framework in [Hod85, 3.4, pp. 72-82].

6.1 Finite forcing structures

A forcing structure used in finite forcing will always have the following two features:

- Every condition is finite, i.e. contains finitely many sentences
- The conditions contain only atomic and negated atomic sentences (also called **basic sentences**).

6.1.1 Definition. Let T be an $\mathcal{L}$ theory and let $\mathcal{L}^+$ be an extension by infinitely many witnesses C (as usual, all languages are countable). The **finite forcing structure for T in $\mathcal{L}^+$** is defined as follows:

$$\{P \mid P \text{ is a finite set of basic } \mathcal{L}^+\text{-sentences such that } P \cup T \text{ is consistent}\}$$

We will usually use N to denote this forcing structure, but if there is any ambiguity of which theory is in use, we may use $N(T)$ instead.

6.1.2 Proposition ([Hod85, Lemma 3.4.1 (a)]). *The finite forcing structure for T in $\mathcal{L}^+$ is indeed a forcing structure satisfying (C1)-(C8) and (F1)-(F2).*

Proof. Each P is consistent so (C1) is satisfied. If P is consistent with T then so is $P \cup \{t = t\}$ and (C2) is satisfied. Since $\{\varphi(s), s = t\} \models \varphi(t)$, if P is a condition containing $\varphi(s)$ and $s = t$ is consistent then so is $P \cup \{\varphi(t)\}$ and (C3) is satisfied. Observe that any $P \in N$ will not contain any double negations, conjunctions, disjunctions, or quantified sentences, and so (C4)-(C8) are satisfied vacuously.

As for the requirements specifically for forcing structures, if t is a closed term and

c is a constant appearing nowhere in P then $P \cup \{t = c\}$ is again a finite set of basic sentences consistent with T , and we have $P \cup \{t = c\} \in N$. Thus (F1) is satisfied. Finally since each P is finite, it contains finitely many witnesses, so (F2) is satisfied. $\square$

6.1.3 Remark. We now make some observations about the compiled structure of finite forcing.

- (i) First, we observe that a compiled theory T^+ will be an infinite set of basic $\mathcal{L}^+$ -sentences such that $T \cup T^+$ is consistent - as by compactness, every finite $\Sigma \subseteq T^+$ must be contained in some condition P in the construction chain of T^+ . Therefore every Σ is consistent with T , and thus so is T^+ .

In finite forcing, the compiled theory T^+ will contain only basic $\mathcal{L}^+$ -sentences. Therefore, in order to ensure that T^+ is a Hintikka theory, it suffices to ensure T^+ satisfies (H1)-(H3), as every set of basic sentences satisfies (H4)-(H8) vacuously. That is, the compiled theory T^+ constructed in finite forcing is a Hintikka theory if and only if T^+ is consistent and $=$ -closed.

We have already shown in 4.3.4 that it is enforceable in any forcing structure that T^+ is a Hintikka theory and that the compiled structure, the term algebra $\mathcal{A}^+$ of T^+ , is a model of T^+ . In addition, it is enforceable in any forcing structure that every element of $\mathcal{A}^+$ is named by infinitely many witnesses. Since finite forcing for a theory T is a particular instance of forcing in general, these features of the compiled structure continue to be enforceable in finite forcing.

- (ii) In finite forcing with a theory T , we usually want the compiled structure to be a model of T . We have seen that it is enforceable that the compiled structure is a model of T^+ . However, if T contains sentences which are not atomic or negated atomic, we will not be able to ensure $T \subseteq T^+$, or even that $T^+ \models T$. However, we will soon see that when T is $\forall\exists$, it is enforceable that $\mathcal{A}^+ \models T \cup T^+$.

In fact, to enforce $\mathcal{A}^+ \models T \cup T^+$, we do not need to enforce the stronger claim that $T^+ \models T$. The latter statement holds if every model of T^+ is also a model of T . However, the compiled structure is not an arbitrary model of T^+ , but its term algebra in particular. Its key feature is that every element is named by at least one witness, and so the atomic formulas satisfied by each element of $\mathcal{A}^+$ can be determined entirely by the basic sentences of $\mathcal{L}^+$.

For example, if we take $\theta(x)$ to be an atomic formula with $T = \{\forall x \theta\}$, the non-basic sentence $\forall x \theta$ can never appear in any construction chain in finite forcing. We can get around this by ensuring that for every witness c we have $\theta(c) \in T^+$. This will then entail that $\mathcal{A}^+ \models \forall x \theta$. However, there may be $\mathcal{L}^+$ -structures $\mathcal{M}^+ \models T^+$ which extend $\mathcal{A}^+$ (these exist by Löwenheim-Skolem) and which therefore contain elements not named by any constant. It is then possible for there to be $a \in \mathcal{M}^+ \setminus \mathcal{A}^+$ with $\mathcal{M}^+ \models \neg\theta(a)$, and thus $\mathcal{M}^+ \not\models T$, even though $\mathcal{A}^+ \models T$.

The rest of this section will focus on illuminating several properties which are enforceable in finite forcing. As we proceed, we will always use the [Conjunction Lemma](#) to assume that, in addition to enforcing a particular property $\mathcal{P}$ of interest, we are additionally enforcing that the compiled structure satisfies $\mathcal{A}^+ \models T^+$ and that each element of $\mathcal{A}^+$ is named by infinitely many witnesses.

For the remainder of this section, we fix a countable language $\mathcal{L}$ and extension by witnesses $\mathcal{L}^+$. Initially, we will prove some results which hold for finite forcing with a general $\mathcal{L}$ -theory T in its finite forcing structure N in $\mathcal{L}^+$.

Then, we will narrow our focus to finite forcing with $\forall\exists$ -theories. As we have already begun to explore in section 2, $\forall\exists$ -theories often have interesting properties in relation to existentially closed structures, and this connection will be explored further within the context of finite forcing.

The next few results, which we prove for arbitrary T , will only be applied when finite forcing with an $\forall\exists$ -theory. This is also the case in the literature used for this section ([Hod85] and [Hod93]). However, it is interesting to see which properties are enforceable of any theory T , and which require T to be $\forall\exists$ (or are at least required to be $\forall\exists$ in order to remain within the scope of this dissertation).

For the rest of this section, we will simply say without ambiguity that properties are ‘enforceable’ rather than N -enforceable, as the forcing structure N in use is always the finite forcing structure in $\mathcal{L}^+$ for a given $\mathcal{L}$ -theory T .

The following exploration of forcing mixes the approaches taken in both [Hod85] and [Hod93], which both take a similar path to proving the Omitting Universal Types theorem.

6.1.4 Lemma ([Hod85, Lemma 3.4.1 (b)]). *Let T be an $\mathcal{L}$ -theory. It is enforceable that the compiled structure is a model of $T_\forall$.*

Proof. By 4.3.4, we can assume that we have a well-defined compiled structure $\mathcal{A}^+$ in which every element is named by infinitely many witnesses.

Let $\forall \bar{x} \chi \in T_\forall$ for quantifier-free $\chi(\bar{x}) \in \text{Fml}(\mathcal{L})$. For every tuple of witnesses $\bar{c}$, we enforce $\mathcal{A}^+ \models \chi(\bar{c})$. Let $X \subset \mathbb{N}$ be infinite and coinfinite and suppose we are given P_i with $i \in X$.

Since $T \models \forall \bar{x} \chi$ we know $T \models \chi(\bar{c})$, and thus that $T \cup P_i \cup \chi(\bar{c})$ is consistent. However, we cannot put $\chi(\bar{c})$ into P_{i+1} directly, as though it is quantifier-free, it is not basic.

We express χ in disjunctive normal form:

$$\bigvee_{j=1}^n \bigwedge_{k=1}^m \theta_{jk}(\bar{x})$$

where each of the θ_{jk} are basic $\mathcal{L}$ -formulas. Then for some $1 \leq j \leq n$, we know that $T \cup P_i \cup \{\theta_{jk}(\bar{c}) \mid 1 \leq k \leq m\}$ is consistent and, for that choice of j , we can put $P_{i+1} = P_i \cup \{\theta_{jk}(\bar{c}) \mid 1 \leq k \leq m\}$, so that $P_{i+1} \models \chi(\bar{c})$.

Thus for every $\forall \bar{x} \chi \in T_\forall$ with χ quantifier-free, and every tuple of witnesses $\bar{c}$, it is enforceable that $T^+ \models \chi(\bar{c})$. An arbitrary tuple of elements $\bar{a}$ from $\mathcal{A}^+$ is always named by some tuple $\bar{c}$ of witnesses, thus $\mathcal{A}^+ \models \chi(\bar{a})$. We have shown how to enforce $\mathcal{A}^+ \models \forall \bar{x} \chi$, and thus that $\mathcal{A}^+ \models T_\forall$ is enforceable. $\square$

6.1.5 Lemma ([Hod93, p. 370]). *It is enforceable that for every finite set of basic $\mathcal{L}^+$ -formulas, the compiled theory satisfies that either $T \cup T^+ \cup \Phi$ is inconsistent or otherwise there are infinitely many disjoint tuples $\bar{c}$ of distinct witnesses such that $\Phi(\bar{c}) \subseteq T^+$.*

Proof. By the conjunction lemma (4.2.11), it suffices to show that, for some finite set $\Phi(\bar{x})$ of basic $\mathcal{L}^+$ -formulas, it is enforceable that either $T \cup T^+ \cup \Phi$ is inconsistent or there are infinitely many disjoint tuples $\bar{c}$ of distinct witnesses such that $\Phi(\bar{c}) \subseteq T^+$. Once we have shown this, since there are only countably many finite sets of basic $\mathcal{L}^+$ -sentences in the countable language $\mathcal{L}^+$, we can use the conjunction lemma to conclude that this is enforceable for all such Φ simultaneously.

Given an infinite and coinfinite set $X \subset \mathbb{N}$ and P_i with $i \in X$, we set:

$$P_{i+1} = \begin{cases} P_i \cup \{\Phi(\bar{c})\} & \text{if } T \cup P_i \cup \Phi \text{ is consistent, where } \bar{c} \text{ is an } |\bar{x}|\text{-tuple of} \\ & \text{distinct witnesses which do not occur in } T \cup P_i \\ P_i & \text{otherwise} \end{cases}$$

Note that in the first case, where $T \cup P_i \cup \Phi$ is consistent, there will always be a suitable tuple of witnesses $\bar{c}$ available. This is because P_i is a finite set which contains at most finitely many witnesses, while there are infinitely many witnesses available in $\mathcal{L}^+$.

To show this strategy works, note that for each tuple of distinct witnesses $\bar{c}$, the compiled theory will either satisfy $\Phi(\bar{c}) \subseteq T^+$ or not.

If $\Phi(\bar{c}) \subseteq T^+$, there must be some condition in the chain P_i with $\Phi(\bar{c}) \subseteq P_i$ (i.e. we must have added $\Phi(\bar{c})$ after a finite number of steps). Therefore, there will be infinitely many more $j \in X$ with $j > i$ where we are able to choose P_{j+1} . Since a new tuple of witnesses is chosen each time, it follows that there will be infinitely many distinct tuples of witnesses $\bar{c}$ such that $\Phi(\bar{c}) \subseteq T^+$.

If $\Phi(\bar{c}) \not\subseteq T^+$ it must be because $T \cup P_i \cup \Phi$ is inconsistent for every $i \in X$ (otherwise we would have added $\Phi(\bar{c})$ to P_{i+1}). As $(T \cup P_i \cup \Phi) \subseteq (T \cup T^+ \cup \Phi)$, it follows that $T \cup T^+ \cup \Phi$ is inconsistent. $\square$

6.1.6 Theorem ([Hod85, Corollary 3.4.3, pp. 73-74] and [Hod93, Theorem 8.2.5, pp. 370-371]). *Let T be an $\forall\exists$ -theory in $\mathcal{L}$. It is enforceable that the compiled structure $\mathcal{A}^+$ is an existentially closed model of T .*

Proof. By 6.1.5, we can assume that for every finite set $\Phi(\bar{x})$ of basic $\mathcal{L}^+$ -formulas, the compiled theory T^+ satisfies that either $T \cup T^+ \cup \Phi$ is inconsistent or there are infinitely many disjoint tuples $\bar{c}$ of distinct witnesses such that $\Phi(\bar{c}) \subseteq T^+$. Additionally, by 4.3.4 and 6.1.4, we can assume that the compiled structure $\mathcal{A}^+ \models T^+ \cup T_\forall$.

Let $\varphi(\bar{x})$ be a primitive $\mathcal{L}$ -formula and let $\bar{a} \in \mathcal{A}^{\bar{x}}$, and suppose that $\mathcal{A} \models \neg\varphi(\bar{a})$. We show that there is a primitive $\mathcal{L}$ -formula $\psi(\bar{x})$ with $\mathcal{A} \models \psi(\bar{a})$ and $T \models \forall\bar{x}(\psi \rightarrow \neg\varphi)$. Then, by 2.4.8, as we have $\mathcal{A} \models T_\forall$, this suffices to show that $\mathcal{A}$ is an e.c. model of T .

Suppose $\bar{a}$ is named in $\mathcal{A}^+$ by the tuple of witnesses $\bar{c}$. Write the primitive formula $\varphi(\bar{x})$ as $\exists \bar{y} \bigwedge \Theta(\bar{x}, \bar{y})$ for some finite set $\Theta(\bar{x}, \bar{y})$ of basic $\mathcal{L}$ -formulas.

Then from $\mathcal{A} \models \neg \varphi(\bar{a})$ we have:

$$(\dagger) \quad \mathcal{A} \models \neg \exists \bar{y} \bigwedge \Theta(\bar{a}, \bar{y}) \quad \text{and} \quad \mathcal{A}^+ \models \neg \exists \bar{y} \bigwedge \Theta(\bar{c}, \bar{y})$$

Then for every tuple of witnesses $\bar{d}$, we have a finite set $\Theta(\bar{x}, \bar{d})$ of basic $\mathcal{L}^+$ -formulas, which by $(\dagger)$ must have $\Theta(\bar{c}, \bar{d}) \not\subseteq T^+$. By the properties we have already enforced concerning finite sets of basic formulas, such as Θ , it follows that $T \cup T^+ \cup \Theta(\bar{x}, \bar{y})$ is inconsistent, and therefore:

$$T \cup T^+ \models \neg \exists \bar{y} \bigwedge \Theta(\bar{c}, \bar{y})$$

By the compactness theorem, there is a finite subset $\Sigma \subseteq T^+$ of basic $\mathcal{L}^+$ -formulas such that $T \cup \Sigma \models \neg \exists \bar{y} \bigwedge \Theta(\bar{c}, \bar{y})$. Then we can write $\bigwedge \Sigma$ as the $\mathcal{L}^+$ -sentence $\sigma(\bar{c}, \bar{d})$, where $\bar{d}$ is a tuple of all the witnesses which appear in Σ but not $\bar{c}$, and $\sigma(\bar{x}, \bar{y})$ is a conjunction of basic $\mathcal{L}$ -sentences. Then we have:

$$\begin{aligned} & T \cup \{\sigma(\bar{c}, \bar{d})\} \models \neg \exists \bar{y} \bigwedge \Theta(\bar{c}, \bar{y}) \\ \implies & T \models \sigma(\bar{c}, \bar{d}) \rightarrow \neg \exists \bar{y} \bigwedge \Theta(\bar{c}, \bar{y}) \\ \implies & T \models \exists \bar{z} \sigma(\bar{c}, \bar{z}) \rightarrow \neg \exists \bar{y} \bigwedge \Theta(\bar{c}, \bar{y}) \quad (\text{as } \bar{d} \text{ does not appear in } \Theta(\bar{c}, \bar{y})) \\ \implies & T \models \forall \bar{x} \left(\exists \bar{z} \sigma(\bar{x}, \bar{z}) \rightarrow \neg \exists \bar{y} \bigwedge \Theta(\bar{x}, \bar{y}) \right) \quad (\text{by 2.1.9}) \end{aligned}$$

Finally, since $\Sigma \subseteq T^+$ we have $T^+ \models \sigma(\bar{c}, \bar{d})$, and so also $\mathcal{A} \models \exists \bar{z} \sigma(\bar{a}, \bar{z})$. Thus we have found $\exists \bar{z} \sigma(\bar{x}, \bar{z})$, the required primitive $\mathcal{L}$ -formula for our proof. We conclude that it is enforceable that the reduct $\mathcal{A}$ of the compiled structure is an e.c. model of T . $\square$

6.1.7 Corollary. *If T is an $\forall\exists$ -theory in $\mathcal{L}$ then it is enforceable that both the compiled structure $\mathcal{A}^+$ and its reduct $\mathcal{A}$ to $\mathcal{L}$ are models of T .*

6.2 Omitting universal types

In [Mar25] we proved the Omitting Types theorem, which we recall now:

6.2.1 Theorem. *Let $\mathcal{L}$ be a countable language and T a consistent $\mathcal{L}$ -theory. For each $n \in \mathbb{N}$, let $\Phi_n(x_1, \dots, x_{k_n})$ be a set of $\mathcal{L}$ -formulas in at most the free variables*

$x_1, \dots, x_{k_n}$. Suppose for every Φ_n that whenever there is $\psi(x_1, \dots, x_{k_n}) \in \text{Fml}(\mathcal{L})$ such that $T \cup \{\psi\}$ is consistent, there is some $\varphi \in \Phi_n$ such that $T \cup \{\psi, \neg\varphi\}$ is consistent. Then T has a model which omits all Φ_n .

The proof of this theorem used techniques similar to forcing. We constructed a chain of sets of sentences, the union of which produced a model of T which simultaneously omitted countably many sets of formulas (types). Also in [Mar25] we stated, without proof, a variation on the Omitting Types theorem called the **Omitting Universal Types** theorem. We will now state this theorem and give a proof using finite forcing. In fact, we will first show that, under the right circumstances, it is enforceable that the compiled structure omits just one universal type. This will then be extended to the full theorem.

6.2.2 Lemma. *Let N be the finite forcing structure in $\mathcal{L}^+$ for the $\mathcal{L}$ -theory T . Let $P \in N$ and let $\sigma \in \exists\text{-Sen}(\mathcal{L}^+)$. Then if $T \cup P \cup \{\sigma\}$ is consistent then there is $Q \supseteq P$ such that $Q \models \sigma$.*

Proof. Write σ as follows:

$$\exists \bar{x} \bigvee_{i=1}^n \bigwedge \Theta_i(\bar{x}) \quad \text{or equivalently} \quad \bigvee_{i=1}^n \exists \bar{x} \bigwedge \Theta_i(\bar{x})$$

where each Θ_i is a finite set of basic $\mathcal{L}^+$ -formulas.

Then it cannot be the case that $T \cup P \cup \Theta_i(\bar{x})$ is inconsistent for all $1 \leq i \leq n$, as this would imply that $T \cup P \cup \{\sigma\}$ is inconsistent. Therefore, for some $1 \leq i \leq n$ we have that $T \cup P \cup \Theta_i(\bar{x})$ is consistent.

Then, there is a tuple of witnesses $\bar{c}$, none of which appear anywhere in P or Θ_i , as both contain at most finitely many witnesses. Thus, $T \cup P \cup \Theta_i(\bar{c})$ is a consistent set of $\mathcal{L}^+$ -sentences, and furthermore as $P \cup \Theta_i(\bar{c})$ is finite, we get that $Q := P \cup \Theta_i(\bar{c}) \supseteq P$ is an N -condition.

Furthermore, as we have:

$$\Theta_i(\bar{c}) \models \exists \bar{x} \bigwedge \Theta_i(\bar{x}) \text{ and } \exists \bar{x} \bigwedge \Theta_i(\bar{x}) \models \exists \bar{x} \bigvee_{i=1}^n \bigwedge \Theta_i(\bar{x})$$

It follows that $Q \models \sigma$. □

6.2.3. Omitting One Universal Type. *Let T be an $\forall\exists$ -theory in $\mathcal{L}$, and let $\Phi(\bar{x})$*

be a set of $\forall$ -formulas in $\mathcal{L}$. Suppose that whenever there is $\psi(\bar{x}) \in \exists\text{-Fml}(\mathcal{L})$ such that $T \cup \{\psi\}$ is consistent, there is some $\varphi \in \Phi$ such that $T \cup \{\psi, \neg\varphi\}$ is consistent.

Then it is enforceable that both the compiled structure $\mathcal{A}^+$ and its reduct $\mathcal{A}$ omit Φ .

Proof. As usual, we extend the language $\mathcal{L}$ to $\mathcal{L}^+$ by the addition of countably many new witnesses $c \in C$, and work within the finite forcing notion N for T in $\mathcal{L}^+$. We enforce that the compiled theory T^+ is a Hintikka theory, that the compiled structure has $\mathcal{A}^+ \models T^+$, and that every element of $\mathcal{A}^+$ is named by infinitely many witnesses.

We show that the following is enforceable:

(*) For every tuple of distinct witnesses $\bar{c} \in C^{\bar{x}}$ there is $\varphi \in \Phi$ with $T^+ \models \neg\varphi(\bar{c})$

Given infinite and coinfinite $X \subset \mathbb{N}$, let all of the countably many tuples of distinct witnesses (of the same length as $\bar{x}$) be indexed by X , so we have $\{\bar{c}_i \mid i \in X\}$. Then, given P_i with $i \in X$ we choose P_{i+1} as follows.

Write $\chi(\bar{c}_i, \bar{d})$ for the $\mathcal{L}^+$ -sentence $\bigwedge P_i$, where $\bar{d}$ is a tuple consisting of all the witnesses which appear in P_i but not in $\bar{c}_i$, and $\chi(\bar{x}, \bar{y})$ is a conjunction of basic $\mathcal{L}$ -sentences. Then we have an existential $\mathcal{L}$ -formula $\exists \bar{y} \chi(\bar{x}, \bar{y})$. Since $T \cup P_i$ is consistent, we know that $T \cup \{\exists \bar{y} \chi(\bar{x}, \bar{y})\}$ is consistent.

We apply our assumption to obtain some $\varphi \in \Phi$ such that $T \cup \{\exists \bar{y} \chi(\bar{x}, \bar{y}), \neg\varphi(\bar{x})\}$ is consistent. Since none of the witnesses in $\bar{c}_i$ appear in $T \cup \{\exists \bar{y} \chi(\bar{x}, \bar{y}), \neg\varphi(\bar{x})\}$, it follows that also $T \cup \{\exists \bar{y} \chi(\bar{c}_i, \bar{y}), \neg\varphi(\bar{c}_i)\}$ is consistent. That is, we have found $\varphi \in \Phi$ such that $T \cup P_i \cup \{\neg\varphi(\bar{c}_i)\}$ is consistent.

We want $T \cup P_{i+1} \models \neg\varphi(\bar{c}_i)$. However, as we only know that φ is a universal formula, and thus $\neg\varphi$ is existential and not necessarily basic, we cannot actually add $\neg\varphi(\bar{c}_i)$ to the chain directly. Instead, we must add some basic $\mathcal{L}^+$ -sentences to achieve the same goal.

We write the existential formula $\neg\varphi(\bar{x})$ as follows, as the disjunction of primitive formulas, and where the Θ_j are finite sets of basic $\mathcal{L}$ -formulas:

$$\bigvee_{j=1}^m \exists \bar{y} \bigwedge \Theta_j(\bar{x}, \bar{y})$$

Then $\neg\varphi(\bar{c}_i)$ is logically equivalent to:

$$\bigvee_{j=1}^m \exists \bar{y} \bigwedge \Theta_j(\bar{c}_i, \bar{y})$$

Then as $T \cup P_i \cup \{\neg\varphi(\bar{c}_i)\}$ is consistent, there must be some $1 \leq j \leq m$ such that the following is consistent:

$$T \cup P_i \cup \{\exists \bar{y} \bigwedge \Theta_j(\bar{c}_i, \bar{y})\}$$

We choose a $|\bar{y}|$ -tuple of witnesses $\bar{d}$, none of which appear in P_i or $\exists \bar{y} \bigwedge \Theta_j(\bar{c}_i, \bar{y})$. Then $T \cup P_i \cup \Theta_j(\bar{c}_i, \bar{d})$ is a consistent set of $\mathcal{L}^+$ -sentences. Thus $P_i \cup \Theta_j(\bar{c}_i, \bar{d})$ is a finite set of basic $\mathcal{L}^+$ -sentences consistent with T . Furthermore, we have:

$$\Theta_j(\bar{c}_i, \bar{d}) \models \bigvee_{j=1}^m \exists \bar{y} \bigwedge \Theta_j(\bar{c}_i, \bar{y})$$

Thus by setting $P_{i+1} = P_i \cup \Theta_j(\bar{c}_i, \bar{d})$ we achieve $T^+ \models \neg\varphi(\bar{c}_i)$ as required.

Now, we show that $\mathcal{A}^+$ omits Φ . Let $\bar{a}$ be a tuple from $\mathcal{A}^+$. Then $\bar{a}$ is named by some $\bar{c}_i$ for some $i \in X$. Then, we have enforced that there is some $\varphi \in \Phi$ with $T^+ \models \neg\varphi(\bar{c}_i)$, and thus $\mathcal{A}^+ \not\models \Phi(\bar{a})$. That is, $\mathcal{A}^+$ omits Φ . $\square$

6.2.4. Omitting $\forall$ -Types Theorem ([Hod85, Corollary 3.4.5, pp. 75-76] and [Hod93, Theorem 8.2.6, pp. 371-372]).

Let T be an $\forall\exists$ -theory in a countable language $\mathcal{L}$. Let $\Phi_n(x_1, \dots, x_{k_n}) \subseteq \forall\text{-Fml}(\mathcal{L})$ for each $n \in \mathbb{N}$ have the property that whenever there is $\psi(x_1, \dots, x_{k_n}) \in \exists\text{-Fml}(\mathcal{L})$ such that $T \cup \{\psi\}$ is consistent, there is some $\varphi \in \Phi_n$ such that $T \cup \{\psi, \varphi\}$ is consistent.

Then T has an existentially closed model which omits Φ_n for all $n \in \mathbb{N}$.

Proof. This follows from a combination of many of the key results that we have proved so far. By 6.1.6, it is enforceable that the reduct $\mathcal{A}$ of the compiled structure $\mathcal{A}^+$ is an e.c. model of T .

Furthermore, we have shown that it is enforceable that $\mathcal{A}$ omits $\Phi(\bar{x})$ for any set $\Phi(\bar{x})$ of $\forall$ -formulas in $\mathcal{L}$ such that if $\psi(\bar{x}) \in \exists\text{-Fml}(\mathcal{L})$ with $T \cup \{\psi\}$ consistent, then there is $\varphi \in \Phi$ such that $T \cup \{\psi, \neg\varphi\}$ is consistent. From the [conjunction lemma](#),

it follows that it is enforceable that $\mathcal{A}$ omits countably many such $\Phi_n(x_1, \dots, x_{k_n})$ simultaneously.

Therefore, again using the conjunction lemma, we suppose that $\mathcal{A}^+$ has been constructed with these properties enforced. Then its reduct $\mathcal{A}$ is the required existentially closed model of T which omits Φ_n for all $n \in \mathbb{N}$. $\square$

6.3 The finite forcing companion

The material on model completeness, model companions, and companion operators can be found in [CK73].

6.3.1 Definition. Let T be an $\mathcal{L}$ -theory. Then T is **model complete** if for any two models $\mathcal{M}, \mathcal{N}$ of T such that $\mathcal{M} \leq \mathcal{N}$, we have $\mathcal{M} \prec \mathcal{N}$.

We quote without proof the following property of model complete theories:

6.3.2 Proposition. *Let T be a model complete theory. Then T is an $\forall\exists$ -theory.*

Proof. We show that if T is model complete then the class of models of T is closed under unions of chains. Suppose $\mathcal{M}_1 \leq \mathcal{M}_2 \leq \dots$ is a chain of models of T . Then we do not just have $\mathcal{M}_i \leq \mathcal{M}_{i+1}$ for all $i \in \mathbb{N}$, but also $\mathcal{M}_i \prec \mathcal{M}_{i+1}$. It follows from 2.3.4 that we have $\mathcal{M}_i \prec \bigcup \mathcal{M}_i$ for all $i \in \mathbb{N}$. Thus $\bigcup \mathcal{M}_i$ is elementarily equivalent to all of the $\mathcal{M}_i$, and it follows that $\bigcup \mathcal{M}_i \models T$. Thus the class of models of T is closed under unions of chains. Thus, by 2.3.6, T is both an inductive theory and an $\forall\exists$ -theory. $\square$

For some theories, called **companionable** theories, there is a special counterpart theory called a **model companion**. The model companion of a companionable theory T is a particularly well-behaved counterpart of T . We give the definition of a model companion before looking at some examples.

6.3.3 Definition. Let T be an $\mathcal{L}$ -theory. An $\mathcal{L}$ -theory T^* is a **model companion** of T if:

- (i) $T_{\forall} = (T^*)_{\forall}$
- (ii) T^* is model-complete

The statement in part (i) above means that any model of T can be embedded in a model of T^* , and any model of T^* can be embedded in a model of T .

6.3.4 Proposition ([CK73, Proposition 3.5.13, p. 198]). *An $\mathcal{L}$ -theory T has at most one model companion up to logical equivalence.*

Proof. We show that if T^* and T^{**} are model companions of an $\mathcal{L}$ -theory T then $T^* \models T^{**}$ and $T^{**} \models T^*$. Since both are a model companion of T , we know that $T_{\forall}^* = T_{\forall} = T_{\forall}^{**}$ and that both T^* and T^{**} are model complete.

Let $\mathcal{M}_1 \models T^*$. Then, since this property ensures that every model of one of T^* and T^{**} can be embedded into a model of the other, there is a chain of $\mathcal{L}$ -structures:

$$\mathcal{M}_1 \leq \mathcal{N}_1 \leq \mathcal{M}_2 \leq \mathcal{N}_2 \leq \dots$$

such that the $\mathcal{M}_i$ are a model of T^* and the $\mathcal{N}_i$ are a model of T^{**} . Let $\mathcal{A}$ be the union of the chain, as defined in 2.3.3.

Then as T^* is model complete, the $\mathcal{M}_i$ form an elementary chain. Similarly for T^{**} we get that the $\mathcal{N}_i$ form an elementary chain. By the elementary chain lemma (2.3.4), it follows that for all $i \in \mathbb{N}$ we have $\mathcal{M}_i \prec \mathcal{A}$ and $\mathcal{N}_i \prec \mathcal{A}$.

Then, we have $\mathcal{A} \models T^{**}$. Since $\mathcal{M}_1 \prec \mathcal{A}$ also $\mathcal{M}_1 \models T^{**}$. Thus $T^* \models T^{**}$. The same argument adapts to prove $T^{**} \models T^*$. $\square$

6.3.5 Examples. Here are some examples of companionable theories with their model companion, which by 6.3.4, is unique up to logical equivalence. These examples can be found in [CK73, Examples 3.5.12, p. 198]

- The theory of algebraically closed fields is the model companion of the theory of fields, in the language of rings.
- The theory of real closed fields is the model companion of the theory of ordered fields, in the language of ordered rings.
- The theory of dense linear orders without endpoints is the model companion of the theory of linear orders, in the language of orders.

An example of a non-companionable theory is the theory of groups in the language $\mathcal{L}_{\text{Group}} = \{*, ^{-1}, e\}$, which has no model companion. This is proved in [CK73, Example 3.5.16, pp. 199-200]. The axioms of the theory of groups can be found in [Hod93, p. 37].

The following definition of a **companion operator** generalises the notion of the relationship between a theory and its model companion. However, we will see that

if a theory is companionable, a companion operator will always send it to its model companion. The advantage of a companion operator is that it also produces a desirable and well-behaved counterpart for a non-companionable theory.

6.3.6 Definition. Let $\mathcal{L}$ be a countable language and let Δ be the set of all deductively closed $\mathcal{L}$ -theories. A function $\Delta \rightarrow \Delta$ with $T \mapsto T^*$ is called a **companion operator** if it satisfies all the following:

- (i) $T_{\forall} = (T^*)_{\forall}$
- (ii) If $T_{\forall} = U_{\forall}$ then $T^* = U^*$
- (iii) $T^* \models T_{\forall\exists}$

6.3.7 Proposition ([CK73, Proposition 3.5.21, pp. 205-206]). *Suppose $*$: $\Delta \rightarrow \Delta$ is a companion operator on $\mathcal{L}$ -theories. Then, for every companionable theory T , the theory T^* is the model companion of T .*

Proof. Let T' be the model companion of T . Then by 6.3.2, $(T')_{\forall\exists}$ is logically equivalent to T' . Writing $\equiv$ for the logical equivalence of theories, and noting that $(T')_{\forall} = T_{\forall}$, we have.

$$T^* = (T')^* \models (T')_{\forall\exists} \equiv T'$$

Thus $T^* \models T'$. Since T' is model-complete, this implies that T^* is model complete as well. Then, as $(T^*)_{\forall} = T_{\forall}$, it follows that T^* is the model of companion of T . $\square$

6.3.8 Remark. As we saw in the case of the theory of groups, not all theories are companionable. However, with a companion operator, we can construct a theory with similar properties to the model companion for any theory at all, not just companionable ones. One example of this is the **finite forcing companion**, which exists for any theory T .

6.3.9 Definition. Let T be an $\mathcal{L}$ -theory and let N be the finite forcing structure for T in $\mathcal{L}^+$. We define the **finite forcing companion** of T as follows:

$$T^f = \{\sigma \in \text{Sen}(\mathcal{L}) \mid \sigma \text{ is } N\text{-enforceable}\} = \{\sigma \in \text{Sen}(\mathcal{L}) \mid \Vdash_N \sigma\}$$

We will need the following proposition about $\forall$ -sentences within finite forcing structures.

6.3.10 Proposition ([Hod85, Theorem 3.4.2, p. 73]). *Let N be the finite forcing structure in $\mathcal{L}^+$ for an $\mathcal{L}$ -theory T . Let $P \in N$ and let $\sigma \in \forall\text{-Sen}(\mathcal{L})$. Then:*

$$P \Vdash \sigma \iff T \cup P \models \sigma$$

Proof. First, suppose $T \cup P \not\models \forall \bar{x} \chi$. Then for a tuple of witnesses $\bar{c}$ which do not occur in P , it follows that $T \cup P \cup \{\neg \chi(\bar{c})\}$ is consistent. Then as $\neg \chi(\bar{c}) \in \exists\text{-Sen}(\mathcal{L}^+)$ by 6.2.2 there is some $Q \supseteq P$ with $Q \models \neg \chi(\bar{c})$. Then, it must be the case that $P \not\models \chi(\bar{c})$. Thus, also $P \not\models \forall \bar{x} \chi(\bar{x})$.

On the other hand, suppose $T \cup P \models \forall \bar{x} \chi$. Write $\psi(\bar{d})$ for the conjunction $\bigwedge P$ where $\psi(\bar{y})$ is an $\mathcal{L}$ -formula and $\bar{d}$ is a tuple of the witnesses appearing in P . Then we have $T \models \psi(\bar{d}) \rightarrow \forall \bar{x} \chi$ and thus also, by the lemma on constants, $T \models \exists \bar{y} \psi \rightarrow \forall \bar{x} \chi$. Now, by the same reasoning as in 2.4.3, we have that $\exists \bar{y} \psi \rightarrow \forall \bar{x} \chi$ is a $\forall$ -sentence of $\mathcal{L}$. Thus, also $T_{\forall} \models \exists \bar{y} \psi \rightarrow \forall \bar{x} \chi$. We apply the lemma on constants again to obtain $T_{\forall} \models \psi(\bar{d}) \rightarrow \forall \bar{x} \chi$, and so $T_{\forall} \cup P \models \forall \bar{x} \chi$.

Now, we know that it is always enforceable in finite forcing that the compiled structure has $\mathcal{A}^+ \models T_{\forall}$. Thus, since we also know it is N/P -enforceable that $\mathcal{A}^+ \models P$, also $\mathcal{A}^+ \models \forall \bar{x} \chi$ is N/P -enforceable. It follows that $P \Vdash \forall \bar{x} \chi$. $\square$

6.3.11 Lemma. *Let T be an $\mathcal{L}$ -theory. Recall that $T_{\forall}$ is the set of all $\forall$ -consequences of T (in $\mathcal{L}$), and $T_{\forall\exists}$ is the set of all $\forall\exists$ -consequences of T (in $\mathcal{L}$). Consider the following finite forcing structures:*

$$\begin{aligned} N(T) &= \{P \mid P \text{ is a finite set of basic } \mathcal{L}^+ \text{-sentences, } P \cup T \text{ is consistent}\} \\ N(T_{\forall}) &= \{P \mid P \text{ is a finite set of basic } \mathcal{L}^+ \text{-sentences, } P \cup T_{\forall} \text{ is consistent}\} \\ N(T_{\forall\exists}) &= \{P \mid P \text{ is a finite set of basic } \mathcal{L}^+ \text{-sentences, } P \cup T_{\forall\exists} \text{ is consistent}\} \end{aligned}$$

Then $N(T) = N(T_{\forall}) = N(T_{\forall\exists})$.

Proof. Note that for any $\mathcal{L}$ -theory T , every universal consequence of T is contained in both $T_{\forall}$ and $T_{\forall\exists}$ as $\forall\text{-Sen}(\mathcal{L}) \subset \forall\exists\text{-Sen}(\mathcal{L})$. Thus, $(T_{\forall\exists})_{\forall} = T_{\forall}$. It follows that we only need to show $N(T) = N(T_{\forall})$ for any $\mathcal{L}$ -theory T . We show that for any finite set P of $\mathcal{L}^+$ sentences, $P \cup T_{\forall}$ is consistent iff $P \cup T$ is consistent.

Since $T \models T_{\forall}$, if $P \cup T$ is consistent then $P \cup T_{\forall}$ is consistent. Conversely, suppose $P \cup T_{\forall}$ is consistent. Then we can write $\chi(\bar{c})$ for $\bigwedge P$ where $\chi(\bar{x})$ is an $\mathcal{L}$ -formula and $\bar{c}$ consists of all the witnesses which appear in P , and we have a model $\mathcal{M}^+ \models$

$T_{\forall} \cup \{\chi(\bar{c})\}$. Then the reduct of $\mathcal{M}^+$ is an $\mathcal{L}$ -structure $\mathcal{M}$ which has $\bar{a} \in \mathcal{M}^{\bar{x}}$ (the elements in $\mathcal{M}^+$ named by $\bar{c}$) with $\mathcal{M} \models T_{\forall} \cup \{\chi(\bar{a})\}$.

Then by 2.4.5 every model of $T_{\forall}$ can be embedded into a model of T , and so there is an $\mathcal{L}$ -structure $\mathcal{N} \geq \mathcal{M}$ with $\mathcal{N} \models T$. Then, as $\mathcal{M} \models \chi(\bar{a})$ and quantifier-free formulas are preserved by embeddings, it follows that $\mathcal{N} \models \chi(\bar{a})$, and therefore $T \cup \{\chi(\bar{x})\}$ is a consistent set of $\mathcal{L}$ -formulas. Since none of the $\bar{c}$ appear anywhere in T or $\chi(\bar{x})$, we can conclude that $T \cup \{\chi(\bar{c})\}$ is consistent (in particular, it is true in some expansion $\mathcal{N}^+$ of $\mathcal{N}$ where we make sure to interpret the constants $\bar{c}$ as the same elements $\bar{a}$ mentioned above). Therefore also $T \cup P$ is consistent. $\square$

6.3.12 Proposition. *The function $T \mapsto T^f$ which sends a theory to its finite forcing companion is a companion operator.*

Proof. We prove that the map which sends T to its finite forcing companion T^f satisfies all of 6.3.6 (i)-(iii).

(i) We want to show for every $\sigma \in \forall\text{-Sen}(\mathcal{L})$ we have $T \models \sigma$ if and only if $T^f \models \sigma$.

By 6.3.10 for every $\sigma \in \forall\text{-Sen}(\mathcal{L})$ and every $P \in N$ we have that $P \Vdash \sigma$ if and only if $T \cup P \models \sigma$. Therefore:

$$T \models \sigma \iff \Vdash \sigma \iff \sigma \in T^f$$

(ii) By 6.3.11, we have that $N(T) = N(T_{\forall}) = N(U_{\forall}) = N(U)$. Therefore, if $T_{\forall} = U_{\forall}$, we have $T^f = U^f$ because:

$$\sigma \in T^f \iff \Vdash_{N(T)} \sigma \iff \Vdash_{N(U)} \sigma \iff \sigma \in U^f$$

(iii) By 6.1.7 it is $N(T_{\forall\exists})$ -enforceable that the compiled structure is a model of $T_{\forall\exists}$. Thus, by 5.2.14, for every $\sigma \in T_{\forall\exists}$ we have that $\Vdash \sigma$. Thus, $T_{\forall\exists} \subseteq T^f$, from which we get $T^f \models T_{\forall\exists}$. $\square$

6.3.13 Definition. Let K be a class of $\mathcal{L}$ -structures. Then K is said to have the **Joint Embedding Property** (JEP) if for every pair $\mathcal{M}_1, \mathcal{M}_2 \in K$ there is $\mathcal{N} \in K$ such that $\mathcal{M}_1 \hookrightarrow \mathcal{N}$ and $\mathcal{M}_2 \hookrightarrow \mathcal{N}$.

An $\mathcal{L}$ -theory T has JEP if the class of its models in $\mathcal{L}$ has JEP.

6.3.14 Examples.

- (i) Any class of structures K which is a chain of substructures has JEP, as for any $\mathcal{M}, \mathcal{N} \in K$ we have $\mathcal{M} \leq \mathcal{N}$ or the converse.
- (ii) The theory of groups has JEP, as any two groups G and H can both be embedded into the direct product $G \times H$. By the same reasoning, the theory of rings also has JEP.
- (iii) The theory of fields T does not have JEP. This is because no two fields of different characteristics can be embedded into the same field, as embeddings of fields preserve their characteristic.

6.3.15 Observation. Suppose π is a permutation of a set of witnesses C . For $\varphi \in \text{Fml}(\mathcal{L}^+)$ write $\pi(\varphi)$ for the $\mathcal{L}^+$ -formula which we obtain by replacing each witness $c \in C$ which occurs in φ with $\pi(c)$. For a set of $\mathcal{L}^+$ -formulas Φ write $\pi(\Phi)$ for the set which results from replacing each $\varphi \in \Phi$ with $\pi(\varphi)$.

Now, let T be an $\mathcal{L}$ -theory and let N be the finite forcing structure for T in $\mathcal{L}^+$. Then, if P is an N -condition, then so is $\pi(P)$. This follows from the observation that, as no witnesses occur in T , and π is just a relabelling of the witnesses, $T \cup P$ is consistent if and only if $T \cup \pi(P)$ is consistent.

Furthermore, if σ is an $\mathcal{L}$ -sentence, so contains no occurrences of witnesses, we have $P \Vdash \sigma$ if and only if $\pi(P) \Vdash \sigma$.

6.3.16 Theorem. *Let T be an $\mathcal{L}$ -theory. Then if T has JEP, then T^f is a complete theory.*

Proof. We show that, in N the finite forcing structure for T in $\mathcal{L}^+$, for every $\sigma \in \text{Sen}(\mathcal{L})$ we have either $\Vdash \sigma$ or $\Vdash \neg\sigma$.

Suppose for contradiction that σ is an exception. Since no condition can force both σ and $\neg\sigma$, this must be because $\emptyset \nVdash \sigma$ and $\emptyset \nVdash \neg\sigma$.

Recall that a condition does not force τ if and only if there is a stronger condition which forces $\neg\tau$. We get $P, Q \in N$ with $P \Vdash \neg\sigma$ and $Q \Vdash \neg\neg\sigma$ - that is, $Q \Vdash \sigma$.

Since no witnesses occur in σ , we can assume by 6.3.15 that the witnesses occurring in P and Q are disjoint. If not, we can simply replace P with $\pi(P)$ for some permutation which maps all witnesses in P to witnesses not in Q - such a permutation must exist, as only finitely many witnesses occur in each of P and Q .

Then as P and Q are N -conditions there are $\mathcal{L}^+$ -structures $\mathcal{M}^+ \models P \cup T$ and $\mathcal{N}^+ \models Q \cup T$. As T has JEP in $\mathcal{L}$, there is an $\mathcal{L}$ -structure $\mathcal{A}$ with $\mathcal{M} \hookrightarrow \mathcal{A}$ and $\mathcal{N} \hookrightarrow \mathcal{A}$, where $\mathcal{M}$ and $\mathcal{N}$ are the reducts to $\mathcal{L}$ of $\mathcal{M}^+$ and $\mathcal{N}^+$. Without loss of generality, we can identify $\mathcal{M}$ and $\mathcal{N}$ with their images inside $\mathcal{A}$ and assume $\mathcal{M} \leq \mathcal{A}$ and $\mathcal{N} \leq \mathcal{A}$.

Then, as we were careful to specify that the witnesses in P and Q are disjoint, there is an expansion $\mathcal{A}^+$ of $\mathcal{A}$ to $\mathcal{L}^+$ with $\mathcal{M}^+ \leq \mathcal{A}^+$ and $\mathcal{N}^+ \leq \mathcal{A}^+$. By the fact that quantifier-free sentences are preserved by embeddings, we get $\mathcal{A}^+ \models P \cup Q$.²²

Therefore, $P \cup Q$ is a condition. However, by the monotonicity property we then get that $P \cup Q \Vdash \sigma \wedge \neg\sigma$, which is not possible. Thus T^f is complete. $\square$

6.3.17 Corollary. *Let $\mathcal{L} = \{*, ^{-1}, e\}$ and let T be the theory of groups. The finite forcing companion T^f of the theory of groups is a complete $\mathcal{L}$ -theory.*

Proof. This follows from theorem 6.3.16, and our earlier observation that the theory of groups has JEP, as any two groups G, H can be embedded into their direct product $G \times H$. $\square$

6.3.18 Remark. One of the next steps in studying finite forcing is to study particularly interesting structures called **finite generic structures**. One property of a finite generic $\mathcal{L}$ -structure $\mathcal{A}$ is that every $\mathcal{L}$ -sentence that is enforceable in the finite forcing companion for T in $\mathcal{L}^+$ is true in $\mathcal{A}$. It follows that a finite generic structure will be a model of T^f .

6.3.19 Definition. Let $\mathcal{L}$ be a countable language and $\mathcal{L}^+$ an extension by witnesses. Let T be an $\mathcal{L}$ -theory and let N be the finite forcing structure for T in $\mathcal{L}^+$. An $\mathcal{L}^+$ -structure is **finite generic⁺ for T** if and only if it satisfies both of the following:

- (i) Every element of $\mathcal{A}^+$ is named by infinitely many witnesses
- (ii) For every $\sigma \in \text{Sen}(\mathcal{L}^+)$, if $\mathcal{A}^+ \models \sigma$ then there is a condition $P \subseteq \text{Th}(\mathcal{A}^+)$ in $\mathcal{L}^+$ such that $P \Vdash_N \sigma$.

²²If there were a witness c which occurred in both P and Q , it could be the case that $c^{\mathcal{M}^+} = a$ and $c^{\mathcal{N}^+} = b$ but $\mathcal{A} \models a \neq b$. Then, it would not be possible to expand $\mathcal{A}$ to an $\mathcal{L}^+$ -structure $\mathcal{A}^+$ satisfying both $\mathcal{M}^+ \leq \mathcal{A}^+$ and $\mathcal{N}^+ \leq \mathcal{A}^+$, as this would entail the contradiction:

$$\mathcal{A} \models a = c^{\mathcal{M}^+} = c^{\mathcal{A}^+} = c^{\mathcal{N}^+} = b \text{ but } \mathcal{A} \models a \neq b$$

6.3.20 Observation. It is clear from the definition that any finite generic⁺ structure $\mathcal{A}^+$ for an $\mathcal{L}$ -theory T will satisfy $\mathcal{A}^+ \models T^f$, as will its reduct $\mathcal{A}$ to $\mathcal{L}$.

6.3.21 Theorem ([Hod85, Theorem 4.3.1, p. 106]). *Let N be the finite forcing structure in $\mathcal{L}^+$ for an $\mathcal{L}$ -theory T . Then it is N -enforceable that the compiled structure $\mathcal{A}^+$ is finite-generic⁺ for T .*

Proof. We have already seen that it is N -enforceable that every element of the compiled structure is named by infinitely many witnesses. Our strategy is as follows.

For each $\mathcal{L}^+$ -sentence σ and each $P \in N$, we know from 4.3.8 that there must be some $Q \supseteq P$ which forces σ or some $R \supseteq P$ which forces $\neg\sigma$. Therefore, for each sentence σ we can play to make sure that the chain contains a condition P which forces either σ or $\neg\sigma$. Since we know it is enforceable that $\mathcal{A}^+ \models T^+$, this gives us $P \subseteq \text{Th}(\mathcal{A}^+)$. We can then play to make sure that same choice of σ or $\neg\sigma$ is true in the compiled structure. $\square$

Finally, we define finite generic structures, and present without proof the theorem linking finite generic structures with the finite forcing companion. This theorem and its proof can be found in [Hod85, Theorem 4.3.2, pp. 106-107].

6.3.22 Definition. An $\mathcal{L}$ -structure $\mathcal{A}$ is **finite generic for T** if and only if $\mathcal{A} = \mathcal{A}^+ \upharpoonright_{\mathcal{L}}$ for some $\mathcal{L}^+$ -structure $\mathcal{A}^+$ which is finite generic⁺ for T .

6.3.23 Theorem. *An $\mathcal{L}$ -structure $\mathcal{A}$ is finite generic for an $\mathcal{L}$ -theory T if and only if $\mathcal{A} \models T^f$ and for every $\mathcal{L}$ -structure $\mathcal{B}$, if $\mathcal{A} \leq \mathcal{B}$ and $\mathcal{B} \models T^f$ then $\mathcal{A} \prec \mathcal{B}$.*

Proof. See [Hod85, Theorem 4.3.2, pp. 106-107]. $\square$

In [Hod85, p. 77], when introducing the finite forcing companion of a theory, the following statement is made: ‘by forcing with the theory of groups we shall never be able to construct two groups which are not elementarily equivalent’.

We conclude by discussing this. Though we have not explored this in depth, there is an intuitive concept of an ‘enforceable structure’ - an $\mathcal{L}$ -structure $\mathcal{B}$ is **enforceable** if it is enforceable that the reduct of the compiled structure is isomorphic to $\mathcal{B}$. It follows that the enforceable groups are all models of the finite forcing companion T^f of the theory of groups, a complete theory. Thus all the enforceable groups have the same complete theory, and are thus all elementarily equivalent.

In a way, this represents a limitation of finite forcing with the theory of groups (and other theories with JEP), as only a very limited class of groups can be constructed with finite forcing. However, the finite generic groups stand out from this class, as any embedding of a finite generic group into another enforceable group is guaranteed to be elementary.

The finite forcing companion of a non-companionable theory can serve some of the same purposes as the model companion. Even though T^f for a non-companionable $\mathcal{L}$ -theory T will not be model complete, and thus not *all* embeddings of models of T^f are guaranteed to be elementary, we at least know that the embeddings of the finite generic models of T^f are elementary.

References

- [Bia18] Ricardo Bianconi. *Lectures on Model-Theoretic Forcing*. 2018. URL: http://www.ime.usp.br/~matweb/5865/Lectures_Forcing.pdf (visited on 09/14/2025) (cit. on p. 17).
- [CK73] Chen Chung Chang and Howard Jerome Keisler. *Model Theory*. 3rd Edition (1990). Vol. 73. Studies in Logic and the Foundations of Mathematics. North-Holland Publishing Co., 1973, pp. xvi+650 (cit. on pp. 16, 105–107).
- [Fra14] John B. Fraleigh. *A First Course in Abstract Algebra*. New International Edition, Seventh Edition. Pearson, 2014, pp. ii+456 (cit. on p. 15).
- [Hod85] Wilfrid Hodges. *Building Models By Games*. Vol. 2. London Mathematical Society Student Texts. Cambridge University Press, 1985, pp. iv+311 (cit. on pp. 8, 10, 11, 16, 17, 19, 32–34, 40, 43, 49, 52, 55, 58, 59, 61, 67, 68, 72, 77, 81, 82, 85, 86, 91, 94, 96, 98, 100, 104, 108, 112).
- [Hod93] Wilfrid Hodges. *Model Theory*. Vol. 42. Encyclopedia of Mathematics and its Applications. Cambridge University Press, 1993, pp. xiv+772 (cit. on pp. 15, 16, 18, 19, 23, 26, 32, 34, 35, 40, 49, 50, 98–100, 104, 106).
- [JT24] Gareth O. Jones and Marcus Tressl. *Mathematical Logic (MATH33021 Lecture Notes) 2025-26*. 2024. URL: <https://personalpages.manchester.ac.uk/staff/Marcus.Tressl/teaching/MathematicalLogic/MathematicalLogic.pdf> (visited on 09/14/2025) (cit. on pp. 15, 40, 41).
- [Koe23] Peter Koellner. “The Continuum Hypothesis”. In: *The Stanford Encyclopedia of Philosophy*. Ed. by Edward N. Zalta and Uri Nodelman. Winter 2023. Metaphysics Research Lab, Stanford University, 2023. URL: <https://plato.stanford.edu/archives/win2023/entries/continuum-hypothesis/> (visited on 09/15/2025) (cit. on p. 7).
- [Mar02] David Marker. *Model Theory: An Introduction*. Vol. 217. Graduate Texts In Mathematics. Springer, 2002, pp. viii+345 (cit. on p. 15).
- [Mar24] Sophie Marshall. *The Algebraic Closure of a Field (MATH61000 Mini Project)*. 2024 (cit. on p. 18).
- [Mar25] Sophie Marshall. *Omitting Types (MATH61000 Main Project)*. 2025 (cit. on pp. 12, 15, 19, 20, 23, 24, 26, 55, 101, 102).

- [Rob71] Abraham Robinson. “Forcing in Model Theory”. In: *Symposia Mathematica* 5 (1971), pp. 69–82 (cit. on pp. 12, 16, 77, 81, 82, 85, 91, 94, 96).
- [Sar71] Daniel M. Saracino. “Generic Structures and Model Companions”. PhD thesis. Princeton University, 1971 (cit. on pp. 17, 54).
- [Tre10] Marcus Tressl. *Introduction to Model Theory*. 2010. URL: <https://personalpages.manchester.ac.uk/staff/Marcus.Tressl/papers/ModelTheory.pdf> (visited on 09/14/2025) (cit. on pp. 15, 19, 23, 28–30, 32, 33).
- [Zie80] Martin Ziegler. “Algebraisch Abgeschlossene Gruppen”. In: *Studies in Logic and the Foundations of Mathematics* 95 (1980), pp. 449–476 (cit. on pp. 17, 62, 67).